Local exploration laws, initial germs, and affine metrics

Autonomous mathematical investigation, with independent critical checks

29 October 2026

Abstract

A pair of remote signed-energy comparisons controls both driving characteristics. On two bounded physical calibration states this yields the polynomial coercivity bound $\kappa(A)^{p-1} \leq C(1+\mathcal{B}_p(A))$, so signed calibration estimates would also compactify the unknown affine structure. A hereditary Borel arc certificate, stable under local conformal charts in the isotropic case, strengthens the known simple-SLE tensor decoder. These additions do not supply the missing signed estimate from the exploration axioms. Literal strong DMP at time zero makes the initial trace germ trivial. Consequently any family represented using one common prior on smooth global coordinate diffeomorphisms collapses, under translation covariance, to a deterministic affine-SLE_6 metric. No such representation is assumed for the unknown family. A quantitative necessity estimate at visited states identifies the inverse-linear drift and inverse-cubic variance anomalies that the signed-energy bound must eliminate. The regularity bridge is now expressed through exact exit-sign criteria: finite aggregate predictable sign budget is equivalent to integrable finite-variation drift; subquadratic weighted sign energies characterize L^p absolutely continuous drift for $1 < p < 2$. A new conditional rigidity theorem identifies SLE_6 from source-linear expected L^p exit-energy bounds whose constants grow subcritically along mirrored shrinking-slit comparisons; no bounded drift density or initial tangent is assumed. Conditional crossing moments also give bounded cluster clocks along deterministic meshes and finite-mesh analytic-locality characteristic covariance without semimartingality. None of these criteria supplies the missing signed estimate from the exploration axioms. A singular analytic clock example shows why clock-path Taylor regularity alone cannot settle finite-affine clock absolute continuity; a dyadic uniform-integrability criterion identifies a precise increment-measure target. Signed-test drift energy is lower semicontinuous in the affine tensor; fixed-state affine driving laws are continuous under the standing admissibility and lifetime conditions. A compact finite-test selection principle can therefore select a common tensor jointly with signed-energy estimates, if common finite envelopes and a compact ellipticity class can be established. The 29 October priority is to understand affine-SLE_6 driving in arbitrary coordinates before imposing coordinate-sensitive stochastic regularity. Matched-prefix affine driving and capacity admit uniform first variations even for nonsimple filled hulls; the linear capacity response has every temporal exponent below one under the crossing assumptions, and the linear driver response retains every temporal exponent below one-half. The continuum crossing/locality bounds give nondegenerate subsequential adaptive quadratic-activity clocks without semimartingality. Wrong-coordinate semimartingality would force absolute continuity of the affine capacity clock, which remains open. The current priority is continuum classification under explicit RSW-type crossing assumptions, with verification for lattice limits deferred. Corrected source locality and a ray-deletion comparison give a diffusive lower driving tail; combined with the upper crossing estimates, this forces variation index two without assuming semimartingality. Once the signed restart budget is finite, the bracket is absolutely continuous and uniformly elliptic, while singular drift remains a separate issue. Analytic locality transports finite signed budgets and characteristic measures. A Schwarzian test excludes the nontrivial one-force-point SLE families without requiring initial drift tangents. None of these results proves finiteness of the signed budget or the classification. Independent-color resampling and RSW source shields imply an initial ordered trace-germ zero-one law,

with quantitative decorrelation between small prefixes and the full exploration curve. This conclusion does not assume affine-SLE marginals or a limiting domain Markov theorem. Exact coalescence also gives total-variation prefix–suffix decoupling and exponential mixing of annular source-crossing statistics, with a centered strong law that does not require dilation invariance. A universal tensor decoder at analytic regular endpoints, exact suffix coalescence, and source splicing then show that a coherent family of affine-SLE_6 mixtures must use one deterministic common affine metric. The affine-mixture representation itself remains an unproved hypothesis. Separately, a conditional Loewner-driving classification permits arbitrary continuous finite-variation drift with a uniform first-order initial tangent in probability; tightness in law along mirrored slit tests can replace deterministic tangents. Locality and compatible strong DMP then force Brownian driving of variance six. A wrong-affine-coordinate obstruction holds under the weakened deterministic tangent assumptions. A separate signed conditional-bias criterion supplies semimartingality and the required initial tangents without postulating them; its discrete version passes to continuous driver limits without filtration convergence. The requisite bias cancellation and compatible limiting DMP remain unproved for general percolation.

Strong DMP in the usual natural driver filtration automatically transports canonical drift and bracket. Vanishing adapted microscopic correctors can replace raw one-step driving-bias bounds. Annular scale sums admit an exponentially integrable martingale correction and a functional Gaussian limit conditional on a deterministic variance rate. A spherical empirical-law decoder handles arbitrary sample-dependent local C^1 charts; a stationary-cylinder argument supplies dilation ergodicity for the relevant whole-plane SLE model. A canonical tube-cluster measure supplies an affine-stable intrinsic measure class. Whole-plane reversal and natural-length Palm laws then give a Borel tensor decoder from every unlabelled compact subarc of an affine simple SLE, simultaneously under all affine maps and subarc choices. Identifying such arcs on general exploration frontiers remains unresolved. These results leave the corrected three-axiom classification and critical percolation universality open.

1 Current locality and scope

Two colored marked domains that agree near their common source must have identical initial exploration laws until exit from that neighborhood, even if their remote targets differ. This is the user's corrected locality axiom at commit `0e47018`. We fix one color orientation throughout. The terminal-germ construction for the earlier fixed-target axiom is excluded; its original proofs are retained in `archive/original-locality-results.tex` and the detailed research record.

The first theorem below needs only a deletion consequence of corrected locality. It does not claim that locality supplies regular driving drift. For rough restart domains we state the prime-end neighborhood or admissible test-deletion requirement explicitly. All driving clocks use $\partial_t g = 2/(g - W)$ and hence $\mathrm{hcap}(K_t) = 2t$. The standard conformal-map comparison for Loewner chains is derived below; ordinary SLE facts used for affine examples are described in [6, 7]. Continuous Loewner generation and infinite lifetime of affine SLE images were proved in the companion research record [18].

Current priority: continuum classification under RSW. Following the user's clarification of 28 October 2026, the immediate problem is to classify continuum exploration families satisfying strong DMP, corrected source-neighborhood locality, physical translation covariance, and explicitly specified uniform curve-level RSW estimates. Verifying these axioms for percolation subsequential limits is deferred. The expected conclusion permits one fixed affine normalization; no conformal covariance or scale invariance is assumed.

The exploration is an ordered uniform limit of simple curves, not a simple curve. Dense self-touching, boundary contacts, and swallowing are allowed. No embedded nontrivial time interval

is presumed. Ordered prefixes, filled hulls, and unlabelled accessible frontiers are distinct objects. The decoder for known simple SLE arcs with $0 < \kappa < 4$ is a separate model theorem and supplies no simple subarc of the unknown exploration. Continuous Loewner admissibility and stochastic driving regularity are separate questions. At rough restart tips, prime-end access and inherited colored boundary sides must be retained; the source-locality neighborhood convention must be stated explicitly.

2 A near-source drift-boundedness criterion

Status. This is a new conditional rigidity result. It does not assume conformal invariance, Brownian driving, or local-martingale driving. Instead it shows that a locally bounded *initial driving drift*, together with an absolutely continuous drift part and locality, already forces zero drift and variance six. No regularity of quadratic variation is assumed. Strong DMP then identifies the whole driving law. The boundedness and characteristic hypotheses are not derived from RSW or from the three project axioms. A finite-dimensional bulk-jet closure is given as one explicit sufficient hypothesis.

Only a particular deletion consequence of the corrected source-neighborhood locality is needed. Arbitrary target changes and the full force of corrected locality are not used in this theorem. Thus the result is compatible with, but does not exhaust, the corrected research objective.

2.1 Chart-dependent initial characteristics and their covariance

Let $f : \mathbb{H} \to D$ be an inverse chordal conformal chart, with source $f(0) = p$ and target $f(\infty) = q$, both interpreted as prime ends. For the physical ensemble in D, let W^f be its driving process in this chart, with half-plane capacity normalization

$$\partial_t g_t(z) = \frac{2}{g_t(z) - W_t^f}, \qquad W_0^f = 0.$$

Assume that it is a continuous semimartingale in its natural trace filtration, with decomposition

$$W_t^f = M_t^f + \int_0^t b_s^f \, ds, \qquad C_t^f = \langle M^f \rangle_t. \tag{DR1}$$

The drift-density paths b^f are assumed finite and right-continuous, after local stopping if necessary. No absolute continuity or regularity of C^f is assumed. At the initial time suppose

$$b_0^f = B(f) \in \mathbb{R} \tag{DR2}$$

is deterministic. It may depend on the entire physical marked domain and its conformal chart. In particular no Markov property of the single real driver is assumed. A finite deterministic initial variance rate $V(f)$ will be *derived* below from locality and (DR1)–(DR2).

Let $A \subset \overline{\mathbb{H}}$ be a half-plane hull disjoint from 0, let $h : \mathbb{H} \setminus A \to \mathbb{H}$ be hydrodynamically normalized, and set

$$\bar{h}(z) = h(z) - h(0), \qquad f_A = f \circ \bar{h}^{-1}. \tag{DR3}$$

The source is still the same physical prime end, since $\bar{h}(0) = 0$; the target is also unchanged. Put $d = h'(0) > 0$, $e = h''(0)$. Source locality, together with the usual deletion-chart calculation, will give a finite deterministic initial variance rate $V(f)$ and the covariance

$$\boxed{V(f_A) = V(f), \qquad B(f_A) = \frac{B(f)}{d} + \left(\frac{V(f)}{2} - 3\right)\frac{e}{d^2}.} \tag{DR4}$$

These are identities for *initial* characteristics and do not identify the two complete driving laws.

Here are the details without assuming a quadratic-variation density. If h_t is the evolving deletion map, then $du = h_t'(W_t)^2 dt$, and Itô's formula gives

$$d\widehat{W}_{u(t)} = h_t'(W_t)\, dM_t + h_t'(W_t) b_t\, dt + \frac{1}{2} h_t''(W_t)\, dC_t - 3h_t''(W_t)\, dt. \tag{DR5}$$

The -3 comes from

$$\partial_t h_t(z) = \frac{2h_t'(W_t)^2}{h_t(z) - h_t(W_t)} - \frac{2h_t'(z)}{z - W_t}, \qquad (\partial_t h_t)(W_t) = -3h_t''(W_t).$$

The comparison of filtrations is justified by geometric invertibility: from a deleted-domain initial trace one reconstructs its original trace and capacity, so the inverse capacity clock is adapted and its inverse is a stopping-time clock. Locality therefore compares the same trace filtration, not an unrelated enlargement. Adaptedness of an arbitrary increasing clock would not suffice for this step.

Local bracket-regularization lemma (proved). Suppose $h''(0) \neq 0$. Then on a positive initial interval, C_t is absolutely continuous and has a finite right-continuous density. In particular its initial right derivative $V(f)$ exists and is deterministic.

Proof. The deleted-domain driver also has an absolutely continuous drift, $\int \widehat{b}_u\, du$, by (DR1). Uniqueness of the continuous semimartingale decomposition applied to (DR5), after the justified capacity time change, gives the signed-measure identity

$$h_t''(W_t)(dC_t - 6\, dt) = 2\big[h_t'(W_t)^2 \widehat{b}_{u(t)} - h_t'(W_t) b_t\big] dt. \tag{DR5a}$$

The second derivative starts nonzero and remains so on a positive initial interval. On that interval the bracket is absolutely continuous, with

$$v_t = 6 + 2\frac{h_t'(W_t)^2 \widehat{b}_{u(t)} - h_t'(W_t) b_t}{h_t''(W_t)}. \tag{DR5b}$$

The derivative paths of h_t along the continuous driving point are continuous, while the two drift-density paths are right-continuous and finite. Thus v is right-continuous and finite there. In particular

$$V(f) = v_0 = 6 + 2\frac{d^2 B(f_A) - dB(f)}{e} \tag{DR5c}$$

is deterministic. This is an intrinsic initial rate of C, so it does not depend on which nonzero-e deletion was used to derive it. Applying the same time change to bracket gives $V(f_A) = V(f)$. Rearranging (DR5c) gives the drift identity (DR4). □

There is always an available nonzero-e test deletion, for example one vertical slit at any fixed real point different from 0. Right-continuity makes the initial drift and variance rates intrinsic, rather than arbitrary versions of densities at a single time. The bracket density has only been proved locally at the start; its global regularity is not being presumed.

Source-neighborhood convention. In half-plane coordinates these slit deletions preserve a full neighborhood of the source end. Under f this is a prime-end neighborhood. Thus (DR4) is justified on arbitrary rough domains if the corrected locality assumption is formulated for such neighborhoods, or if these particular deletions are explicitly admissible. For regular analytic source boundaries it is ordinary Euclidean neighborhood locality. If locality is instead stated only using Euclidean balls at non-singleton source impressions, the all-rough-domain extension must remain an explicit additional hypothesis.

2.2 Two vanishing slits force both characteristics

For $\epsilon > 0$ take the reflected vertical slits

$$A_\epsilon^\pm = \{\, \pm\epsilon^2 + iy : 0 \le y \le 2\epsilon \,\}. \tag{DR6}$$

Their hydrodynamic maps are

$$h_\epsilon^\pm(z) = \pm\epsilon^2 + \sqrt{(z \mp \epsilon^2)^2 + 4\epsilon^2}, \tag{DR7}$$

with the square-root branch asymptotic to $z \mp \epsilon^2$ at infinity. Let $f_\epsilon^\pm = f \circ (h_\epsilon^\pm - h_\epsilon^\pm(0))^{-1}$. Both sequences are conformal charts of bounded simply connected domains and

$$f_\epsilon^\pm \longrightarrow f \quad \text{locally uniformly in } \mathbb{H}. \tag{DR8}$$

Indeed the slits have diameter 2ϵ and approach 0; directly from (DR7), their maps and inverse maps converge to the identity on compact subsets of $\mathbb{H}$, while $h_\epsilon^\pm(0) \to 0$. Crucially their boundary derivatives do *not* converge to the identity jet:

$$d_\epsilon = (h_\epsilon^\pm)'(0) = \frac{\epsilon}{\sqrt{\epsilon^2+4}} \longrightarrow 0, \qquad e_\epsilon^\pm = (h_\epsilon^\pm)''(0) = \mp\frac{4}{\epsilon(\epsilon^2+4)^{3/2}}. \tag{DR9}$$

In particular

$$\frac{e_\epsilon^\pm}{d_\epsilon^2} = \mp\frac{4}{\epsilon^3\sqrt{\epsilon^2+4}}. \tag{DR10}$$

Thus compact convergence of inverse charts permits very large second-jet changes precisely at the source boundary.

Initial rigidity theorem (proved). Fix f and assume (DR1)–(DR4) for f and the charts $f_\epsilon^\pm$. If

$$\sup_{0<\epsilon<\epsilon_0} \max\{|B(f_\epsilon^+)|, |B(f_\epsilon^-)|\} < \infty \tag{DR11}$$

for some $\epsilon_0 > 0$, then

$$B(f) = 0, \qquad V(f) = 6. \tag{DR12}$$

No continuity of B is required.

Proof. Write $b = B(f)$, $v = V(f)$ and $b_\epsilon^\pm = B(f_\epsilon^\pm)$. Equation (DR4) gives

$$b_\epsilon^\pm = \frac{b}{d_\epsilon} + \left(\frac{v}{2} - 3\right)\frac{e_\epsilon^\pm}{d_\epsilon^2}.$$

Since $e_\epsilon^- = -e_\epsilon^+$,

$$b_\epsilon^+ + b_\epsilon^- = \frac{2b}{d_\epsilon}, \qquad b_\epsilon^+ - b_\epsilon^- = (v-6)\frac{e_\epsilon^+}{d_\epsilon^2}. \tag{DR13}$$

The first right side is bounded only if $b = 0$, since $d_\epsilon \to 0$. The second is bounded only if $v = 6$, since its multiplier has magnitude asymptotic to $2\epsilon^{-3}$. This proves (DR12). □

Exact instability and weaker rate conditions. For real numbers U, C one has $\max\{|U + C|, |U - C|\} = |U| + |C|$. Hence the two drift covariances give the exact identity

$$\boxed{\max\{|b_\epsilon^+|, |b_\epsilon^-|\} = \frac{|b|\sqrt{\epsilon^2+4}}{\epsilon} + \frac{4|v/2-3|}{\epsilon^3\sqrt{\epsilon^2+4}}.} \tag{DR14}$$

There is no possible cancellation between the two reflected tests. In particular, if the displayed maximum is $o(\epsilon^{-3})$, then $v = 6$. If it is $o(\epsilon^{-1})$, then $b = 0$ and $v = 6$. Boundedness is more than sufficient. Conversely, if $v \neq 6$, its leading blow-up is $|v - 6|\epsilon^{-3}$; if $v = 6$ and $b \neq 0$, its leading blow-up is $2|b|\epsilon^{-1}$. This is a concrete, quantitative obstruction to any alternative semimartingale driving law with tame initial drift under source perturbations.

2.3 Local boundedness in marked-chart topology suffices

Equip inverse marked charts with locally uniform convergence on $\mathbb{H}$, retaining the fixed preimages $0, \infty$ of source and target. If B is locally bounded at every f in this topology, then (DR8) implies (DR11). In particular continuity of B is sufficient, but unnecessarily strong. This is a hypothesis about the driving characteristic itself; weak convergence of curve laws alone does not imply it.

Conditional classification theorem (proved). Suppose a Borel family of continuous Loewner-generating exploration laws on all bounded simply connected marked prime-end domains has strong DMP and source-neighborhood locality (including the test deletions (DR6) in each chart). Suppose its drivers satisfy (DR1)–(DR2), with finite right-continuous drift-density paths and the usual restart compatibility of characteristic kernels, with no initial assumption on quadratic variation. If the initial drift map B is locally bounded in marked inverse- chart topology, then every driving process is

$$W_t^f = \sqrt{6}\, B_t,$$

and the family is chordal SLE_6 in these conformal coordinates.

Proof. The initial rigidity theorem gives $B(f) = 0$ and the derived $V(f) = 6$ for every marked inverse chart. At any deterministic rational time r, condition on the initial exploration and use strong DMP in its remaining marked domain and current conformal chart. The centered future driver is exactly the restarted driver; its initial drift is b_r^f, by right-continuity. The restarted law has initial drift zero, so $b_r^f = 0$ almost surely. Countably many rational times and right-continuity give $b_t^f = 0$ throughout each localized lifetime. Therefore all domain chart drivers are continuous local martingales.

For completeness the no-density local-martingale locality argument then identifies their brackets. With both drifts zero, (DR5a) becomes $h_t''(W_t)(dC_t - 6\,dt) = 0$. A nonzero-$e$ vertical deletion yields $C_t = 6t$ on a positive initial interval. Let

$$T = \inf\{t \geq 0 : C_t - 6t \neq 0\}.$$

This is an entrance stopping time of a continuous adapted process into an open set, in the usual right-continuous filtration. The initial zero interval gives $T > 0$. If $T < \infty$, continuity gives $C_T = 6T$ and the first-deviation definition gives nonzero values arbitrarily soon after T. Strong DMP restarted at T gives another positive zero interval, a contradiction. Thus $C_t = 6t$ throughout. Lévy's characterization gives Brownian driving, and Loewner generation identifies SLE_6. □

The Borel-kernel condition means that the family admits the standard strong conditional-kernel interpretation on random slit domains, and that the characteristics can be chosen consistently on that family. It is not a claim that deterministic-domain convergence of discrete explorations proves this property. Translation invariance is not needed in this conditional theorem. If its hypotheses hold only after one fixed physical affine normalization, the conclusion is the corresponding fixed-affine SLE_6 family; an affine change of physical coordinates is not presumed to preserve semimartingale driving in an arbitrary preselected complex structure.

2.4 A finite-dimensional closure that is completely classified

For example, fix finitely many interior points $z_j \in \mathbb{H}$ and finitely many nonnegative derivative orders m_j. Let

$$J(f) = \big(f^{(k)}(z_j) : 1 \le j \le N, 0 \le k \le m_j\big).$$

Suppose the initial drift has a representation

$$B(f) = \mathcal{B}(J(f)), \tag{DR15}$$

where $\mathcal{B}$ is locally bounded at every realized jet vector. The bracket has no assumed density or finite-dimensional representation; its initial rate is supplied by the local regularization lemma. Locally uniform convergence of conformal maps implies convergence of all these interior jets by Cauchy's formula, so (DR15) implies local boundedness of B in marked-chart topology. The conditional classification theorem therefore applies: within (DR1)–(DR2), every source-local strong-DMP family with such a locally bounded finite-dimensional *bulk-jet drift closure* is SLE_6.

This statement does not cover a closure depending on unbounded source boundary derivatives such as $f'(0)$ or a boundary pre-Schwarzian. Those quantities can change without control under (DR8), and can carry exactly the singular terms in (DR13). Nor does it show that true percolation limits have semimartingale driving in a fixed physical complex structure. The new rigorous reduction is that any non-SLE candidate with regular characteristic rates must violate the displayed near-source drift bound, in the explicit ϵ^{-1} or ϵ^{-3} manner above.

2.5 The criterion also detects a wrong affine coordinate system

Corollary (proved). Let A be an orientation-preserving real-linear map which is not a Euclidean similarity, and let the physical ensemble be the fixed-affine pullback

$$P_{D,p,q} = (A^{-1})_* \,\mathrm{Law}(\mathrm{SLE}_6(AD; Ap, Aq)).$$

Although this family satisfies corrected source-neighborhood locality and strong DMP, in the original Euclidean complex structure it cannot satisfy both the regular semimartingale characteristic assumptions (DR1)–(DR2) in all marked domains and the locally bounded initial-drift condition above.

Proof. If both properties held, the conditional classification theorem would make this ensemble ordinary SLE_6 in the original complex structure. The single-domain affine identifiability theorem would then make A a similarity, a contradiction. □

Thus the drift criterion selects a conformal structure. It is not a coordinate-free consequence of coarse crossing bounds: bounded affine changes preserve such bounds, whereas a wrong affine structure necessarily violates at least one of the criterion's additional analytic hypotheses. The

corollary does not decide whether the failure is semimartingality, right-continuity of the drift density, deterministic initial drift, or boundedness of the initial drift; distinguishing these failures is a separate question.

2.6 What corrected locality says at a regular analytic source germ

Here is a separate necessary local structure statement that uses the full corrected locality, including changes of the distant target. It concerns initial characteristics only, not existence of a process from arbitrary coefficient choices.

Suppose the physical boundary is analytic near p and the source colors are fixed there. Let G denote this embedded colored source germ. For a chart f extending conformally across 0, define

$$a(f) = |f'(0)| > 0, \qquad P(f) = \operatorname{Re} \frac{f''(0)}{f'(0)}. \tag{DR16}$$

If another physical marked domain has the same source germ but any distant target, let k be its inverse chordal chart with $k(0) = p$. The local map $h = k^{-1} \circ f$ is analytic across 0, takes the real boundary interval to a real interval, and has $h(0) = 0$, $d = h'(0) > 0$, $e = h''(0) \in \mathbb{R}$. The two initial physical traces agree by corrected locality. The local Loewner coordinate comparison is exactly (DR4), with the same derivation from pole cancellation and Itô's formula; a globally defined deletion map is unnecessary. Consequently $V(f) = V(k)$ depends only on G.

Proposition (proved). Under the semimartingale assumptions above, corrected locality at analytic source germs is equivalent at the level of initial characteristic covariance to the existence of two source-germ quantities $V(G)$ and $\beta(G)$ such that

$$\boxed{V(f) = V(G), \qquad B(f) = a(f)\,\beta(G) + \left(3 - \frac{V(G)}{2}\right) P(f).} \tag{DR17}$$

Here "equivalent at the level of covariance" means precisely that every covariant initial-characteristic assignment has this form, and every assignment of this form obeys the local chart covariance identity. It does not mean that arbitrary V, β yield a domain Markov exploration family.

Proof. Since $k = f \circ h^{-1}$, the chain rule gives

$$a(k) = \frac{a(f)}{d}, \qquad P(k) = \frac{P(f)}{d} - \frac{e}{d^2}.$$

Write $c = V(G)/2 - 3$. Combining these transformations with $B(k) = B(f)/d + ce/d^2$ yields

$$\frac{B(k) + cP(k)}{a(k)} = \frac{B(f) + cP(f)}{a(f)}.$$

This common value is the source-germ invariant $\beta(G)$, proving (DR17). Conversely direct substitution gives the required covariance. □

Physical translation invariance makes $V(G)$ and $\beta(G)$ invariant under translations of the colored germ. If dilation covariance is separately assumed, then $V(\lambda G) = V(G)$ and $\beta(\lambda G) = \lambda^{-1}\beta(G)$. For a straight source-boundary germ, which is unchanged up to translation by dilation, this would force $\beta(G) = 0$; the dimensionless variance can still depend on the boundary orientation before an affine metric has been identified.

The boundary pre-Schwarzian term in (DR17) explains the quantitative anomaly in the vanishing-slit test. Even when the physical source germ is unchanged, $P(f_\epsilon)$ may grow as ϵ^{-3} while $f_\epsilon \to f$ on every interior compact set. Thus source locality by itself gives a connection-type transformation law, rather than vanishing drift. The locally bounded bulk closure theorem removes exactly this uncontrolled boundary dependence. The missing global step is to understand how DMP constrains these local assignments at rough, dynamically created source germs, where the analytic boundary jet in (DR16) no longer exists.

3 Rigidity from an averaged initial finite-variation characteristic

Scope. This conditional theorem weakens the preceding driving-regularity premise. The canonical finite-variation characteristic need not be absolutely continuous and need not possess a right-continuous density. We assume only a deterministic first-order tangent in probability, uniformly on shrinking initial intervals. No such property is derived from RSW or the three axioms.

Let the driver in inverse chart f have canonical decomposition

$$W^f = M^f + A^f, \qquad W_0^f = M_0^f = A_0^f = 0, \tag{FV1}$$

where M^f is a continuous local martingale and A^f is continuous, adapted, and locally of finite variation. Write $C^f = \langle M^f \rangle$. Assume for every deterministic state and inherited chordal chart that a finite deterministic number $B(f)$ satisfies

$$\frac{1}{t} \sup_{0 \le s \le t} |A_s^f - B(f)s| \xrightarrow[t \downarrow 0]{\mathbb{P}} 0. \tag{FV2}$$

An almost-sure derivative $A_t^f/t \to B(f)$ implies (FV2), but (FV2) is weaker. No bound on $\mathrm{TV}(A^f; [0,t])/t$ is assumed.

Lemma 3.1 (A small finite-variation displacement in an analytic field)**.** *Let $W = M + A$ as in (FV1), let $C = \langle M \rangle$, and suppose $\sup_{s \le t} |A_s| = O_{\mathbb{P}}(t)$ and $A_t/t \to B$ in probability. Let $H_t(x)$ be a random field analytic in x on a fixed neighborhood of zero for a positive initial time interval. Suppose it has absolutely continuous finite variation in t, and, after localization, its first three spatial derivatives and the first spatial derivative of $\partial_t H_t$ are bounded there. Assume $H_t'(x) \to d$ as $(t,x) \to (0,0)$. Then*

$$\frac{1}{t} \int_0^t H_s'(W_s)\, dA_s \longrightarrow dB \quad \textit{in probability.} \tag{FV3}$$

Proof. Localize so M, A, W stay in the spatial neighborhood. Apply the finite-variation Itô–Wentzell formula to $F_t(x) = H_t(x + A_t) - H_t(x)$ at $x = M_t$. It gives

$$\begin{aligned} \int_0^t H_s'(W_s)\, dA_s = {} & H_t(M_t + A_t) - H_t(M_t) \\ & - \int_0^t [H_s'(M_s + A_s) - H_s'(M_s)]\, dM_s \\ & - \frac{1}{2} \int_0^t [H_s''(M_s + A_s) - H_s''(M_s)]\, dC_s \\ & - \int_0^t [(\partial_s H_s)(M_s + A_s) - (\partial_s H_s)(M_s)]\, ds. \end{aligned}$$

The first term divided by t converges to dB by the mean-value formula. The second term is a continuous local martingale N_t with bracket at most $K^2 \sup_{s \le t} |A_s|^2 C_t$. Thus $N_t/t \to 0$ in

probability: stop when $|A_s| > Lt$ or $C_s > \eta$, apply the martingale maximal inequality, then let $t \downarrow 0$, $\eta \downarrow 0$, and $L \uparrow \infty$. Here $C_t \to 0$ almost surely and $\sup |A_s|/t$ is tight. The third term divided by t is bounded by $K \sup |A_s| C_t/t = o_{\mathbb{P}}(1)$; the fourth is bounded by $K \sup |A_s| = o_{\mathbb{P}}(1)$. No total-variation estimate was used. □

Proposition 3.2 (Initial covariance without a drift density)**.** *Suppose source locality allows one test deletion with hydrodynamic map h, $d = h'(0) > 0$, $e = h''(0) \neq 0$, and both original and deleted drivers obey (FV1)–(FV2), with their canonical characteristics in the same trace filtration. Then*

$$C_t^f/t \longrightarrow V(f) \in [0, \infty) \quad \textit{in probability,}$$

where $V(f)$ is deterministic, and

$$V(f_A) = V(f), \qquad B(f_A) = \frac{B(f)}{d} + \left(\frac{V(f)}{2} - 3\right)\frac{e}{d^2}. \tag{FV4}$$

Proof. Use the same evolving deletion maps and stopping-time capacity clocks as in (DR5), now without assuming a drift density. Put $p_t = h_t'(W_t)$, $q_t = h_t''(W_t)$ and $u(t) = \int_0^t p_s^2 ds$. Uniqueness of the canonical semimartingale decomposition gives

$$\widehat{A}_{u(t)} = \int_0^t p_s \, dA_s + \frac{1}{2}\int_0^t q_s \, dC_s - 3\int_0^t q_s \, ds. \tag{FV5}$$

The deletion field satisfies the analytic-field lemma after localization; its Loewner field derivative and spatial derivatives are bounded near the uncollided source. Therefore the first integral divided by t tends to $dB(f)$, without assuming C_t/t is tight. Also $u(t)/t \to d^2$. Condition (FV2) for the deleted state implies $\widehat{A}_{u(t)}/t \to d^2 B(f_A)$: on the event $u(t) \leq Kt$, bound the error by $t^{-1} \sup_{s \leq Kt} |\widehat{A}_s - B(f_A)s|$, and choose fixed $K > d^2$.

Since $q_t \to e \neq 0$, (FV5) first implies C_t/t is tight: with probability tending to one q_s has fixed sign and magnitude at least $|e|/2$ on $[0, t]$, so $|\int q_s dC_s| \geq |e| C_t/2$. Consequently

$$\frac{1}{t}\int_0^t q_s \, dC_s - e\frac{C_t}{t} \longrightarrow 0 \quad \text{in probability.}$$

Taking (FV5) divided by t now gives

$$V(f) = 6 + \frac{2\{d^2 B(f_A) - dB(f)\}}{e}.$$

This is intrinsic, nonnegative, and deterministic. The bracket relation $\widehat{C}_{u(t)} = \int_0^t p_s^2 dC_s$ gives $V(f_A) = V(f)$. For this last step the tangents of $C, \widehat{C}$ may be evaluated at asymptotically deterministic clocks by monotonic squeezing; convergence in probability of an arbitrary signed process would not justify that step. □

Lemma 3.3 (A finite-variation process with zero right tangent at every stopping time)**.** *Let $A_0 = 0$ be a continuous adapted locally finite-variation process. If for every bounded stopping time τ before lifetime*

$$\frac{A_{\tau+h} - A_\tau}{h} \longrightarrow 0 \quad \textit{in probability as } h \downarrow 0, \tag{FV6}$$

then A is indistinguishably zero.

Proof. Work on a compact localized interval. Pathwise put $V_t = \mathrm{TV}(A;[0,t])$, and use its generalized inverse $\theta(u) = \inf\{t : V_t \geq u\}$ for $0 < u < V_T$. The usual arclength parametrization of a continuous finite-variation path is $G(u) = A_{\theta(u)}$: it is 1-Lipschitz, satisfies $A_t = G(V_t)$, and has $|G'(u)| = 1$ for Lebesgue almost every u. Indeed the variation of G on every interval is its interval length. The monotone function θ has a finite derivative $\theta'(u) \in [0,\infty)$ for almost every u. Exclude also the countably many levels corresponding to nontrivial flat intervals of V.

At a level where these properties hold, let $s = \theta(u)$ and $v(h) = V_{s+h} - u$. Then $v(h) > 0$ and $v(h) \to 0$ as $h \downarrow 0$. The generalized-inverse bounds $\theta((u+v(h))-) \leq s+h \leq \theta((u+v(h))+)$ and differentiability of θ at u give $h/v(h) \to \theta'(u)$. Combining this with differentiability of G gives

$$\lim_{h\downarrow 0} \frac{A_{s+h} - A_s}{h} = \begin{cases} G'(u)/\theta'(u), & \theta'(u) > 0, \\ \mathrm{sgn}(G'(u))\,\infty, & \theta'(u) = 0. \end{cases} \tag{FV7}$$

Thus the extended right derivative is nonzero at almost every variation level, including levels whose inverse belongs to a singular part of dV.

For deterministic $u > 0$ use the stopping times $\tau_u = \inf\{t : V_t \geq u\}$, truncated at a deterministic endpoint T. Fubini and the preceding pathwise statement show that for Lebesgue almost every deterministic u, on $\{\tau_u < T\}$ the quotient in (FV6) has a nonzero or infinite pathwise limit almost surely. But (FV6) gives convergence in probability to zero at this stopping time, which forbids such an event of positive probability. Thus $\mathbb{P}(\tau_u < T) = 0$ for Lebesgue almost every u, and $\mathbb{E}[V_T] = \int_0^\infty \mathbb{P}(\tau_u < T)du = 0$ after localizing V_T to be integrable. Hence $V_T = 0$ almost surely. Localization and a countable exhaustion of the lifetime prove the claim. □

Theorem 3.4 (Conditional classification from a finite-variation initial tangent)**.** *Suppose a Borel family on all relevant marked slit states has strong DMP and source locality, including the same mirrored test deletions (DR6). Suppose canonical decompositions (FV1), compatible with conditional restart in the inherited chordal charts, obey (FV2). If $B(f)$ is locally bounded in inverse-chart topology, or just satisfies the paired slit bound (DR11) at every state, the family is chordal SLE$_6$ in these coordinates.*

Proof. Proposition 3.2 gives exactly (DR4), hence the reflected-slit sum and difference argument gives $B(f) = 0$, $V(f) = 6$ at every deterministic state.

At a stopping time τ, the centered future driver is the driver of the remaining physical state in the inherited conformal chart. Its capacity parameter is exactly $s = t - \tau$ in the original half-plane coordinate: recentering by W_τ introduces no additional scale clock. Strong DMP and conditional-characteristic compatibility identify its canonical finite-variation part with $A_{\tau+s} - A_\tau$ under the regular conditional law. Applying (FV2) with $B = 0$, and integrating these conditional probabilities over the random state, gives (FV6). Lemma 3.3 therefore makes $A = 0$.

The same conditional restart argument gives $(C_{\tau+h} - C_\tau)/h \to 6$ in probability at every stopping time. Apply Lemma 3.3 to the continuous finite-variation process $C_t - 6t$: it is identically zero. Thus $W = M$ and $\langle M\rangle_t = 6t$ globally. Lévy's characterization and continuous Loewner generation identify SLE$_6$. □

Strict weakening and remaining limitation. Even an absolutely continuous drift with density $b_s = B + \sin(1/s)$, $s > 0$, has no right limit at zero, whereas $\int_0^t b_s ds = Bt + O(t^2)$ and therefore satisfies (FV2). The theorem also allows an a priori singular finite-variation drift; its exclusion is a conclusion, not an assumption. A claim that an actual exploration family has these initial characteristic tangents remains unproved. The theorem still depends on choosing one correct physical conformal structure and on tame behavior under the source slit tests.

3.1 The wrong-coordinate obstruction also only needs finite-variation tangents

Scaling lemma with probability tangents. The exact-scaling lemma (SM1)–(SM4) remains true if its drift assumption is replaced by continuous finite variation and (FV2), and its bracket assumption by $C_t/t \to v$ in probability for deterministic v. Indeed (FV2) makes $\sup_{s\le T} |c^{-1/2} A_{cs}| = O_{\mathbb{P}}(\sqrt{c}) \to 0$. Since C is increasing, a finite deterministic mesh and monotonic squeezing show

$$\sup_{s\le T} |C_{cs}/c - vs| \longrightarrow 0 \quad \text{in probability.}$$

The continuous-martingale functional central limit argument in the earlier scaling lemma then applies unchanged. Exact scaling makes the whole law Brownian with variance v.

Local transfer of the tangent. In Proposition 3.2, the analytic-field argument actually gives the uniform version

$$\frac{1}{t} \sup_{s\le t} \left| \int_0^s h_r'(W_r)\, dA_r - dB(f)s \right| \longrightarrow 0 \quad \text{in probability.}$$

Use the martingale maximal inequality in its proof, and the same uniform initial bound for A. Monotonic squeezing similarly makes $C_{ts}/t \to V(f)s$ uniformly on compact s intervals in probability. Equation (FV5) therefore transfers (FV2) through a local analytic chart change, once the initial bracket tangent has been obtained; it also transfers its inverse, since the first chart derivative is positive. No total-variation rate bound is introduced by this transfer.

Corollary 3.5 (Wrong-affine obstruction under finite-variation tangents)**.** *Consider the canonical fixed-affine SLE_6 family with matrix A on all bounded marked domains. If every Euclidean chart driver admits (FV1)–(FV2), then A is a Euclidean similarity. No local boundedness of $B(f)$ is needed.*

Proof. Choose the bounded flat-source domain and one test deletion used in the bounded-domain obstruction theorem. Proposition 3.2 gives its deterministic bracket tangent in probability. Couple its initial physical trace by known affine-SLE locality to the canonical half-plane affine-SLE trace. The local analytic chart transfer just proved gives a finite-variation decomposition and uniform deterministic drift tangent for the half-plane driver, and the same deterministic bracket tangent. Its exact Brownian scaling and the extended scaling lemma make its law ordinary Brownian driving. The boundary-swallowing-race and affine identification arguments in the half-plane obstruction theorem then give variance six and force A to be a similarity. □

3.2 A tightness-in-law variant without deterministic initial tangents

Proposition (proved). There is a useful alternative formulation of the preceding theorem. Allow (FV2) to have a finite random tangent β^f rather than a deterministic $B(f)$. Assume that for every fixed chart f the family of initial tangents in its two shrinking slit domains is uniformly tight:

$$\lim_{K\to\infty} \sup_{0<\epsilon<\epsilon_0} \max_{\sigma\in\{+,-\}} \mathbb{P}\{|\beta^{f_\epsilon^\sigma}| > K\} = 0. \tag{FV8}$$

Then the same classification follows, with all other family and restart hypotheses unchanged.

The analytic-field lemma and initial covariance proof apply verbatim to finite random tangents, producing an intrinsic finite random bracket tangent V^f. For each fixed ϵ, source locality and the invertible local coordinate changes put the original and both deleted initial traces on one probability

space. Only their initial germs are needed, so no simultaneous extension of this coupling past their exit times is required. On that space the covariances give

$$\beta^{f_\epsilon^+} + \beta^{f_\epsilon^-} = 2\beta^f/d_\epsilon, \qquad \beta^{f_\epsilon^+} - \beta^{f_\epsilon^-} = (V^f - 6)e_\epsilon^+/d_\epsilon^2.$$

Uniform tightness of the two marginals makes their sum and difference uniformly tight under any coupling. Since $d_\epsilon \to 0$ and $|e_\epsilon^+|/d_\epsilon^2 \to \infty$, these identities force $\beta^f = 0$ and $V^f = 6$ almost surely. Formally, for $a > 0$,

$$\mathbb{P}\{|\beta^f| > a\} \le \sum_{\sigma=\pm} \mathbb{P}\{|\beta^{f_\epsilon^\sigma}| > a/d_\epsilon\} \longrightarrow 0,$$

and the identical argument with threshold $a|e_\epsilon^+|/(2d_\epsilon^2)$ gives $V^f = 6$. The stopping-time lemma then kills the finite-variation drift and identifies the bracket as before. Local boundedness of a deterministic drift tangent is one special case of (FV8); no deterministic initial-tangent premise is otherwise necessary for this version.

4 Canonical characteristics under a genuine strong restart law

The strong DMP required above is substantive. Once it is an identity of regular conditional laws in the usual natural driver filtration, however, compatibility of the continuous canonical characteristics follows from uniqueness. It is useful to separate these two issues.

Proposition 4.1 (Strong DMP transports continuous characteristics)**.** *Let $\{P_f\}$ be a Borel family of laws on $C_0([0,\infty),\mathbb{R})$. In each state the coordinate process has a continuous canonical decomposition $W = M^f + A^f$ in its usual natural filtration, where A^f has finite variation on compact intervals. Write $C^f = \langle M^f\rangle$. Fix an initial state f and a bounded stopping time τ. Suppose f_τ is measurable at τ and the strong restart identity is literally*

$$\mathcal{L}_{P_f}\big((W_{\tau+t} - W_\tau)_{t\ge 0} \mid \mathcal{F}_\tau^f\big) = P_{f_\tau}. \tag{KR1}$$

Then, for almost every conditioned history, the shifted finite-variation process $A^f_{\tau+t} - A^f_\tau$ and bracket $C^f_{\tau+t} - C^f_\tau$ are the canonical characteristics of the restarted law. This is equality up to indistinguishability under that conditional law. No continuity of $f \mapsto P_f$ is required.

The same conclusion applies to an inherited Loewner chart when the future coordinate in (KR1) is exactly that inherited driving process. Changing to a different chart or clock still requires the appropriate invertible adapted coordinate transformation.

Proof. Write $P = P_f$, $M = M^f$, $A = A^f$, and $C = C^f$. The continuous adapted processes M, A, C have raw Borel adapted versions, up to one P-evanescent set. Indeed, at a positive time approximate from strictly earlier rational times; a variable measurable in the usual filtration at $r < t$ has a version measurable in the raw filtration at some time strictly between r and t. Make these choices on the countable rational approximation grid and take the continuous limits. Their values at zero are fixed. On exceptional paths retain the raw adapted pathwise-limit definitions, rather than zeroing a nonadapted exceptional set. After conditioning, freeze the stopped past, τ, and any additional information in $\mathcal{F}_\tau$. Thus the shifted processes are adapted Borel functionals of the future coordinate $U_t = W_{\tau+t} - W_\tau$ alone.

Choose stopping times $\sigma_n \uparrow \infty$ such that M^{σ_n} and $(M^2 - C)^{\sigma_n}$ are bounded martingales; for example stop at time n or at the first time $|M| + C$ reaches n. For rational $0 \le r < s$ and a bounded cylinder test H of U at rational times at most r, optional sampling and the tower property give

$$\mathbb{E}_P\Big[(M_{(\tau+s)\wedge\sigma_n} - M_{(\tau+r)\wedge\sigma_n})H \mid \mathcal{F}_\tau^f\Big] = 0. \tag{KR2}$$

Use a countable cylinder class, for example indicators of finite intersections of rational half-lines. Disintegrating all these identities at once gives, outside one null set of histories, their martingale identities under the conditional future law. Monotone classes and continuity extend them to every time and bounded past test. The bounded martingales remain martingales after usual augmentation by conditioning at times decreasing to the desired time.

Under any such conditional law put $T_n = (\sigma_n - \tau)^+$. The stopped shifted martingale $M_{\tau+t\wedge T_n} - M_\tau$ is the process in (KR2) with its initial value subtracted: if $\sigma_n \le \tau$ both are identically zero; otherwise they coincide. The T_n are stopping times in the conditional future natural filtration, increase to infinity, and hence $M_{\tau+t} - M_\tau$ is a continuous local martingale there. The same argument applied to $M^2 - C$, subtracting the frozen cross term $2M_\tau(M_{\tau+t} - M_\tau)$, identifies its bracket as $C_{\tau+t} - C_\tau$.

The shifted A is adapted, continuous, and of finite variation, so it is predictable. Consequently

$$U_t = (M_{\tau+t} - M_\tau) + (A_{\tau+t} - A_\tau)$$

is a canonical decomposition in the conditional future natural filtration. By (KR1) this is the usual natural filtration of the coordinate process under P_{f_τ}. Uniqueness of the continuous canonical decomposition and bracket proves the claim. A separate jointly measurable choice of characteristics over every deterministic state is unnecessary for this almost-every-history conclusion. □

Consequence for the driving criteria. Under the genuine strong restart identity (KR1), the explicit canonical-restart-compatibility premises in the earlier conditional driving theorems are redundant. The locality identification of natural trace filtrations and the existence of the strong conditional kernels remain necessary premises of those arguments. This reduction does not deduce strong DMP from deterministic-domain convergence.

4.1 Pointwise convergence, even of continuous kernels, is insufficient

Here is a small counterexample to an otherwise tempting kernel-transfer argument. Let V take the values $-1, 1$ with equal probability and put $I_n = V$, $D_n = V/n$, and $Y_n = V$. Choose continuous functions $g_n : \mathbb{R} \to [-1, 1]$ supported in the union of intervals of radius $1/(4n^2)$ around $-1/n$ and $1/n$, with $g_n(1/n) = 1$ and $g_n(-1/n) = -1$. Define $K_n(d) = \delta_{g_n(d)}$. These are weakly continuous probability kernels, and for every fixed deterministic d,

$$K_n(d) \longrightarrow \delta_0 =: K(d). \tag{KR3}$$

The limiting kernel K is continuous. Moreover the exact conditional identity $\mathcal{L}(Y_n \mid I_n, D_n) = K_n(D_n)$ holds. Nevertheless $(I_n, D_n, Y_n) \to (V, 0, V)$ almost surely, and $\mathcal{L}(V \mid V, 0) = \delta_V \neq K(0)$. Thus pointwise convergence at every deterministic state, and continuity of every kernel including the limit, do not justify a conditional limit at random states. Uniform equicontinuity, sequential convergence along the actually visited states, or another quantitative substitute is essential.

Lemma 4.2 (An annealed substitute for sequential kernel convergence)**.** *Let the past, state, and future variables take values in fixed Polish spaces. Suppose $(I_n, D_n, Y_n) \Rightarrow (I, D, Y)$ and $\mathcal{L}(Y_n \mid I_n, D_n) = K_n(D_n)$. Suppose K is a Borel kernel such that $d \mapsto K\varphi(d)$ is continuous at D almost surely, for each member φ of a countable bounded continuous convergence-determining class on the future space. It suffices that*

$$\mathbb{E}\big|K_n\varphi(D_n) - K\varphi(D_n)\big| \longrightarrow 0 \tag{KR4}$$

for each such φ. Then $\mathcal{L}(Y \mid I, D) = K(D)$.

Proof. For any bounded continuous test h of (I_n, D_n), replace $K_n\varphi(D_n)$ in the exact finite-n conditional identity by $K\varphi(D_n)$. The error is bounded by $\|h\|_\infty$ times (KR4). The continuous mapping theorem, with its almost-sure-continuity version, then passes the resulting identity to the limit. A monotone class in h, followed by the countable measure-determining class in φ, identifies the conditional law. □

Condition (KR4) only asks for a stability estimate on domains sampled by the exploration. It can therefore be more economical than a theorem uniform over every rough slit domain. It has not been proved here for percolation explorations. For strong DMP it must hold for the relevant stopping times, or be combined with an independent strong-Markov extension theorem.

5 Rigidity from conditional first-moment bounds

Scope. The next criterion replaces the a priori semimartingale and initial-tangent premises by conditional first-moment bounds. It does not derive those bounds from RSW. They must hold in one fixed physical conformal structure, with the stated control along the specified shrinking slit tests. No quadratic-moment bound or initial drift derivative is assumed.

Lemma 5.1 (Conditional bias produces a bounded drift density)**.** *Let X be a bounded continuous adapted real process, with $X_0 = 0$, in a usual filtration. Suppose a finite deterministic K satisfies*

$$|\mathbb{E}[X_t - X_s \mid \mathcal{F}_s]| \le K(t-s) \quad \text{almost surely,} \qquad 0 \le s \le t. \tag{BI1}$$

Then $X = M + \int_0^{\cdot} b_r dr$, where M is a continuous local martingale and b is predictable with $|b| \le K$ almost everywhere.

Proof. On a fixed finite time interval, $X_t + Kt$ and $-X_t + Kt$ are class-D submartingales. Their Doob–Meyer decompositions have increasing predictable parts A^+ and A^-. Adding the decompositions and using uniqueness gives $A^+ + A^- = 2Kt$. Thus A^+ and A^- are absolutely continuous, with densities in $[0, 2K]$. The decomposition of X has drift density $b = dA^+/dt - K \in [-K, K]$. Since X and its drift are continuous, its martingale part is continuous. Uniqueness patches the decompositions over finite time intervals. □

Proposition 5.2 (Controlled rate bounds force the tangents to exist)**.** *Use the mirrored shrinking slits $A_\epsilon^\pm$ in (DR6), with initial derivatives d_ϵ and $e_\epsilon^\pm$ in (DR9). Assume the driver in a state f, and the drivers in all these deleted states, have canonical continuous-semimartingale decompositions with absolutely continuous drift. In each state the drift density is bounded on some strictly positive initial stopping interval. The interval may depend on the state and sample. Let $K_\epsilon < \infty$ be a deterministic bound common to the two deleted states. If $d_\epsilon K_\epsilon \to 0$ along a sequence tending to zero, then in the original state*

$$\frac{1}{t}\sup_{s \le t}|A_s| \longrightarrow 0, \qquad \frac{C_t}{t} \longrightarrow 6 \quad \text{almost surely as } t \downarrow 0. \tag{BI2}$$

Here A is canonical drift and C is the martingale bracket. No right continuity or initial tangent of the drift densities is presumed.

Proof. By locality, compare the original and each deleted driver using the same trace filtration and the stopping-time capacity clocks. With $p_t^\sigma = (h_t^\sigma)'(W_t)$ and $q_t^\sigma = (h_t^\sigma)''(W_t)$, uniqueness of canonical characteristics gives

$$\frac{1}{2}q_t^\sigma(dC_t - 6dt) = \big[(p_t^\sigma)^2 \widehat{b}^\sigma_{u_\sigma(t)} - p_t^\sigma b_t\big]dt. \tag{BI3}$$

One fixed nonzero-q_0 test makes C absolutely continuous on a positive initial interval; write $dC_t = v_t dt$ there. For each fixed ϵ, both mirrored identities hold almost everywhere on a positive initial interval, as a two-by-two system in b_t and $(v_t - 6)/2$. Its coefficients converge as $t \downarrow 0$ to

$$\begin{pmatrix} d_\epsilon & e_\epsilon^+ \\ d_\epsilon & -e_\epsilon^+ \end{pmatrix},$$

whose determinant is nonzero. The right sides are $(p_t^\sigma)^2 \widehat{b}^\sigma_{u_\sigma(t)}$, with each transformed density bounded by K_ϵ. Continuity of the coefficients and matrix inversion therefore give the pathwise bounds

$$\operatorname*{ess\,limsup}_{t\downarrow 0} |b_t| \le d_\epsilon K_\epsilon, \qquad \operatorname*{ess\,limsup}_{t\downarrow 0} |v_t - 6| \le \frac{2 d_\epsilon^2 K_\epsilon}{|e_\epsilon^+|}. \tag{BI4}$$

These assertions concern the same original densities. Intersect the probability-one sets for a countable sequence $\epsilon \downarrow 0$. By assumption $d_\epsilon K_\epsilon \to 0$; also $d_\epsilon / |e_\epsilon^+| \to 0$, so both right sides tend to zero. Integrating the resulting essential right limits proves (BI2). The positive comparison intervals may shrink with ϵ: the essential limsup is taken at fixed ϵ first. □

Theorem 5.3 (SLE from controlled conditional driving bias)**.** *Let a Borel exploration family have continuous Loewner drivers on all $t \ge 0$, source locality including the test deletions, and strong DMP. Require locality to identify the completed natural trace filtrations via the invertible capacity clocks. Require strong DMP to be the regular-conditional-law identity (KR1) for the inherited driver in its usual natural filtration. Proposition 4.1 then supplies canonical characteristic compatibility; it is not a separate premise. Suppose that, in every deterministic state f, there is an increasing sequence of stopping times $\rho_{f,m}$ increasing almost surely to infinity, with $\rho_{f,1} > 0$ almost surely, such that each $X_t^{f,m} = W^f_{t\wedge\rho_{f,m}}$ is bounded and satisfies (BI1), with a finite deterministic bound $K(f,m)$. Require*

$$\lim_{n\to\infty} \epsilon_n \max_{\sigma=\pm} K(f^\sigma_{\epsilon_n}, 1) = 0 \tag{BI5}$$

at every state, for some sequence $\epsilon_n \downarrow 0$. In particular uniformly bounded constants suffice; growth $o(\epsilon_n^{-1})$ is allowed. Then the family is chordal SLE_6 in these coordinates.

Proof. Lemma 5.1, applied to the localizing sequence, gives global continuous-semimartingale decompositions with locally bounded absolutely continuous drift. Uniqueness patches the decompositions on overlapping stopped intervals. Proposition 5.2 then gives the zero drift tangent and bracket tangent six in every state.

Use the conditional-kernel compatibility of canonical decompositions in the inherited natural trace filtration, as in the earlier conditional characteristic theorems. At every bounded stopping time τ, conditional DMP and (BI2) give

$$\frac{A_{\tau+h} - A_\tau}{h} \longrightarrow 0, \qquad \frac{C_{\tau+h} - C_\tau}{h} \longrightarrow 6 \quad \text{in probability.}$$

The zero-stopping-tangent lemma (Lemma 3.3) applied to A and $C - 6t$ gives $A = 0$ and $C_t = 6t$. Lévy's characterization identifies the driver as $\sqrt{6}B$, hence the trace as SLE_6. □

Rates in the slit test. Since $d_\epsilon \sim \epsilon/2$ and $|e_\epsilon^+| \sim 1/(2\epsilon)$, the two bounds in (BI4) are of orders ϵK_ϵ and $\epsilon^3 K_\epsilon$. The weaker condition $\epsilon^3 K_\epsilon \to 0$ alone forces the initial bracket tangent to be six. The full classification above also needs the stronger drift condition in (BI5).

What this reduction gains and what it leaves open. Conditional bias bounds concern first moments of observable driver increments, rather than unknown canonical decompositions. They produce semimartingality, bounded drift densities, and the missing initial tangents without assuming quadratic variation estimates. They are nevertheless substantial cancellation estimates: a bound on $\mathbb{E}|W_{t+h} - W_t|$ of Brownian order $\sqrt{h}$, or tightness from RSW, does not imply (BI1). The rate control in (BI5) is coordinate sensitive and is not justified by convergence of curve laws in the shrinking slit domains. Thus this is a conditional route toward the desired regularity, not a proof of percolation universality.

5.1 An observable discrete criterion that passes to curve limits

There is no need to identify the discrete or limiting canonical characteristics in order to test the preceding criterion. Signed conditional first moments on a discrete exploration clock suffice. The following elementary transfer also avoids convergence of filtrations. It does not establish the separate strong-DMP premise of the rigidity theorem.

Proposition 5.4 (Discrete conditional bias passes to the natural limit filtration)**.** *Fix $T > 0$. For each n, let $0 = t_0^n < t_1^n < \cdots < t_{N_n}^n = T$ be a deterministic partition with maximum spacing tending to zero. Let X^n be its continuous linear interpolation of grid values, all bounded by one deterministic constant L. Suppose the grid values are adapted to filtrations $(\mathcal{G}_j^n)_j$ and*

$$\left|\mathbb{E}\left[X_{t_{j+1}^n}^n - X_{t_j^n}^n \mid \mathcal{G}_j^n\right]\right| \le (K + \delta_n)(t_{j+1}^n - t_j^n), \qquad \delta_n \downarrow 0. \tag{DB1}$$

Suppose X^n converges in distribution in $C([0,T])$ to X. Then in the usual augmentation of the natural filtration of X,

$$|\mathbb{E}[X_t - X_s \mid \mathcal{F}_s]| \le K(t - s), \qquad 0 \le s \le t \le T. \tag{DB2}$$

Consequently X has a continuous local-martingale part and an absolutely continuous canonical drift with density bounded by K.

Proof. Summing the conditional inequalities and using the tower property shows that the grid processes $X_{t_j^n}^n \pm (K + \delta_n)t_j^n$ are respectively a submartingale and a supermartingale. In particular, for grid points $s_n \le t_n$ and every bounded nonnegative $\mathcal{G}_{j(s_n)}^n$-measurable H_n,

$$\left|\mathbb{E}[(X_{t_n}^n - X_{s_n}^n)H_n]\right| \le (K + \delta_n)(t_n - s_n)\mathbb{E}H_n. \tag{DB3}$$

Choose grid points $s_n \to s$ and $t_n \to t$, with $s_n \le t_n$, and set $H_n = H(X_{\cdot\wedge s_n}^n)$, where H is any bounded nonnegative continuous function on $C([0,T])$. The stopped interpolated path is measurable at the grid time s_n. Convergence in $C([0,T])$ and uniform continuity of each limiting path give joint convergence of the evaluations and stopped paths. The deterministic bound L permits passage of their expectations to the limit in (DB3).

Bounded continuous tests determine finite measures on the stopped-path space. Thus the resulting two inequalities extend to all bounded nonnegative tests measurable in the raw natural filtration at s. This proves (DB2) there. Equivalently $X_t + Kt$ and $-X_t + Kt$ are continuous class-D submartingales. These remain submartingales under the usual augmentation: for $u \downarrow s$, apply the inequality at $u < t$, then use reverse-martingale convergence and continuity of X to condition on $\bigcap_{u>s} \mathcal{F}_u^{\mathrm{raw}}$. Completion changes only null sets. Lemma 5.1 now supplies the asserted canonical decomposition. □

Exploration-time variant. An equivalent usable sufficient hypothesis allows a random discrete clock. Let the successive capacity times t_j^n and grid values X_j^n be adapted to the discrete revealed-color filtration, with uniform deterministic bounds on the drivers and clock horizons. Assume the times are nondecreasing, interpolation is well defined, and $J_n(s) = \inf\{j : t_j^n \geq s\} < \infty$ almost surely for each $s < T$. Suppose their largest spacing tends to zero in probability and

$$\left|\mathbb{E}[X_{j+1}^n - X_j^n \mid \mathcal{G}_j^n]\right| \leq (K + \delta_n)\mathbb{E}[t_{j+1}^n - t_j^n \mid \mathcal{G}_j^n]. \tag{DB4}$$

After continuous interpolation, assume the same uniform-path convergence. Then (DB2) again follows. Indeed $X_j^n \pm (K + \delta_n)t_j^n$ are bounded discrete sub/supermartingales. Use optional sampling at $J_n(s) = \inf\{j : t_j^n \geq s\}$ and $J_n(t)$, with a test of the path stopped at $t_{J_n(s)}^n$. The indices are stopping times because $\{J_n(s) \leq j\} = \{t_j^n \geq s\}$. The clock overshoots tend to zero in probability; tightness in the continuous-path topology makes the stopped paths and sampled values converge to their deterministic-time counterparts. Boundedness then passes the inequalities to the limit as above. At T, use continuity from times below T.

Fixed-interval block estimates also suffice. The per-reveal condition (DB4) is sufficient, not necessary for this transfer argument. It is enough that for each fixed $s < t$ and every bounded nonnegative continuous stopped-path test H, the tested inequalities (DB3), using approximations of s, t, hold up to an error $o(1)\|H\|_\infty$. The same weak-convergence proof then gives (DB2). Thus one may instead seek signed cancellation over mesoscopic blocks, allowing microscopic biases to cancel before a fixed capacity interval. No per-reveal estimate is asserted for lattice percolation.

Thus a possible lattice-level research target is the signed cancellation estimate (DB4), after suitable bounded stopping, in an unknown but fixed affine normalization. The constants must satisfy the shrinking-slit control in Theorem 5.3, and the limit must separately have its stated strong-DMP and conditional-characteristic compatibility. Neither a Brownian-size absolute increment estimate nor RSW alone is being substituted for this signed conditional estimate.

6 A wrong affine metric cannot have tame driving drift in every bounded domain

Status. The conclusion here is stronger than the earlier wrong-coordinate consequence of drift boundedness. For a fixed non-similarity affine deformation of SLE_6, regular semimartingale driving with absolutely continuous right-continuous drift cannot hold in every bounded domain in the wrong Euclidean complex structure. No local boundedness of the initial drift functional is needed. The proof uses exact scale invariance only for the *known fixed-affine SLE_6 half-plane law*; it does not assume that arbitrary critical subsequential limits are scale invariant. All characteristic regularity assumptions remain explicit.

6.1 Exact Brownian scaling and a regular initial bracket

Scaling lemma (proved). Let W be a continuous real process on $[0, \infty)$ with $W_0 = 0$, whose law has exact Brownian scaling:

$$(c^{-1/2}W_{ct})_{t\geq 0} \overset{\text{law}}{=} (W_t)_{t\geq 0} \qquad (c > 0). \tag{SM1}$$

Suppose up to a strictly positive stopping time it is a continuous semimartingale

$$W_t = M_t + \int_0^t b_s \, ds, \tag{SM2}$$

where $M_0 = 0$ is a continuous local martingale and b is finite and right-continuous at 0. Suppose its bracket has a finite deterministic initial right derivative

$$\lim_{t\downarrow 0} \frac{\langle M\rangle_t}{t} = v \in [0,\infty) \quad \text{almost surely.} \tag{SM3}$$

Then the entire law of W is that of $\sqrt{v}B$. This includes the identically zero process if $v = 0$.

Proof. Fix $T < \infty$ and stop the local decomposition at a positive initial localizing time. As $c \downarrow 0$, with probability tending to one the whole interval $[0, cT]$ lies before that time. Right-continuity of b gives

$$\sup_{t\le T}\left|c^{-1/2}\int_0^{ct} b_s\,ds\right| \le \sqrt{c}\,T \sup_{s\le cT}|b_s| \longrightarrow 0$$

almost surely on the localized event. Writing $C_t = \langle M\rangle_t$, (SM3) implies

$$\sup_{t\le T}|C_{ct}/c - vt| \longrightarrow 0. \tag{SM4}$$

Indeed $|C_{ct}/c - vt| \le t\sup_{0<s\le cT}|C_s/s - v|$. The continuous-martingale functional central limit theorem now gives $c^{-1/2}M_{c\cdot} \Rightarrow \sqrt{v}B$ on $C[0,T]$. One may see this directly from Dambis–Dubins–Schwarz: extend its Brownian motion if necessary and write

$$c^{-1/2}M_{ct} = B^{(c)}_{C_{ct}/c}, \qquad B^{(c)}_u = c^{-1/2}\widetilde{B}_{cu}.$$

Each $B^{(c)}$ has standard Brownian law. Uniform bracket convergence to the deterministic clock and the Brownian modulus of continuity imply the stated limit; independence between $B^{(c)}$ and the bracket is unnecessary. Adding the negligible drift gives the same limit for W. By exact scaling, however, each rescaled process already has the law of W on $[0,T]$. That law is therefore Brownian with variance v. Arbitrariness of T proves the assertion. □

The regular initial bracket is not automatic for a general self-similar continuous process. In the affine-SLE application below it follows from locality and absolutely continuous drift in two domain charts, using the local bracket-regularization identity.

6.2 The half-plane affine-SLE obstruction

Let A be an invertible orientation-preserving real-linear map and let η be chordal SLE_6 in $A\mathbb{H}$ from 0 to ∞. Put $\gamma = A^{-1}\eta \subset \overline{\mathbb{H}}$ and let W be its half-plane-capacity driving function. Its continuous Loewner generation and infinite capacity lifetime are supplied by the affine-admissibility result already proved in the research record. Its law has exact Brownian scaling (SM1): ordinary SLE is dilation covariant, A commutes with positive scalar dilations, and capacity scales by the square of physical length.

Proposition (proved). If this W satisfies (SM2)–(SM3), then A is a Euclidean similarity.

Proof. The scaling lemma makes W ordinary Brownian driving of variance v. Thus the trace is ordinary chordal SLE_v in $\mathbb{H}$. We identify v without a Hausdorff-dimension theorem.

For $\kappa \le 4$ the nonzero real boundary points are never swallowed by SLE_κ: their Loewner distances divided by $\sqrt{\kappa}$ are Bessel processes of dimension $1 + 4/\kappa \ge 2$ (the case $\kappa = 0$ is immediate).

An affine image of SLE_6 swallows boundary points, so $v > 4$. For any $\kappa > 4$, the boundary-race calculation gives

$$\mathbb{P}_\kappa(\tau_y < \tau_x) = F_\kappa\Big(\frac{-x}{y-x}\Big), \qquad F_\kappa(r) = \frac{\displaystyle\int_0^r [u(1-u)]^{-4/\kappa}\,du}{\displaystyle\int_0^1 [u(1-u)]^{-4/\kappa}\,du}, \quad x < 0 < y. \tag{SM5}$$

This includes $\kappa \geq 8$. To check it, the ratio $R = -X/(Y-X)$ has generator

$$\frac{\kappa}{2}\frac{d^2}{dr^2} + \frac{2(1-2r)}{r(1-r)}\frac{d}{dr}$$

after $ds = dt/(Y-X)^2$. Its scale density is $[r(1-r)]^{-4/\kappa}$, integrable at both ends exactly when $\kappa > 4$. The first swallowing is finite almost surely by the Bessel description; simultaneous swallowing is impossible since $Y - X$ is positive and nondecreasing. Bounded optional stopping gives (SM5).

Rotate $A\mathbb{H}$ conformally back to $\mathbb{H}$. The resulting map has matrix

$$T = \begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix}, \qquad \alpha, \delta > 0.$$

It acts on the real boundary by $x \mapsto \alpha x$, so it preserves the ratio $-x/(y-x)$ and the topological swallowing order. The law of γ therefore has boundary-race function F_6, while its Brownian driving gives F_v. Equality for every $r \in (0,1)$ forces $v = 6$: compare the derivatives in (SM5), proportional respectively to $[r(1-r)]^{-4/v}$ and $[r(1-r)]^{-2/3}$.

Finally $T^{-1}\mathrm{SLE}_6(\mathbb{H}; 0, \infty)$ is ordinary SLE_6. Half-plane affine identifiability now implies T is a similarity. For completeness, the strictly increasing SLE_6 left-passage function

$$L(u) = \frac{\int_{-\infty}^u (1+s^2)^{-2/3}\,ds}{\int_{-\infty}^\infty (1+s^2)^{-2/3}\,ds}$$

and preservation of topological side give $L(u) = L((\alpha u + \beta)/\delta)$ for all real u. Thus $\alpha = \delta$ and $\beta = 0$. Rotation does not affect whether A is a similarity, proving the proposition. □

6.3 Transfer from bounded-domain regularity to the half-plane germ

The preceding proposition is already a complete unbounded-domain statement. The following additional argument shows why requiring the tame drift structure in *every bounded domain* is enough to trigger it. We spell out the local coordinate transfer, rather than silently assume semimartingality is preserved by a general physical affine map.

Bounded-domain obstruction theorem (proved). Consider the canonical fixed-affine SLE_6 family

$$P_{D,p,q} = (A^{-1})_* \,\mathrm{Law}(\mathrm{SLE}_6(AD; Ap, Aq))$$

on bounded simply connected marked domains. Suppose in the original Euclidean complex structure every domain-chart driver is a continuous semimartingale whose finite-variation part is $\int_0^t b_s\,ds$, with finite right-continuous drift density and deterministic initial value $B(f)$, consistently under the usual conditional kernel interpretation. Then A is a Euclidean similarity. No boundedness or continuity of the map $f \mapsto B(f)$ is assumed.

Proof. Choose a bounded half-disk $D \subset \mathbb{H}$ with 0 in the interior of its real diameter, and a distinct target on that diameter away from 0. Choose the boundary colors so they agree near 0 with the half-plane problem $(\mathbb{H}, 0, \infty)$. Let $\phi : D \to \mathbb{H}$ be its chordal chart sending these two marks to $0, \infty$. It extends analytically across 0, with $\phi'(0) > 0$.

The local bracket-regularization lemma applies to this bounded-domain driver: delete one further vertical slit in its ϕ chart, away from its marks, with nonzero $h''(0)$. Both domains have tame absolutely continuous drift by hypothesis. The signed-measure identity

$$h_t''(W_t)(dC_t - 6\,dt) = 2[h_t'(W_t)^2 \widehat{b}_{u(t)} - h_t'(W_t) b_t]\,dt$$

therefore gives a finite right-continuous initial bracket density in D, with deterministic initial value v_D.

Corrected source-neighborhood locality of ordinary SLE_6, preserved by the fixed map A^{-1}, couples this bounded-domain curve to the canonical affine-SLE_6 half-plane curve so that their physical initial segments are identical until leaving a sufficiently small half-neighborhood of 0. This uses ordinary SLE_6 locality including distant target changes; it is not a new hypothesis about an unknown nonconformal family. On that common initial trace, the coordinate change is the analytic map ϕ at the source.

A local analytic coordinate change and its inverse preserve the tame semimartingale structure just obtained. More explicitly, if t is physical half-plane capacity and u is the bounded-chart capacity, the evolving local map satisfies

$$W_D(u(t)) = h_t(W_H(t)), \qquad \dot{u}(t) = h_t'(W_H(t))^2.$$

Initially $h_0 = \phi$, so $h_0'(0) > 0$ and all required local derivatives are finite. Before the trace leaves the common neighborhood, localize so the first derivative stays bounded away from zero and the derivatives stay bounded. Both capacity clocks are adapted functions of the same initial trace; their inverses are stopping-time clocks. The inverse field h_t^{-1} has finite variation in time, as follows by differentiating the Loewner composition. Applying the finite-variation Itô–Wentzell formula to the semimartingale W_D, and changing the capacity clock, produces a continuous-semimartingale decomposition of W_H on a positive initial interval. Its drift is absolutely continuous and right-continuous, and its initial bracket density is the same scalar v_D. These conclusions use only a *conformal local chart transition*, not the nonconformal map A.

For an explicit initial check, writing $d = \phi'(0)$, $e = \phi''(0)$, the transferred rates obey

$$V_H = v_D, \qquad B_H = dB_D - \left(\frac{v_D}{2} - 3\right)\frac{e}{d}, \tag{SM6}$$

so both are finite and deterministic. Thus the canonical half-plane driver satisfies (SM2)–(SM3). Its exact scaling and the half-plane proposition force A to be a similarity. □

Only a bounded domain flat at its source and one of its source-preserving slit deletions are needed for the bracket part of this contradiction. The full all-bounded-domain assumption is a convenient sufficient formulation. The obstruction does not identify the precise failure for a particular wrong affine metric: semimartingality, absolute continuity or right-continuity of the drift, or deterministic initial drift may fail. It does show that at least one must fail somewhere. It supplies no argument that an unknown exploration family admits a correct affine normalization with tame drift.

7 An affine metric is identifiable from one chordal law

For a bounded Jordan domain D, distinct $p, q \in \partial D$, and an orientation-preserving invertible real-linear map A, put

$$\nu^A_{D,p,q} = (A^{-1})_* \operatorname{Law}\big(\mathrm{SLE}_6(AD; Ap, Aq)\big). \tag{1}$$

Translations in an affine normalizer are immaterial. The tensor $A^{\mathsf{T}}A$ is its Euclidean conformal metric, determined only up to a positive scalar.

7.1 A boundary-swallowing race

Lemma 7.1 (Strict boundary-race probability)**.** *Let η be chordal SLE_6 in $\mathbb{H}$ from 0 to ∞. For $x < 0 < y$, let τ_x, τ_y denote their Loewner swallowing times. Then*

$$\mathbb{P}(\tau_y < \tau_x) = F\left(\frac{-x}{y-x}\right), \qquad F(r) = \frac{\int_0^r [u(1-u)]^{-2/3}\,du}{\int_0^1 [u(1-u)]^{-2/3}\,du}. \tag{2}$$

In particular F is strictly increasing on $(0,1)$. The event in (2) is invariant under orientation-preserving homeomorphisms of marked Jordan domains, when formulated as the order of separation from the terminal prime end by the filled exploration hull.

Proof. Use $\partial_t g_t(z) = 2/(g_t(z) - W_t)$ with $W_t = \sqrt{6}B_t$. Until the first swallowing time put

$$X_t = g_t(x) - W_t < 0, \quad Y_t = g_t(y) - W_t > 0, \quad T_t = Y_t - X_t, \quad R_t = -X_t/T_t \in (0,1).$$

Then

$$dX_t = \frac{2}{X_t}\,dt - \sqrt{6}\,dB_t, \qquad dY_t = \frac{2}{Y_t}\,dt - \sqrt{6}\,dB_t,$$
$$dT_t = \frac{2T_t}{-X_tY_t}\,dt > 0, \qquad dR_t = \frac{2(1-2R_t)}{R_t(1-R_t)T_t^2}\,dt + \frac{\sqrt{6}}{T_t}\,dB_t.$$

With $ds = dt/T_t^2$, the generator of R is

$$\mathcal{L} = 3\frac{d^2}{dr^2} + \frac{2(1-2r)}{r(1-r)}\frac{d}{dr}.$$

The increasing solution of $\mathcal{L}F = 0$, $F(0) = 0$, $F(1) = 1$ has $F'(r) = C[r(1-r)]^{-2/3}$, giving (2) by bounded optional stopping. To justify the terminal values, $|X|/\sqrt{6}$ and $Y/\sqrt{6}$ are Bessel processes of dimension $1 + 4/6 = 5/3$ up to their first zero; thus the first swallowing time is finite almost surely. Alternatively, the accessible endpoints of the displayed diffusion give the same stopping argument. There is no simultaneous swallowing, since $T_t \geq y - x > 0$, whereas both $X_t, Y_t \to 0$ would force $T_t \to 0$. At the first swallowing, $R = 1$ exactly when y is swallowed first. ODE solutions and the continuous driver stay finite up to that first time.

Swallowing means that the marked boundary end has been cut off from the target by the explored filled hull (including the boundary swallowed at first contact). This definition uses only the domain, the oriented path, and its order of traversal. A homeomorphism carries filled target components to filled target components. Capacity parametrization therefore has no role in the comparison event. □

7.2 Boundary rigidity and affine uniqueness

Lemma 7.2 (A single SLE law identifies its conformal boundary map)**.** *Let $h : \overline{D} \to \overline{D'}$ be an orientation-preserving homeomorphism between closed bounded Jordan domains, with $p' = h(p)$, $q' = h(q)$. If $h_*\mathrm{SLE}_6(D; p, q) = \mathrm{SLE}_6(D'; p', q')$, then $h|_{\partial D}$ agrees with the boundary extension of a conformal map $f : D \to D'$.*

Proof. Choose conformal maps $\phi : D \to \mathbb{H}$ and $\phi' : D' \to \mathbb{H}$ sending their marked endpoints to $0, \infty$. Their Jordan boundary extensions give an increasing homeomorphism

$$H = \phi' \circ h \circ \phi^{-1} : \overline{\mathbb{R}} \to \overline{\mathbb{R}}$$

fixing $0, \infty$. For every $x < 0 < y$, equality of the corresponding boundary-race probabilities and strictness in Lemma 7.1 give

$$\frac{-H(x)}{H(y) - H(x)} = \frac{-x}{y - x}, \quad \text{hence} \quad \frac{H(x)}{H(y)} = \frac{x}{y}.$$

Fixing one $x < 0$ and one $y > 0$ shows $H(s) = cs$ for all real s, with $c > 0$. Consequently $f = (\phi')^{-1} \circ (c\phi)$ has exactly the required boundary values. Equality of interior maps is not asserted for an arbitrary h. □

Theorem 7.3 (Single-domain affine identifiability)**.** *For even one bounded Jordan D and one pair of distinct boundary points p, q, the equality $\nu^A_{D,p,q} = \nu^B_{D,p,q}$ holds if and only if BA^{-1} is a Euclidean similarity. Equivalently,*

$$B^\mathsf{T} B = cA^\mathsf{T} A \qquad (c > 0).$$

The statement also allows orientation-reversing normalizers by composing one map with a reflection.

Proof. The equality of laws means $M = BA^{-1}$ carries the SLE law in AD to the SLE law in BD, at the corresponding endpoints. By Lemma 7.2, a conformal map $f : AD \to BD$ has boundary values equal to M. Both real coordinate functions of the real-linear map M are harmonic. The real and imaginary parts of f are also harmonic, continuous on the closure, with the same boundary values. The maximum principle on the bounded domain gives $f = M$ throughout AD. Thus M is complex-linear and invertible, hence a similarity. The converse is ordinary SLE similarity covariance. Reflection covariance reduces the remaining orientation case to this one. □

8 An affine tensor is readable from a regular endpoint germ

Status. The argument below extends the initial half-plane trace-germ decoder to analytic regular source germs. By ordinary SLE_6 reversibility, it gives a decoder from the ordered terminal trace germ at any analytic regular target, even when the other endpoint and the rest of the domain are rough. It does not decode the unlabelled accessible frontier at a random tip. All almost-sure assertions are made separately under each fixed affine tensor; no uncountable intersection of full-probability events is used.

Let $\mathcal{X}$ be the usual Polish space of oriented unparameterized continuous finite curves, with the uniform Fréchet metric. Curves are identified under increasing reparameterization, including deletion of pauses. For a continuous path Y starting at 0 and exiting every disk, write $\tau_a(Y)$ for its first hit of $\{|z| = a\}$ and $\mathrm{Stop}_a Y = Y[0, \tau_a(Y)]$. The operations used here are Borel in standard continuous-curve coding; first hits can equivalently be defined by rational-time approximation. For a rough opposite prime end the full curve need not land continuously there. Only finite prefixes

near the decoded regular endpoint are coded in $\mathcal{X}$; the unbounded half-plane curves exit every finite disk. This avoids any assumption of a singleton impression at the other mark.

Lemma 8.1 (Unit-radius stopping is almost surely continuous)**.** *Let Y have a law invariant under positive scalar spatial dilation and suppose it is continuous, starts at 0, and exits every disk. Then almost surely*

$$\tau_{1-\delta}(Y) \uparrow \tau_1(Y), \qquad \tau_{1+\delta}(Y) \downarrow \tau_1(Y) \quad (\delta \downarrow 0).$$

Consequently the diameter of $Y[\tau_{1-\delta}, \tau_{1+\delta}]$ tends to zero.

Proof. The first assertion on the left holds pathwise by continuity. The nondecreasing function $a \mapsto \tau_a(Y)$ has only countably many right jumps. Thus the set of radii with $\tau_{a+}(Y) > \tau_a(Y)$ has Lebesgue measure zero for every sample. Spatial dilation invariance implies that the probability of such a jump is the same at every fixed radius. Fubini's theorem on $[1, 2]$ makes this probability zero. Continuity of Y on a compact time interval containing these exit times then gives the diameter assertion. The argument is independent of the chosen continuous strictly increasing curve clock. □

For an orientation-preserving real linear map L carrying $\mathbb{H}$ onto itself, let μ_L be the law of $L\eta$, where η is chordal SLE_6 in $\mathbb{H}$ from 0 to ∞. Scalar multiples of L give the same law. Normalize

$$L = \begin{pmatrix} 1 & u \\ 0 & v \end{pmatrix}, \qquad (u, v) \in \Theta := \mathbb{R} \times (0, \infty). \tag{ED1}$$

The normalized metric represented by this law is $[(L^{-1})^{\mathsf{T}} L^{-1}]$.

Choose a countable bounded Lipschitz determining class $(f_m)_{m \geq 1}$ on $\mathcal{X}$, with values in $[0, 1]$ and Lipschitz constant at most one, allowing constant rescaling of an arbitrary countable bounded Lipschitz determining class. Such a class can be built from truncated distances to a countable dense set and finite rational combinations. Set

$$v_m(L) = \mathbb{E}_{\mu_L} f_m(\mathrm{Stop}_1 Y). \tag{ED2}$$

The vector $v(L) = (v_m(L))_m$ is Borel in L. Indeed, linear image and first-exit stopping are jointly Borel, so integration against the fixed ordinary SLE law preserves Borel measurability.

Brownian spatial scaling is mixing: under the Lamperti transform $Z_s = e^{-s/2} B_{e^s}$ it becomes a shift of a stationary Gaussian process with covariance $e^{-|s-u|/2}$. Its SLE trace factor, and each fixed linear image, are therefore dilation mixing. This gives the ergodicity used below without a separate path-space independence assumption.

The map $L \mapsto v(L)$ is injective. If two unit-radius stopped laws agreed, scaling invariance would identify their stopped laws at every radius. Those restrictions determine the entire ordered unbounded curve, contradicting the half-plane affine identifiability theorem unless the normalized matrices agree. Therefore the Lusin–Souslin theorem makes $v(\Theta)$ Borel and its inverse Borel.

Fix $\rho > 1$ and $r_n = \rho^{-n}$. For a curve γ starting at p define the local observable

$$X_n(\gamma, p) = r_n^{-1}(\gamma[0, \tau_{r_n}(\gamma - p)] - p). \tag{ED3}$$

If a finite curve fails to exit one of the large initial disks, assign a cemetery value to that term. It exits all sufficiently small disks, and finitely many cemetery values do not affect the limits below. Define

$$\ell_m(\gamma, p) = \lim_{N \to \infty} \frac{1}{N} \sum_{n=1}^{N} f_m(X_n(\gamma, p)), \tag{ED4}$$

using a cemetery vector if any limit fails to exist. Each limit is a function of every sufficiently small ordered initial trace prefix: deleting finitely many summands changes no Cesàro limit. In particular this is a Borel ordered-trace-germ statistic.

Lemma 8.2 (The decoder survives a quadratic local error)**.** *Let η be ordinary half-plane SLE_6 and let Φ be a local C^1 embedding at 0, defined on a half-neighborhood of 0, with*

$$\Phi(0) = 0, \qquad \Phi(z) = Lz + O(|z|^2), \tag{ED5}$$

where L is invertible. Assume after a fixed rotation that L maps $\mathbb{H}$ onto itself. Let γ agree with $\Phi(\eta)$ up to the first exit from this local chart. Then almost surely

$$\ell_m(\gamma, 0) = v_m(L) \quad \textit{for every } m. \tag{ED6}$$

Scalar normalization of L is understood in (ED2).

Proof. Put $Y_n = r_n^{-1} L\eta$. Spatial dilation of the trace includes the induced capacity reparameterization. Brownian scaling is mixing; its trace factor and every fixed affine image are therefore dilation ergodic, as proved in the initial half-plane decoder theorem. Hence

$$\frac{1}{N}\sum_{n=1}^{N} f_m(\mathrm{Stop}_1 Y_n) \longrightarrow v_m(L) \quad \text{almost surely.} \tag{ED7}$$

One must control the error in this average; pointwise stopping continuity at all the scaled random samples is not asserted.

For $0 < \delta < 1/4$ define

$$h_\delta(Y) = \min\{2, \operatorname{diam} Y[\tau_{1-\delta}(Y), \tau_{1+\delta}(Y)]\}\,. \tag{ED8}$$

The linearization error on a fixed bounded scaled chart is at most Cr_n, uniformly over the sample path until its linear image exits the disk of radius $1+\delta$. This uniformity is deterministic: the inverse linear map bounds the relevant preimage disk, and (ED5) bounds the error there. For all sufficiently large n, the first time that $r_n^{-1}\Phi(\eta)$ reaches radius one lies between the corresponding Y_n exit times at radii $1-\delta$ and $1+\delta$. Thus the two stopped curves consist of matched portions with pointwise error at most Cr_n, together with an unmatched terminal portion whose diameter is bounded by the diameter in (ED8). Increasing reparameterizations, with pauses allowed, give

$$|f_m(X_n(\gamma, 0)) - f_m(\mathrm{Stop}_1 Y_n)| \le Cr_n + h_\delta(Y_n). \tag{ED9}$$

The bounded range of f_m justifies capping the diameter by two.

Take a countable sequence $\delta_k \downarrow 0$. Birkhoff's theorem applied to every h_{δ_k} gives

$$\limsup_N \frac{1}{N}\sum_{n=1}^{N} |f_m(X_n(\gamma, 0)) - f_m(\mathrm{Stop}_1 Y_n)| \le \mathbb{E}_{\mu_L} h_{\delta_k}(Y) \tag{ED10}$$

simultaneously for all m, k. Lemma 8.1 and dominated convergence make the right side tend to zero as $k \to \infty$. Combine this with (ED7). Only the initial segment of η enters the argument, since all sufficiently small physical exits occur inside the fixed local chart. This proves (ED6). □

Theorem 8.3 (A universal analytic-endpoint tensor decoder)**.** *There is a Borel rule, depending only on the ordered initial trace germ, the endpoint, and its oriented tangent half-plane, that recovers the projective affine tensor from a single affine-SLE_6 sample in every bounded simply connected marked domain with analytic regular source. The rest of its boundary and its target may be rough prime-end data. There is also a terminal version of the rule for every domain with analytic regular target; the source may then be rough. The rule is the same for two domains agreeing near that regular endpoint.*

Proof. Let the law be $(A^{-1})_* \mathrm{SLE}_6(AD, Ap, Aq)$ and assume p is an analytic regular source. A conformal map $f : \mathbb{H} \to AD$ with marks $0, \infty$ extends conformally across 0 by analytic boundary reflection, with $f(0) = Ap$ and $f'(0) \neq 0$. In a physical tangent frame Q carrying the interior tangent half-plane to $\mathbb{H}$, the local map is

$$\Phi(z) = Q(A^{-1}f(z) - p) = Lz + O(|z|^2), \qquad L = QA^{-1}M_{f'(0)}, \tag{ED11}$$

where M_c denotes multiplication by the nonzero complex number c. The linear map L carries $\mathbb{H}$ onto itself and has positive diagonal entries in the boundary frame. Apply Lemma 8.2. The Borel inverse of (ED2) recovers the scalar-normalized L, hence the projective tensor

$$[(L^{-1})^{\mathsf{T}} L^{-1}] = [QA^{\mathsf{T}} AQ^{\mathsf{T}}]. \tag{ED12}$$

Conjugating back by the known orthogonal frame Q recovers $[A^{\mathsf{T}}A]$. The derivative of the conformal map contributes only a positive scalar to this tensor. This rule uses the endpoint germ and tangent half-plane, not the rest of the conformal chart.

Ordinary SLE_6 reversibility [8] says the reversed curve has the same affine-SLE form with marks swapped. Apply the source decoder to that reversed curve at the regular target. Reversibility on prime-end domains follows by conformal transport from the half-plane theorem; only the regular target germ needs a Euclidean analytic boundary extension. □

Corollary 8.4 (Terminal singularity and identifiable mixtures)**.** *For one fixed marked domain with analytic regular target, affine-SLE_6 laws with distinct projective affine tensors are mutually singular on the ordered terminal trace-germ sigma-field. More strongly, if π is any Borel probability on normalized affine tensors and $P_\pi = \int P_T \, \pi(dT)$, then the terminal decoder has law π under P_π. Consequently $\pi \mapsto P_\pi$ is injective, and no nontrivial affine mixture can equal one pure affine-SLE law.*

Proof. The same Borel decoder equals T almost surely under P_T. Integrating that equality proves the distribution statement. Distinct pure laws concentrate on disjoint decoder fibers. □

Additional initial-germ obstruction for random mixtures. At an analytic regular source, the initial decoder likewise returns the latent tensor almost surely. Therefore a nontrivial mixture of affine metrics has a nontrivial initial trace-germ sigma-field. At every positive exploration time the metric posterior has already collapsed to a point, even though its prior was nontrivial. This rules out such mixtures under an initial-germ zero-one premise. Under the current literal strong DMP in the usual right-continuous natural filtration, that premise is automatic by Lemma 30.1. The independent-color version remains useful before a limiting strong DMP theorem is available. A deterministic source-germ metric assignment remains unresolved; it requires restart consistency at every rough contact state.

Proposition 8.5 (DMP forces restarted mixing measures to be pure)**.** *Suppose an all-state family has the mixed-affine representation $P_S = \int \nu_S^T \, \pi_S(dT)$, with Borel mixing kernel π_S. Start it in a domain with analytic regular source and target. If it has strong DMP, then at each admissible stopping time $\tau > 0$ before the target is reached,*

$$\pi_{S_\tau} = \delta_{D_s(\gamma)} \qquad \textit{almost surely}, \tag{ED13}$$

where D_s is the initial tensor decoder. In particular a rule that uses a nondegenerate prior at every restarted state cannot satisfy DMP.

Proof. Sample the latent tensor T with law π_S and then sample the fixed-T affine SLE. The source decoder gives $D_s(\gamma) = T$ almost surely, and this variable is measurable from every positive prefix, including the prefix up to τ. Conditional on this prefix, ordinary fixed-T SLE DMP therefore gives the pure continuation law $\nu_{S_\tau}^{D_s(\gamma)}$. The family's DMP gives the mixed law with mixing measure π_{S_τ}. Since τ precedes reaching the target, the surviving target end has the same analytic regular germ: the compact prefix avoids a sufficiently small target neighborhood. Corollary 8.4 applied to this current state makes the two mixing measures equal, proving (ED13). □

9 Microscopic correctors for signed driving cancellation

The per-reveal condition (DB4) is sensitive to bounded microscopic oscillations which disappear in the continuum limit. The following criterion permits exactly this kind of cancellation. It supplies neither a corrector for a percolation exploration nor the required slit-uniform bounds; those remain mathematical tasks.

Proposition 9.1 (Vanishing-corrector transfer on an exploration clock)**.** *Fix a deterministic horizon T. In the random-clock setting of (DB4), assume $|X_j^n| \le L$, $t_0^n = 0$, $0 \le t_j^n \le T_*$, every level $s < T$ is reached almost surely, and the maximum clock spacing tends to zero in probability. Require zero clock increments to have zero driver increments, so continuous interpolation is well defined. Suppose the interpolated X^n converge in law in $C([0,T])$ to X. Let R_j^n be adapted real correctors with $\mathbb{E}\sup_j |R_j^n| < \infty$ for every n, such that*

$$\mathbb{E}\sup_j |R_j^n| \longrightarrow 0. \tag{MC1}$$

Assume also zero clock increments have zero corrector increments. Set $Y_j^n = X_j^n + R_j^n$ and suppose for $K < \infty$ and $\delta_n \downarrow 0$ that

$$\left|\mathbb{E}[Y_{j+1}^n - Y_j^n \mid \mathcal{G}_j^n]\right| \le (K+\delta_n)\mathbb{E}[t_{j+1}^n - t_j^n \mid \mathcal{G}_j^n]. \tag{MC2}$$

Then X satisfies (DB2) in its usual natural filtration. In particular its canonical drift is absolutely continuous with density bounded by K. If $K = 0$, X is a continuous martingale.

Proof. The grid processes $Y_j^n + (K+\delta_n)t_j^n$ and $-Y_j^n + (K+\delta_n)t_j^n$ are submartingales. For each fixed n, their absolute values at all stopping indices are dominated by the integrable variable $L + \sup_j |R_j^n| + (K+\delta_n)T_*$. Consequently optional sampling applies to the almost surely finite indices, for $0 \le s < t < T$, $J_n(s) = \inf\{j : t_j^n \ge s\}$ and $J_n(t)$, even when they have no deterministic upper bound. For every bounded nonnegative $\mathcal{G}_{J_n(s)}^n$-measurable test H_n, it gives

$$\left|\mathbb{E}[(Y_{J_n(t)}^n - Y_{J_n(s)}^n)H_n]\right| \le (K+\delta_n)\mathbb{E}[(t_{J_n(t)}^n - t_{J_n(s)}^n)H_n].$$

Replace Y^n by X^n. The resulting error is at most $2\|H_n\|_\infty \mathbb{E}\sup_j |R_j^n| = o(1)\|H_n\|_\infty$. Take $H_n = H(X^n_{\cdot\wedge t^n_{J_n(s)}})$ for a bounded nonnegative continuous stopped-path test H. Clock overshoots vanish in probability, and tightness in $C([0,T])$ identifies the limiting sampled values and stopped paths. The bound on X^n and on the clock then passes their expectations to the limit and yields

$$|\mathbb{E}[(X_t - X_s)H(X_{\cdot\wedge s})]| \le K(t-s)\mathbb{E}H(X_{\cdot\wedge s}).$$

As in Proposition 5.4, bounded continuous tests determine the corresponding finite signed measures, giving (DB2) in the raw natural filtration. Continuity and reverse-martingale convergence extend it to the usual augmentation. The endpoint T follows by continuity, and Lemma 5.1 identifies the canonical drift. □

A finite-state cell problem. There is a useful sufficient construction of such a corrector. For simplicity let $t_j^n = jh_n^2$ and let E_j^n be an adapted process in a fixed finite set $\mathcal{E}$ whose conditional transition matrix given $\mathcal{G}_j^n$ is a fixed irreducible stochastic matrix P. Suppose

$$\mathbb{E}[X_{j+1}^n - X_j^n \mid \mathcal{G}_j^n] = h_n b(E_j^n) + h_n^2 r_j^n, \qquad |r_j^n| \le K. \tag{MC3}$$

Let π be the unique invariant probability of P. If $\sum_e \pi(e)b(e) = 0$, the finite linear system

$$(I - P)\chi = b \tag{MC4}$$

has a solution χ, unique up to an additive constant. This follows because the kernel of $I - P$ is the constants and the image is the π-mean-zero hyperplane; no aperiodicity assumption is needed. The adapted corrector $R_j^n = h_n \chi(E_j^n)$ then obeys (MC1), and

$$\mathbb{E}[\Delta(X^n + R^n)_j \mid \mathcal{G}_j^n] = h_n\{b + P\chi - \chi\}(E_j^n) + h_n^2 r_j^n = h_n^2 r_j^n.$$

Thus (MC2) holds. The same computation remains valid before an adapted stopping index, provided both the driver and corrector are frozen there. The capacity clock may continue after this stopping index, with zero corrected driver increments. For a random clock the right side of (MC3) can instead be a residual bounded by $K\mathbb{E}[\Delta t_j^n \mid \mathcal{G}_j^n]$.

This is only a finite-state sufficient model. The local conformal geometry of an explored percolation boundary is not represented by the tip orientation alone. No closure to a fixed finite P, and no solution of a percolation cell problem, is asserted.

Remark 9.2 (Raw one-step bias may diverge for Brownian approximants). Let $Z_t = \sqrt{6}B_{t\wedge\tau}$, where $\tau = \inf\{t : |\sqrt{6}B_t| = L\}$, and let $h_n \downarrow 0$. At the deterministic grid times $t_j^n = jh_n^2$, put

$$X_j^n = Z_{t_j^n} + h_n\{(-1)^j - 1\}. \tag{MC5}$$

With the Brownian grid filtration, $X_0^n = 0$ and the conditional signed increment is exactly $-2h_n(-1)^j$. Hence any (DB4) constant must satisfy $K_n \ge 2/h_n \to \infty$. Nevertheless $|X_j^n| \le L + 2h_n$ and continuous interpolation converges uniformly on compact time intervals, almost surely, to Z. The corrector $R_j^n = h_n\{1 - (-1)^j\}$ has supremum at most $2h_n$ and makes $X^n + R^n$ an exact grid martingale. On a fixed positive horizon the sum of the absolute one-step predictable biases diverges like h_n^{-1}, while their cumulative signed sum stays bounded by $2h_n$.

Implication for the universality reduction. A percolation proof need not bound the raw per-reveal conditional bias. It would suffice to construct an adapted corrector satisfying (MC1) and the corrected estimate (MC2), in one fixed affine normalization. The resulting constants must still satisfy the shrinking-slit condition (BI5), and strong continuation and characteristic compatibility remain separate requirements of Theorem 5.3. The example proves only a logical obstruction to treating (DB4) as a necessary property of approximating drivers; it is not a counterexample to (DB4) for a particular percolation lattice.

10 A discrete source shield and stability of two exploration starts

Status and model. The deterministic shielding statement and the ensuing probabilistic estimate are proved here for independent site percolation on a triangular-lattice triangulation of a Jordan disk. Boundary vertices are wired black or white, with exactly two color changes, the marked source

and target. The exploration is the nonbranching face-dual black–white interface between those changes. The same argument applies to a cellular model with the corresponding nonbranching primal/dual geometry and the stated white RSW bounds. It is an actual common-color coupling argument. It is *not* inferred for an arbitrary random-curve family from abstract locality or G2 alone.

Work at mesh δ. Two triangulated domains, their source positions, and their boundary colors may differ inside $B(p, \epsilon)$, but agree outside it. Their common target lies outside $B(p, R_0)$. Outside the source patch and inside this latter ball, assume the common geometry is either a lattice half-plane with straight boundary, or a canonical thin straight slit with its two boundary sides. The lattice approximations are chosen with free first rows adjacent to the wired boundary rows; slit width is $O(\delta)$. The black and white wired sides are the same for the two configurations outside the patch. Couple all free colors outside the patch identically. Those shared colors have the independent percolation law. The colors inside the patch may be coupled arbitrarily, with their correct marginal laws.

10.1 The deterministic one-edge crosscut

In a triangular face, either zero or two of its primal edges have endpoints of different colors. Hence the dual interface graph has degree zero or two at every interior dual vertex. In the Dobrushin disk it has a unique open simple path between the source and target; other components are loops.

Lemma 10.1 (A one-heterochromatic-edge shield)**.** *Let a simple primal crosscut c of a triangulated disk connect a white wired boundary vertex to a black wired boundary vertex. Suppose every edge of c is white–white except its final edge e, which joins a white* interior *vertex to the black boundary vertex. Suppose c separates the entire source patch from the target. Then the exploration must use the dual edge e^*, with the direction determined by its black and white sides. After using e^* it cannot reenter the source side of the crosscut.*

Consequently, if the same such crosscut and its colors occur in both configurations above, their explorations have identical ordered suffixes from e^* to the common target.

Proof. The source–target dual path must cross the primal crosscut. Primal and dual vertices are distinct, and dual edges cross primal edges only at their matched transverse intersections. No interface edge can cross a white–white primal edge. Thus the crossing must be e^*, and matching colors give the same direction in both configurations.

The dual interface path is simple, so it cannot traverse e^* twice. All other primal edges of c remain monochromatic, so they cannot be crossed later either. The path therefore stays on the target side after using e^*. In particular it never returns to the altered source patch. Outside the patch the two triangulations and all colors coincide. At each subsequent dual vertex the outgoing edge is uniquely determined by the local colors and the incoming edge. Starting at the same directed edge e^*, the two paths consequently have the same entire suffix. □

Using boundary *vertices* in this formulation avoids an ambiguity of cell-exterior pictures: e is an interior primal edge incident to a boundary vertex, so its dual edge is an ordinary interior edge. Neither a crossing through a dual/primal common vertex nor an unexplained passage around the end of a physical boundary edge is being used.

10.2 Uniform probability of a white arch

Assume a uniform white RSW lower bound for fixed-aspect rectangles of free vertices. No connection to a prescribed individual vertex is assumed. We give the geometry for a horizontal half-plane

source, translated to $p = 0$, with black boundary on the negative real side and white boundary on the positive side. Let $r \gg \epsilon + \delta$.

Use a white vertical crossing in each of

$$[-3r, -2r] \times [0, 3r], \qquad [2r, 3r] \times [0, 3r],$$

and a white horizontal crossing of

$$[-3r, 3r] \times [r, 2r].$$

At the bottom of a stem, start the rectangle in the first free vertex row, not in the fixed wired row. Rounding by $O(\delta)$ changes the aspect ratios by a uniformly bounded amount. The crossings in the overlapping squares intersect by planarity. Thus their union contains a white path from a free vertex adjacent to the black boundary to a free vertex adjacent to the white boundary. Add a white–white edge at the white endpoint and choose an edge e to a black boundary vertex at the other endpoint. Trim the white path at its last white-wired-boundary contact before reaching the left endpoint, so that all its subsequent white vertices are interior; then erase loops. This gives a proper simple primal crosscut with just the one heterochromatic edge e.

All these vertices lie, allowing lattice rounding, in $A(0, r/2, 6r)$. The crosscut and the boundary arc through the source enclose the source patch. Its source-side component is inside $B(0, 6r)$: the crosscut and that local boundary arc form a Jordan polygon contained there, so their bounded interior is also contained there. The common half-annulus between the patch and crosscut shows that the entire altered domain portion of the source patch is on that side.

Each rectangle event is increasing in white colors of independent free vertices. RSW and FKG therefore give a constant $\vartheta > 0$, independent of r, δ and the contents of the source patch, for this white-arch event. Importantly, the left endpoint is *some* first-row white vertex adjacent to the black boundary, selected after the crossing occurs. Requiring connection to one predetermined microscopic vertex would not have this uniform lower bound.

Straight slit geometry. The same construction works for a finite-width lattice slit along the negative real axis, with black upper lip and white lower lip, by using five crossings around a square annulus: an upper-left vertical stem, a top horizontal crossing, a right vertical crossing, a bottom horizontal crossing, and a lower-left vertical stem. The stems occupy $[-3r, -2r] \times [0, 3r]$ and $[-3r, -2r] \times [-3r, 0]$ on their respective sides of the slit. The other three rectangles may be

$$[-3r, 3r] \times [2r, 3r], \quad [2r, 3r] \times [-3r, 3r], \quad [-3r, 3r] \times [-3r, -2r].$$

The first free upper row is adjacent to the black wired lip; the first free lower row is adjacent to the white wired lip. The resulting crosscut goes around the right side of the slit and separates its tip patch from the distant target. The boundary arc through the tip, including the two separate lattice lips, stays inside $B(0, 6r)$. The finite-width triangulated domain is Jordan; one need not identify its two physical boundary rows. RSW and FKG again give a uniform $\vartheta > 0$ after adjusting constants.

10.3 Polynomial source-patch stability

Theorem 10.2 (Common-color coalescence outside a source patch)**.** *Under the preceding geometric and percolation hypotheses, there are constants $C < \infty$ and $\alpha > 0$ depending only on the RSW bounds and the fixed geometry such that, for $0 < R < R_0/6$, the common-color coupling of the two full exploration curves satisfies*

$$\mathbb{P}\{d_{\mathrm{F}}(\gamma_1, \gamma_2) > 2R\} \le C \left(\frac{\epsilon + \delta}{R}\right)^{\alpha}. \tag{SH1}$$

More concretely, outside an event of the displayed probability the curves coalesce at a directed dual edge inside $B(p, R)$; both prefixes before that edge stay inside this ball, and the entire suffixes are identical.

Proof. Put $r_0 = 16(\epsilon + \delta)$ and $r_j = 16^j r_0$. For every j with $6r_j \le R$, the corresponding white-arch event depends only on shared free vertices in $A(p, r_j/2, 6r_j)$. These vertex sets are disjoint for different j, up to an immaterial lattice enlargement absorbed by increasing r_0. The events are therefore independent and each has probability at least ϑ. If there are N available bands, the probability of no shield is at most $(1-\vartheta)^N$.

On any shield event select a white crosscut by a deterministic finite-graph rule using only shared colors, for example a lexicographic simple path in the crossing cluster. The two configurations select the same crosscut and the same single heterochromatic edge. Lemma 10.1 gives a common suffix. Before crossing the shield, each path stays in its source-side component, contained in $B(p, 6r_j) \subset B(p, R)$. Synchronize the two prefixes by any increasing reparameterization and then align their common suffix. The prefixes are in a ball of diameter $2R$, so their oriented Fréchet distance is at most $2R$.

The number of bands is a logarithmic function of $R/(\epsilon + \delta)$. For instance with $\alpha = -\log(1 - \vartheta)/\log 16 > 0$, enlarging a fixed constant C gives $(1-\vartheta)^N \le C((\epsilon+\delta)/R)^\alpha$. This also covers the case with no available band by making its right-hand side at least one. □

Passage to marginal curve limits. Suppose along a common mesh subsequence the two exploration laws converge in the usual Polish oriented uniform-Fréchet curve space. The coupled pairs are tight because their two marginals are tight. Choose a joint subsequential limit. The set $\{d_{\mathrm{F}} > 2R\}$ is open in the product curve space; the Portmanteau inequality transfers (SH1), with $\delta \downarrow 0$, to this limiting coupling. Thus the same quantitative source-patch stability holds for the two marginal curve limits. It requires neither convergence of random slit-domain kernels nor a limiting strong DMP theorem.

If source patches themselves shrink and the mesh is negligible relative to their radii, this gives continuity in law for these particular local source perturbations. One may first fix R and let $\epsilon \downarrow 0$, then let $R \downarrow 0$, or balance the coupling estimate to obtain an explicit Prokhorov rate.

What is not proved. This is not a consequence of abstract unforced-crossing estimates for one curve. It uses a common independent coloring and the planar nonbranching interface to force *coalescence at one edge*. It also is not the full marked inverse-conformal-chart stability condition used in the affine-ansatz support theorems: arbitrary converging charts may create forced thin fjords, noncommon microscopic boundaries, or source approaches outside this shared half-plane/slit geometry. Extending the shield coupling to such approximations requires additional work. No classification of arbitrary source-local exploration ensembles follows from (SH1) alone.

11 Exact suffix coalescence survives curve limits

The source-shield estimate contains more information than a small Fréchet distance. Its exact common suffix is stable under taking a joint curve limit, provided that the prefixes stay in a fixed closed set away from the target. This allows the discontinuous terminal metric decoder to be used without asserting that it is a continuous observable.

Let $K \subset \mathbb{C}$ be closed and consider two oriented continuous curves. Define E_K to be the relation

that there exist cut times s_1, s_2 in continuous representatives on $[0,1]$ such that

$$\gamma_i[0,s_i] \subset K \quad (i=1,2), \qquad \gamma_1[s_1,1] = \gamma_2[s_2,1] \quad \text{as ordered unparameterized curves.} \tag{CB1}$$

The relation is independent of the chosen representatives and allows pauses, as does the usual uniform Fréchet curve space. It implies a common endpoint, which may belong to K in this definition. All applications below have limiting common target $q \notin K$.

Lemma 11.1 (A closed relation of common ordered suffixes)**.** *E_K is closed in the product uniform-Fréchet curve space.*

Proof. Suppose $(\gamma_1^n, \gamma_2^n) \to (\gamma_1, \gamma_2)$ and the pairs belong to E_K. Choose representatives of each convergent sequence converging uniformly to fixed continuous representatives of its limit. This is possible by the definition of the Fréchet metric; reparameterize the cut times along with the curves. Pass to a subsequence on which both cut times s_i^n converge to $s_i \in [0,1]$.

Closedness of K and uniform convergence give $\gamma_i[0,s_i] \subset K$. Linear rescaling of $[s_i^n, 1]$ to $[0,1]$ makes the restricted curves converge uniformly to the restrictions on $[s_i, 1]$, by uniform continuity of γ_i (a cut at $s_i = 1$ gives a constant curve). The two restrictions at stage n have Fréchet distance zero. Their limits therefore also have distance zero and are the same ordered curve. Thus the limiting pair belongs to E_K. □

Proposition 11.2 (Coalescence transfer and comparison of metric priors)**.** *Suppose pairs of discrete curves with a common target converge marginally to laws P_1, P_2 on the oriented curve space, and their couplings satisfy*

$$\mathbb{P}(E_K) \geq 1 - e_n, \qquad \limsup_n e_n \leq e < 1, \tag{CB2}$$

where K is a fixed closed ball not containing the target. Each limiting domain is analytic and regular near that target, and the two colored target germs agree. Suppose

$$P_i = \int P_{i,T}\, \pi_i(dT), \tag{CB3}$$

where $P_{i,T}$ is the affine-SLE$_6$ law with normalized tensor T in domain i, and π_i are Borel probability measures on tensors. Then, with total variation defined as $\sup_B |\pi_1(B) - \pi_2(B)|$,

$$\|\pi_1 - \pi_2\|_{\mathrm{TV}} \leq e. \tag{CB4}$$

In particular if both marginals are pure affine-SLE laws, any positive uniform lower bound on the discrete coalescence probability forces their tensors to be equal.

Proof. Tightness of the marginals makes the coupled pairs tight. Take a joint subsequential limit. Lemma 11.1 and the closed-set Portmanteau inequality give $\mathbb{P}(E_K) \geq 1 - e$ in the limit. Because the target is outside K, the common suffix on E_K includes a nonconstant ordered terminal germ at that target. The universal terminal decoder of Theorem 8.3 is the same for the two domains, since their colored target germs agree. Let T_i be its value on curve i. The laws of these random variables are π_i, and on E_K they are equal. Hence

$$\sup_B |\mathbb{P}(T_1 \in B) - \mathbb{P}(T_2 \in B)| \leq \mathbb{P}(T_1 \neq T_2) \leq e.$$

For two distinct point masses the left-hand side equals one, proving the last assertion. The proof uses exact suffix equality, not continuity of the decoder and not just small curve distance. □

For the source-shield coupling, take $K = \overline{B}(p, R)$ with the regular target outside this ball. Common discrete targets may vary with mesh, provided $q_n \to q \notin K$; the closed-relation proof already allows this. Equation (SH1) then gives

$$\|\pi_1 - \pi_2\|_{\mathrm{TV}} \le C(\epsilon/R)^\alpha \tag{CB5}$$

for two source-patch perturbations with common flat or straight-slit annuli. This is a quantitative stability statement for mixing measures, and yields exact tensor equality for pure marginals once the right side is less than one. It is conditional on the affine-mixture marginal ansatz; no such ansatz is derived from RSW here.

11.1 Stopping an ordered unparameterized curve on a circle

Fix $p \in \mathbb{R}^2$ and $R_0 > 0$. Work in the usual metric space of oriented continuous curves modulo increasing reparameterization, with the uniform Fréchet topology. Curves start at p and end outside $\overline{B(p, R_0)}$. Small changes of the starting points are harmless; for the sequence version they need only lie inside every fixed smaller ball for sufficiently large indices. For a continuous representative $\gamma : [0, 1] \to \mathbb{R}^2$ and $0 < r < R_0$, put

$$\tau_\gamma(r) = \inf\{t : |\gamma(t) - p| \ge r\}, \qquad E_r\gamma = \gamma|_{[0,\tau_\gamma(r)]}. \tag{LT1}$$

The stopped path is regarded as an ordered unparameterized curve, so this definition is independent of pauses or changes of clock.

Lemma 11.3 (Almost every circle is a continuity circle)**.** *For each fixed continuous curve γ, the map $(\eta, s) \mapsto E_s\eta$ is continuous at (γ, r) for all but at most countably many radii $r \in (0, R_0)$. The stopping map is Borel. Consequently, for any probability law μ on these curves, E_r is μ-almost surely continuous for Lebesgue-almost every deterministic r.*

Proof. Choose any continuous representative of γ. The function $r \mapsto \tau_\gamma(r)$ is nondecreasing and hence has at most countably many discontinuities. Suppose it is continuous at r. If $(\gamma_n, r_n) \to (\gamma, r)$ in the product of Fréchet and ordinary metrics, choose increasing reparameterizations for which representatives $\widetilde{\gamma}_n$ converge uniformly to γ. Writing $\varepsilon_n = \|\widetilde{\gamma}_n - \gamma\|_\infty + |r_n - r|$, first-hit times obey, for every slightly larger positive error tending to zero,

$$\tau_\gamma(r - 2\varepsilon_n) \;\le\; \tau_{\widetilde{\gamma}_n}(r_n) \;\le\; \tau_\gamma(r + 2\varepsilon_n). \tag{LT2}$$

If $\varepsilon_n = 0$ the assertion is immediate; using a strictly larger error avoids any immaterial equality convention. Continuity of τ_γ at r gives convergence of the middle times to $\tau_\gamma(r)$. Uniform continuity of γ then makes the stopped representatives, linearly recoded onto $[0, 1]$, converge uniformly. This proves joint continuity at (γ, r).

The stopping map is understood on all curves reaching the selected radius, including already stopped curves reaching a larger radius. For parameterized curves the stopping time is Borel, since

$$\{\tau_\gamma(r) \le t\} = \left\{ \max_{0 \le u \le t} |\gamma(u) - p| \ge r \right\}.$$

Stopping and linear recoding are therefore Borel path operations. They commute with reparameterization and give the usual Borel stopping map on ordered unparameterized curves.

For completeness, the almost-every-radius assertion does not require a measurable choice of representatives. For a map between metric spaces its set of continuity points is a G_δ set, obtained from its oscillations on neighborhoods. Apply this to the joint stopping map. Its discontinuity set in the product of curve space and radius space is Borel, and every vertical section has Lebesgue measure zero by the countability argument above. Fubini's theorem now gives the assertion under μ. □

12 Independent color resampling, initial germs, and a common affine metric

Theorem 12.1 (Initial-germ zero-one law from independent colors and shielding)**.** *Let P be a subsequential exploration-curve limit of the independent color model in a fixed domain with flat source germ. Assume the Polish oriented Fréchet curve topology and the flat-source shield estimate of Theorem 10.2. Then the initial ordered trace-germ sigma-field is P-trivial. Precisely, if $\sigma_r\gamma$ denotes the ordered prefix stopped on the radius-r circle about the source, this field is $\bigcap_{r>0}\overline{\sigma(\sigma_r\gamma)}^{P}$. It is the usual ordered positive-prefix germ, with no clock chosen. No affine ansatz, terminal tensor decoder, limiting DMP theorem, or continuum configuration-measurability theorem is required.*

More quantitatively, for each sufficiently small r, every bounded observable F of the prefix stopped at radius r and every bounded Lipschitz observable G of the entire curve satisfy

$$|\mathbb{E}_P(FG) - \mathbb{E}_P F\,\mathbb{E}_P G| \le C\|F\|_\infty(\operatorname{Lip} G + \|G\|_\infty)r^{\alpha/(1+\alpha)}, \tag{EDG}$$

first at almost every stopping-continuity radius r. Constants and length units are fixed by the source geometry. The zero-one assertion itself holds on the full initial-germ sigma-field.

Proof. Fix a source-patch radius ϵ and couple two discrete configurations with independent colors inside $B(p,\epsilon)$ and shared independent colors outside. For any fixed $r < \epsilon/3$ and all small enough meshes, the first exploration stopped at radius r is determined entirely by its own interior colors. It is therefore independent of the *entire second exploration*, which uses only the shared outer colors and the second interior color set.

Take a joint subsequential curve limit. Both marginals equal P. Choose an almost-sure stopping-continuity radius $r < \epsilon/3$. The continuous mapping theorem preserves the product law, so the stopped prefix of γ_1 is independent of the full curve γ_2. Consequently any bounded initial-germ observable $F(\gamma_1)$ is independent of $G(\gamma_2)$ for every bounded Lipschitz G on the full curve space. This step concerns the limiting product law; it does not attempt to pass an arbitrary discontinuous germ statistic through a discrete limit.

The shield estimate supplies, in this same coupling, for a fixed small R and $\epsilon \ll R$,

$$\mathbb{P}\{d_{\mathrm{F}}(\gamma_1,\gamma_2) > 2R\} \le C(\epsilon/R)^\alpha.$$

Thus

$$\begin{aligned}|\mathbb{E}_P(FG) - \mathbb{E}_P F\,\mathbb{E}_P G| &= |\mathbb{E}F(\gamma_1)(G(\gamma_1) - G(\gamma_2))| \\ &\le \|F\|_\infty\left(2\operatorname{Lip}(G)R + 2C\|G\|_\infty(\epsilon/R)^\alpha\right).\end{aligned} \tag{EDG1}$$

The left side depends only on the marginal law P. Let $\epsilon \downarrow 0$ with R fixed and then $R \downarrow 0$. It vanishes. Bounded Lipschitz functions determine finite signed measures on a Polish space, so every bounded initial-germ observable is independent of the full curve sigma-field. Since it belongs to that sigma-field, it is almost surely constant. This proves initial trace-germ triviality, including its completion.

For the quantitative prefix assertion, start with a fixed good radius r, use $\epsilon = 4r$, and take $R = r^{\alpha/(1+\alpha)}$ after fixing the reference length unit. For sufficiently small r this is an admissible radius with $\epsilon \ll R$. Formula (EDG1) gives (EDG). □

Proposition 12.2 (Pure affine limit at a flat source)**.** *Let one independent site-percolation model satisfy the flat-source shield coupling of Theorem 10.2. Let D have a flat source germ. Suppose*

along one mesh subsequence its exploration converges to

$$P = \int \nu_{D,p,q}^{T}\, \pi(dT), \tag{ED14}$$

where π is a Borel probability on projective affine tensors. Assume the standard Polish oriented curve topology. Then π is a point mass. No limiting domain Markov theorem or continuum configuration-measurability theorem is needed.

Proof. The regular source endpoint decoder has distribution π under (ED14) and is measurable in the initial trace-germ sigma-field. This sigma-field is trivial by Theorem 12.1. Thus the decoder is almost surely one deterministic tensor, and π is a point mass. The target need only meet the curve-convergence assumptions here; analytic target regularity enters the source-comparison theorem below. □

Lemma 12.3 (Splicing a regular colored source germ)**.** *Fix a bounded reference domain D_0 with flat regular source $p = 0$, analytic regular target q, and one fixed boundary-color convention. Given any analytic regular colored source germ G at 0 with that convention, and any sufficiently small $\epsilon > 0$, there is a bounded marked domain $D_{G,\epsilon}$ agreeing with D_0 outside $B(0, \epsilon)$ and having exactly the germ G at 0. Its target germ at q agrees with D_0. The boundary can be chosen smooth except for any corners already present in D_0; only its two marked germs need to be analytic.*

Proof. For any analytic regular colored source germ G at p and any small ϵ, an orientation-preserving smooth diffeomorphism supported in $B(p, \epsilon)$ can map the source germ of one fixed flat-source reference domain to G, fixing p and the distant regular target. For an explicit local construction, use a radial twist to rotate a small core to the required tangent, followed by a cutoff graph shear to realize the analytic curvature in a still smaller core. The graph derivative can be made arbitrarily small, so the cutoff shear remains a diffeomorphism. Call the resulting domain $D_{G,\epsilon}$. It agrees with the reference outside the source patch and agrees with G in a still smaller neighborhood. The support avoids q, and ambient orientation preserves the boundary color order. Applying the diffeomorphism to D_0 proves the claim. □

Theorem 12.4 (One deterministic metric from a coherent affine mixture)**.** *Consider a translation-covariant source-local exploration family arising from one independent Bernoulli nonbranching cellular percolation model admitting the flat-source shield coupling of Theorem 10.2. Suppose each bounded-domain marginal with analytic regular marks is a Borel mixture of fixed affine-SLE_6 laws. Require the family to include the source splices of Lemma 12.3, and require coherent subsequential limits for these comparisons: for every such splice and the reference, compatible common-color discrete approximations along one common mesh subsequence realize the two specified marginal limits.*

Then one deterministic projective affine tensor is shared by every such marginal, and each marginal is the pure affine-SLE_6 law for that tensor. Strong DMP and marked inverse-chart continuity are not premises of this theorem.

Proof. The reference mixing measure π_0 is a point mass by Proposition 12.2. Fix another original domain with analytic regular source germ G. Translation covariance allows its source to be placed at 0 without changing its physical projective tensor. For every small ϵ, source locality identifies the original initial prefix law with that in $D_{G,\epsilon}$ through a common source neighborhood. The source tensor decoder is a function of this prefix and the common germ. Its distribution is the mixing measure in either state, so both have one common mixing measure π_G, independent of ϵ.

Coherence and the common-color shield coupling give a coupling of the two limiting curves in $D_{G,\epsilon}$ and D_0 such that their ordered terminal segments agree outside an event of probability at most $C(\epsilon/R)^\alpha$, for one fixed small R in the shared flat outer geometry. Their target germs coincide. Apply the two terminal decoders; the resulting tensor variables have respective laws π_G, π_0 and are equal on the coalescence event. The coupling inequality for total variation therefore gives

$$\|\pi_G - \pi_0\|_{\mathrm{TV}} \le C(\epsilon/R)^\alpha. \tag{ED16}$$

Here total variation is $\sup_B |\pi_G(B) - \pi_0(B)|$. Letting $\epsilon \downarrow 0$ gives $\pi_G = \pi_0$. Since π_0 is a point mass, the original marginal is pure and shares the same tensor. The argument applies separately to each original regular-source state; no simultaneous uncountable full-probability assertion is required. □

Thus, subject to the explicitly stated coherent independent-color scaling and shielding premises, the mixed-affine ansatz on regular domains already collapses to one deterministic common affine metric. This does not derive the mixed-affine ansatz for an arbitrary curve law, and does not classify arbitrary abstract three-axiom families.

Initial characteristic limits become deterministic. At any source to which the preceding zero-one argument applies, a finite initial limit in probability of an adapted characteristic A_t/t or C_t/t is deterministic. To see germ measurability, stop the characteristic at the first exit from an arbitrary source ball. That exit has strictly positive capacity time; the stopped and unstopped limits therefore agree. A limit in probability of variables measurable from the stopped prefix has a measurable version in its completed sigma-field. This holds for every small radius, so the limit belongs to the trivial initial germ. This removes the deterministic-value premise at these starts once existence of a tangent and natural-trace-filtration compatibility are known; neither tangent existence nor the required slit-uniform bound is obtained here.

13 Total-variation decoupling of an initial prefix and a distant suffix

The closed exact-coalescence relation gives a stronger consequence than bounded-Lipschitz covariance. It controls arbitrary measurable observables of the distant terminal suffix in total variation. This uses the actual independent-color source-shield coupling, not just abstract locality or an unforced-crossing bound.

Fix a bounded domain with source p, target q, flat source geometry, and a marginal curve scaling limit P satisfying the hypotheses of Theorem 12.1. Let $\sigma_r\gamma$ be the initial ordered prefix stopped on the radius-r circle about p. For $R < |q-p|$ define $\lambda_R\gamma$ to be the ordered terminal suffix beginning at the *last* visit to $\overline{B}(p,R)$. For any continuous representative on $[0,1]$ this means restriction from

$$t_R(\gamma) = \max\{t \in [0,1] : |\gamma(t) - p| \le R\}$$

to 1, followed by linear reparameterization. The resulting ordered curve does not depend on the representative or pauses.

Both stopping operations are Borel maps on the assumed Polish oriented Fréchet curve space. For completeness, on the continuous-parameter curve space, t_R is Borel because, for fixed a,

$$\{t_R \ge a\} = \left\{ \min_{a \le t \le 1} |\gamma(t) - p| \le R \right\}$$

is closed. Restriction and rescaling are jointly continuous in a curve and its cut parameter. Hence the terminal-suffix map to the quotient curve space is Borel on representatives and invariant under reparameterization. It descends to a Borel map on the Polish quotient: the preimages of Borel sets are saturated Borel sets, whose images and complements under the continuous quotient map are analytic; both are therefore Borel. First-exit stopping has the same Borel property.

Theorem 13.1 (Prefix–suffix decoupling)**.** *There exist $C < \infty$ and $\alpha > 0$, determined by the source-shield constants, such that for all sufficiently small R and all $0 < r < R/64$,*

$$\|\mathcal{L}_P(\sigma_r\gamma, \lambda_R\gamma) - \mathcal{L}_P(\sigma_r\gamma) \otimes \mathcal{L}_P(\lambda_R\gamma)\|_{\mathrm{TV}} \le C(r/R)^\alpha. \tag{SB1}$$

Here total variation is $\sup_A |\mu(A) - \nu(A)|$ on the product space. In particular this estimate holds at every *small radius r, not merely almost every stopping-continuity radius.*

Proof. Set $\epsilon = 8r$. At the discrete level take two configurations with independent colors in $B(p, \epsilon)$ and identical independent colors outside that ball. Both marginal explorations have the same scaling limit P. Their joint laws are tight. Choose a radius $\rho \in (r, 2r)$ at which first-exit stopping is P-almost surely continuous; such radii have full Lebesgue measure by Lemma 11.3. The first discrete exploration up to radius ρ uses only the first configuration's interior colors, once the mesh is sufficiently small. It is consequently independent of the entire second exploration. Continuous mapping along a joint subsequential limit gives

$$\sigma_\rho\gamma_1 \perp\!\!\!\perp \gamma_2.$$

Stopping the limiting first prefix again, now at radius r, is a Borel operation and requires no continuity assertion at r. Thus $\sigma_r\gamma_1$ is independent of the full curve γ_2.

Apply the source-shield estimate with shield radius $R/2$. Outside an event of probability at most $C(r/R)^\alpha$, both prefixes before coalescence stay in $K = \overline{B}(p, R/2)$ and the ordered suffixes are identical. Lemma 11.1 and the closed-set Portmanteau inequality preserve this statement in the same joint subsequential limit. Since $q \notin \overline{B}(p, R)$, each full curve's last visit to $\overline{B}(p, R)$ is on the common suffix. Therefore

$$\mathbb{P}\{\lambda_R\gamma_1 \ne \lambda_R\gamma_2\} \le C(r/R)^\alpha.$$

The pairs $(\sigma_r\gamma_1, \lambda_R\gamma_1)$ and $(\sigma_r\gamma_1, \lambda_R\gamma_2)$ form a coupling of the joint law in (SB1) with the displayed product law. The usual coupling bound for total variation proves (SB1). □

Corollary 13.2 (Expected conditional total-variation stability)**.** *Let $K_r(x, \cdot)$ be a regular conditional law of $\lambda_R\gamma$ given $\sigma_r\gamma = x$. Then*

$$\int \|K_r(x, \cdot) - \mathcal{L}_P(\lambda_R\gamma)\|_{\mathrm{TV}}\ \mathcal{L}_P(\sigma_r\gamma)(dx) \le C(r/R)^\alpha. \tag{SB2}$$

Consequently for bounded Borel functions F, G of the two respective variables,

$$|\operatorname{Cov}_P(F(\sigma_r\gamma), G(\lambda_R\gamma))| \le 2C\|F\|_\infty\|G\|_\infty(r/R)^\alpha. \tag{SB3}$$

Proof. On standard Borel spaces, total variation between a joint law and the product of its marginals is the integral of the total variations of the corresponding regular conditional laws. One may verify the identity using a countable generating algebra to compute the variation of each signed conditional measure. This gives (SB2). Integrating the bounded product observable against the difference of the two probability laws costs at most twice its supremum norm times the convention for total variation used in (SB1), proving (SB3). □

Variants and limitation. The same proof applies to any standard-Borel-valued observable Y_R that agrees whenever two curves belong to $E_{\overline{B}(p,R/2)}$. For example one may use the compact trace outside the open radius-R ball instead of the ordered distant suffix. Measurability of a given variant should be checked separately. Completed prefix sigma-fields are harmless, since versions may be chosen modulo their common marginal null sets.

This is quantitative spatial loss of memory at a deterministic regular source. It does not identify a conditional continuation kernel in an arbitrary random slit domain: the proof has no shield estimate uniform over rough explored frontiers. In particular it neither proves strong DMP in a curve limit nor the whole-slit stability premise of the driving-rigidity results. The whole-tube approach based on boundary three-arm estimates still needs the deterministic implication from ordered curve disagreement to an appropriate arm event.

14 Annular crossing statistics self-average at a regular source

The same source resampling argument applies to a macroscopic observable whenever the closed exact-coalescence relation forces its equality. It therefore controls annular crossings made *before the first outer exit*, even if the full curve later returns arbitrarily close to the source. No source nonreturn assumption is needed.

Proposition 14.1 (Decoupling for coalescence-invariant observations)**.** *Under the hypotheses of Theorem 13.1, let Y_R be a Borel map from the oriented curve space into a standard Borel space. Assume*

$$(\gamma,\widetilde{\gamma}) \in E_{\overline{B}(p,R/2)} \quad \Longrightarrow \quad Y_R(\gamma) = Y_R(\widetilde{\gamma}). \tag{AS1}$$

Then, for $0 < r < R/64$,

$$\big\|\mathcal{L}_P(\sigma_r\gamma, Y_R(\gamma)) - \mathcal{L}_P(\sigma_r\gamma)\otimes\mathcal{L}_P(Y_R(\gamma))\big\|_{\mathrm{TV}} \le C(r/R)^{\alpha}. \tag{AS2}$$

The same conclusion holds if the first variable is any Borel function of $\sigma_r\gamma$.

Proof. Use exactly the independent-inner/shared-outer coupling in the proof of Theorem 13.1, including a good stopping-continuity radius between r and $2r$. Its limiting curves obey $\sigma_r\gamma_1 \perp\!\!\!\perp \gamma_2$ and

$$\mathbb{P}\big(E_{\overline{B}(p,R/2)}\big) \ge 1 - C(r/R)^{\alpha}.$$

Assumption (AS1) implies $\mathbb{P}\{Y_R(\gamma_1) \ne Y_R(\gamma_2)\} \le C(r/R)^{\alpha}$. Couple the two pairs with common first variable $\sigma_r\gamma_1$, as in the earlier proof. The second pair has the product law by independence. This proves (AS2). Applying a Borel map to the first variable cannot increase total variation. □

Fix $a > 0$ sufficiently small that the source-shield construction applies up to scale a, and $|q-p| > a$. Put

$$r_n = a2^{-n}, \qquad n = 1,2,\dots.$$

For a continuous representative of γ define

$$\tau_n = \inf\{t : |\gamma(t)-p| = 2r_n\}, \qquad \eta_n = \max\{t \le \tau_n : |\gamma(t)-p| \le r_n\}. \tag{AS3}$$

Let $B_n(\gamma)$ be the ordered restriction to $[\eta_n,\tau_n]$, translated by $-p$ and dilated by r_n^{-1}. This is an oriented curve in $\{1 \le |z| \le 2\}$ from its inner to its outer boundary. It is the last inner-to-outer crossing before the first outer hit. Only this crossing is recorded; later returns of the full curve are immaterial.

The map B_n is Borel and invariant under changes of representative. Indeed τ_n is a Borel first-hitting operation. Restrict first to $[0,\tau_n]$, linearly reparameterized on $[0,1]$. On this stopped curve, η_n is a Borel last visit to the closed inner ball, by the last-visit argument in Section 13. Successive restriction, translation, dilation, and passage to the oriented quotient are Borel. In particular B_n is a Borel function of $\sigma_{2r_n}\gamma$.

There is a second property crucial to the proof: if $(\gamma,\widetilde{\gamma}) \in E_{\overline{B}(p,r_n/2)}$, then $B_n(\gamma) = B_n(\widetilde{\gamma})$. The prefixes before the common suffix remain strictly inside the inner circle, and the common suffix starts there. Both first outer hits thus occur on that suffix, and so do their last inner-ball visits before those hits. The corresponding ordered crossing segments agree exactly.

Theorem 14.2 (Exponential absolute regularity across dyadic source scales)**.** *Under the same source-shield and independent-color limit hypotheses, the annular crossing sequence $(B_n)_{n\ge1}$ satisfies*

$$\begin{aligned}\big\|\mathcal{L}_P\big((B_1,\dots,B_m),(B_{m+k},B_{m+k+1},\dots)\big)\\ -\mathcal{L}_P(B_1,\dots,B_m)\otimes\mathcal{L}_P(B_{m+k},B_{m+k+1},\dots)\big\|_{\mathrm{TV}} \le C_0 2^{-\alpha k}, \qquad m,k\ge1.\end{aligned} \tag{AS4}$$

The countable product on the fine-scale side is a standard Borel space. Constants are independent of m and k. No stationarity or dilation covariance of the marginal curve limit is assumed.

Proof. For $k \ge 8$, apply Proposition 14.1 with $R = r_m$ and $r = 2r_{m+k}$. The vector $Y_R = (B_1,\dots,B_m)$ satisfies (AS1): every one of its inner radii is at least r_m, so coalescence inside the radius-$r_m/2$ ball occurs before all the relevant inner and outer visits. The whole countable vector $(B_{m+k},B_{m+k+1},\dots)$ is a Borel function of $\sigma_{2r_{m+k}}\gamma$, because every corresponding outer radius is at most $2r_{m+k}$. Thus (AS2) gives the upper bound

$$C(2r_{m+k}/r_m)^\alpha = C2^\alpha 2^{-\alpha k}.$$

Enlarge C_0 to cover $1 \le k < 8$ by the trivial total-variation bound one. Interchanging the two coordinates of a joint law preserves total variation, giving (AS4) in the displayed order. □

Corollary 14.3 (A strong law for bounded annular observations)**.** *Let f_n be any deterministic sequence of Borel real functions of an annular oriented crossing, with $|f_n| \le 1$. Put $Z_n = f_n(B_n)$. Then*

$$\frac{1}{N}\sum_{n=1}^{N}\{Z_n - \mathbb{E}_P Z_n\} \longrightarrow 0 \qquad P\text{-almost surely.} \tag{AS5}$$

In particular, for a fixed bounded Borel f, if the deterministic Cesàro means

$$\frac{1}{N}\sum_{n=1}^{N}\mathbb{E}_P f(B_n)$$

converge to c_f, then the empirical means $N^{-1}\sum_{n\le N} f(B_n)$ converge almost surely to c_f. These assertions hold simultaneously for any prescribed countable collection of choices of the sequence (f_n).

Proof. The total-variation estimate (AS4) and $|Z_n| \le 1$ imply

$$|\operatorname{Cov}_P(Z_n,Z_j)| \le 2C_0 2^{-\alpha|n-j|} \quad (n \ne j).$$

Consequently, for $S_N = \sum_{n\le N}(Z_n - \mathbb{E}_P Z_n)$,

$$\mathbb{E}_P S_N^2 \le C_1 N.$$

Chebyshev's inequality and Borel–Cantelli give $S_{\ell^2}/\ell^2 \to 0$ almost surely, since for every $\zeta > 0$,

$$\sum_{\ell=1}^{\infty} \mathbb{P}\{|S_{\ell^2}| > \zeta \ell^2\} \le \frac{C_1}{\zeta^2} \sum_{\ell=1}^{\infty} \ell^{-2} < \infty.$$

For $\ell^2 \le N < (\ell+1)^2$, boundedness of the centered summands gives $|S_N - S_{\ell^2}| \le 2(2\ell+1)$. Dividing by N yields (AS5). Add the deterministic expectations for the second assertion, and intersect countably many probability-one events for the last one. □

Meaning and remaining gap. Annular crossing statistics of one limiting exploration self-average over logarithmically many source scales. The centering sequence is deterministic, even when the scaling limit itself has not been shown dilation invariant. Neither the convergence of those deterministic expectations nor their values is asserted. A source law that is invariant under dyadic dilation would make the expectations constant, but such invariance is an additional premise and is not inferred from precompactness of the collection of scaling limits. The result supplies a rigorous scale-statistics diagnostic without an affine-SLE ansatz; it does not classify those deterministic statistics or produce random slit-domain exploration kernels.

14.1 A martingale approximation and a variance criterion for Gaussian scale fluctuations

The exponential absolute regularity in (AS4) yields more than the centered strong law. The following statements require no stationarity or dilation covariance. Their time variable counts source scales, rather than Loewner capacity.

Let $Z_j = f_j(B_j)$ with deterministic Borel $|f_j| \le 1$, put $Y_j = Z_j - \mathbb{E}Z_j$, and write $\mathcal{A}_j = \sigma(B_1, \dots, B_j)$, with $\mathcal{A}_0$ trivial. Then $|Y_j| \le 2$. For constants $C, c > 0$ depending only on (AS4),

$$\|\mathbb{E}[Y_{j+k} \mid \mathcal{A}_j]\|_1 \le Ce^{-ck}, \qquad j \ge 0,\ k \ge 1. \tag{AM1}$$

Indeed, test the joint law of the coarse vector and Y_{j+k} against a bounded sign of this conditional expectation times Y_{j+k}; the product-law expectation vanishes. Boundedness then gives, for every real $p \ge 1$,

$$\|\mathbb{E}[Y_{j+k} \mid \mathcal{A}_j]\|_p \le C_1 e^{-ck/p}. \tag{AM2}$$

Here C_1 can be chosen independent of p, by $\|V\|_p^p \le 2^{p-1}\|V\|_1$ when $|V| \le 2$.

Proposition 14.4 (An exponentially integrable annular corrector)**.** *Under (AS4), define in L^p, for any finite $p \ge 1$,*

$$U_j = \sum_{k=1}^{\infty} \mathbb{E}[Y_{j+k} \mid \mathcal{A}_j], \qquad U_0 = 0, \qquad D_j = Y_j + U_j - U_{j-1}. \tag{AM3}$$

The definitions give common versions for all finite p and all integer j. The process $M_N = \sum_{j=1}^{N} D_j$ is a square-integrable $\mathcal{A}_N$-martingale and

$$S_N := \sum_{j=1}^{N} Y_j = M_N - U_N. \tag{AM4}$$

For constants $a, C_2 > 0$,

$$\sup_j \mathbb{E}e^{a|U_j|} < \infty, \qquad \sup_j \mathbb{E}e^{a|D_j|} < \infty, \qquad \max_{0 \le j \le N} |U_j| = O(\log N) \quad \textit{almost surely.} \tag{AM5}$$

Proof. Minkowski's inequality and (AM2) imply

$$\|U_j\|_p \le C_1 \sum_{k\ge 1} e^{-ck/p} = \frac{C_1}{e^{c/p}-1} \le \frac{C_1}{c} p.$$

Thus the series converges in each L^p, with the same L^1 limit. The tower property and termwise conditioning, justified by L^1 convergence, give

$$\mathbb{E}[Y_j + U_j \mid \mathcal{A}_{j-1}] = U_{j-1},$$

so the D_j are martingale differences. Telescoping gives (AM4). The displayed L^p bound and $\|D_j\|_p \le 2 + \|U_j\|_p + \|U_{j-1}\|_p$ give uniform bounds $\|U_j\|_p + \|D_j\|_p \le C_3 p$. Expanding the exponential into its power series and using $m! \ge (m/e)^m$ proves the two exponential-moment assertions for sufficiently small $a > 0$. In particular $\mathbb{P}(|U_j| > x) \le C_4 e^{-ax}$ uniformly in j. The union bound gives

$$\mathbb{P}\left(\max_{j\le N} |U_j| > A \log N\right) \le C_4 N^{1-aA}.$$

Choose $aA > 2$ and sum over integers $N \ge 2$. Borel–Cantelli proves the last assertion of (AM5). □

Deterministic and conditional variance are distinct. Orthogonality of martingale differences and (AM5) show

$$\left| \mathrm{Var}(S_N) - \sum_{j\le N} \mathbb{E} D_j^2 \right| = O(\sqrt{N}). \tag{AM6}$$

A martingale central limit theorem would normally ask for convergence of conditional variance averages. For the original bounded annular sequence, absolute regularity allows a direct blocking proof using only a deterministic variance limit, as follows.

Theorem 14.5 (A deterministic variance criterion for an annular invariance principle)**.** *Suppose (AS4) holds and, for a deterministic $\sigma^2 \ge 0$,*

$$\frac{1}{N} \mathrm{Var}(S_N) \longrightarrow \sigma^2. \tag{AM7}$$

Let $\mathcal{S}_N$ be the continuous polygonal interpolation with $\mathcal{S}_N(j/N) = S_j/\sqrt{N}$. On every fixed compact time interval,

$$\mathcal{S}_N \implies \sigma B \quad \textit{in the uniform topology.} \tag{AM8}$$

Consequently $S_N/\sqrt{N}$ tends in law to a centered Gaussian of variance σ^2. Condition (AM7) is an additional deterministic premise; it is not inferred from RSW or (AS4).

Proof. We give a blocking and moment proof to separate (AM7) from a hidden stationarity or conditional-variance assumption. The covariance estimate from (AS4) gives

$$|\mathrm{Cov}(Y_i, Y_j)| \le C e^{-c|i-j|}, \qquad i \ne j. \tag{AM9}$$

Thus the variance of a sum over any finite subset of indices is at most a constant times its cardinality. The covariance of sums over any two disjoint consecutive intervals, in either order, is uniformly bounded: sum the geometric bound over the two distances from their common boundary.

First consider a centered bounded triangular array whose row is a deterministic bounded weighting of $O(N)$ successive variables Y_j and whose row-sum variance divided by N has a limit

v. The absolute-regularity and covariance bounds remain uniform. Split the row into big blocks of length $p_N = \lfloor N^{1/3} \rfloor$, separated by gaps of length $q_N = \lceil A \log N \rceil$, and omit the final remainder. The number of omitted indices is $O(Nq_N/p_N + p_N + q_N) = o(N)$, so their sum divided by $\sqrt{N}$ tends to zero in L^2.

The total-variation distance between the joint big-block law and the product of its block marginals is at most $C(N/p_N)e^{-cq_N} = o(1)$ when A is sufficiently large. To see this, separate the final block from the entire preceding block vector using (AS4), and iterate; total variation adds under this telescoping argument. For independent blocks with these marginals, the sum of block variances divided by N tends to v. Indeed replacing the omitted variables changes the original variance by $o(N)$, and the sum of cross-block covariances is at most $CNp_Ne^{-cq_N} = o(N)$, by (AM9). Each independent block divided by $\sqrt{N}$ is deterministically bounded in magnitude by $Cp_N/\sqrt{N} = o(1)$. Taylor expansion of its characteristic function through second order has total remainder bounded by this maximum times the sum of the normalized block variances. Taking the product of characteristic functions therefore gives the limit $e^{-v\theta^2/2}$. The total-variation approximation and the omitted L^2 remainder prove the same Gaussian limit for the original weighted row. This argument also covers $v = 0$; alternatively that case follows from Chebyshev's inequality.

Apply this triangular-array statement to every fixed linear combination of increments of $S_{\lfloor Nt \rfloor}/\sqrt{N}$. Assumption (AM7) gives $N^{-1}\operatorname{Var}(S_{\lfloor Nt \rfloor}) \to \sigma^2 t$. The uniform bound on covariances between consecutive interval sums then shows that a linear combination with increment coefficients a_ℓ and time intervals $(t_{\ell-1}, t_\ell]$ has variance limit $\sigma^2 \sum_\ell a_\ell^2 (t_\ell - t_{\ell-1})$. Cramér–Wold thus gives the Brownian finite-dimensional limits.

For tightness, write an ordered quadruple of indices as $i \le i + a \le i + a + b \le i + a + b + d$. Its centered-product expectation is bounded in absolute value by

$$Ce^{-c\max(a,d)}. \tag{AM10}$$

When $a > 0$, separate the first centered factor from the other three using (AS4); when $d > 0$, separate the last centered factor instead. Use the stronger of these two estimates, or the trivial boundedness estimate if both gaps vanish. This argument permits repeated middle indices and repeated endpoint pairs. Summing (AM10) over quadruples in any interval of m indices gives

$$\mathbb{E}\left|\sum_{j=r+1}^{r+m} Y_j\right|^4 \le C_5 m^2, \tag{AM11}$$

because $\sum_{a,d\ge 0} e^{-c\max(a,d)} < \infty$ while the starting index and the middle gap have $O(m^2)$ choices. For the polygonal process, (AM11) gives $\mathbb{E}|\mathcal{S}_N(t) - \mathcal{S}_N(s)|^4 \le C_6|t-s|^2$ when $|t-s| \ge N^{-1}$: the two fractional endpoint pieces have total magnitude at most $4/\sqrt{N}$, which is absorbed in this fourth-moment bound. For shorter intervals use the deterministic slope bound $2\sqrt{N}$, giving the same estimate. The usual Kolmogorov tightness criterion, together with $\mathcal{S}_N(0) = 0$, proves tightness in the uniform topology. Combining this with the finite-dimensional limits proves (AM8). □

A two-scale criterion for the variance limit. As a more explicit sufficient target, suppose every fixed lag $k \ge 0$ has a deterministic Cesàro covariance limit

$$c_k = \lim_{N\to\infty} \frac{1}{N} \sum_{j=1}^{N-k} \operatorname{Cov}(Y_j, Y_{j+k}).$$

Then (AM9) and dominated summation imply (AM7), with the absolutely convergent formula

$$\sigma^2 = c_0 + 2\sum_{k=1}^{\infty} c_k. \tag{AM12}$$

This provides a condition in terms of two-scale expectations, without requiring exact stationarity. No convergence of these expectations is asserted for the percolation scaling limits under study.

The variance premise cannot be dropped. Even independent bounded observations can have no asymptotic variance rate. For example let $Y_j = a_j\varepsilon_j$, with independent symmetric signs ε_j and deterministic $a_j \in \{0,1\}$. Choose alternating 0 and 1 blocks whose successive lengths dominate all previous lengths. Then (AS4) holds with zero error, but $N^{-1}\operatorname{Var}(S_N) = N^{-1}\sum_{j\le N} a_j$ has subsequential limits 0 and 1. Thus the new invariance principle isolates another deterministic scale-statistics limit; it does not identify universal expectations or fluctuation variances and does not control Loewner-capacity increments.

15 An isotropic whole-plane law identifies its affine tensor

Lemma 15.1 (Only similarities stabilize an isotropic scale-invariant curve law)**.** *Let μ be the law of an almost surely nonconstant oriented continuous curve starting at 0, with compact continuous prefixes up to every finite spatial exit. Suppose its law is invariant under every rotation and every positive scalar dilation. If $L \in GL^+(2,\mathbb{R})$ preserves μ, then L is a similarity. Consequently*

$$A_*\mu = B_*\mu \quad\Longleftrightarrow\quad AA^{\mathsf{T}} = cBB^{\mathsf{T}} \text{ for some } c > 0. \tag{WP1}$$

Proof. If L is not a similarity, singular value decomposition, together with the rotation and scalar symmetries, makes $D = \operatorname{diag}(1,a)$, $0 < a < 1$, another symmetry of μ. For $Y \sim \mu$, the law of $X = \sup|\operatorname{Re} Y|$ is invariant under all positive scalar multiplications; hence $X \in \{0,\infty\}$ almost surely. The event $X = 0$ says the nonconstant curve is contained in one specified line through 0. Its images under countably many distinct rotations are disjoint and have equal probabilities, so each has probability zero. Thus $X = \infty$ almost surely. In particular the first time T with $|\operatorname{Re} Y(T)| = 1$ is finite almost surely, and $M = \sup_{s\le T}|\operatorname{Im} Y(s)| < \infty$.

The first unit-radius exit of D^nY occurs by T; its initial trace stays within distance a^nM of the real axis. Since D^nY has law μ, for every $\epsilon > 0$ the probability that the unit-radius stopped trace leaves the ϵ-strip is at most $\mathbb{P}(a^nM > \epsilon) \to 0$. Thus the unit-radius initial trace lies on the real axis almost surely. Rotation invariance also puts it on the imaginary axis almost surely, a contradiction. Finally $A_*\mu = B_*\mu$ is equivalent to $B^{-1}A$ stabilizing μ, and a direct matrix calculation gives (WP1). □

15.1 A decoder stable under random local differentiable maps

An affine decoder can be constructed with one exceptional set independent of the local coordinate map. The useful change is to average whole curves in the Riemann sphere before extracting the tensor; first-radius stopping is then used only for Borel germ coding, not for continuity of an averaging observable.

We make an additional explicit properness and coding assumption. Let $\mathcal{C}$ be a Polish space of oriented continuous curves in the Riemann sphere from 0 to ∞, modulo increasing reparameterization and pauses, with bounded spherical Fréchet metric $d_{\mathcal{C}}$, induced by the chordal metric on the sphere.

Assume the affine action and scalar dilation are continuous on this space and first-radius stopping is Borel. Require the proper and source-avoiding curve sets to be Borel in this coding. These are properties of the coding, not consequences of the exploration axioms. A curve is *proper* here if it has a continuous planar representative $y : [0, \infty) \to \mathbb{C}$ with $|y(t)| \to \infty$ as $t \to \infty$.

Lemma 15.2 (No later visit to the source)**.** *Let μ be a dilation-invariant law supported on proper curves from 0 to ∞. Then almost surely the curve has no visit to 0 after it first leaves 0.*

Proof. For $R > 0$, let E_R be the event of a visit to 0 after the first radius-R exit. Dilation invariance makes $\mu(E_R)$ independent of R. The events decrease as R increases. For every proper continuous curve, all visits to 0 occur in a finite time interval, whose initial trace has finite radius. Thus $\bigcap_{R>0} E_R$ is empty on the proper-curve set, and $\mu(E_R) = 0$ for every R. A nontrivial later visit to 0 belongs to E_R for some positive rational R. □

Lemma 15.3 (Uniform spherical linearization)**.** *Let ψ be an orientation-preserving C^1 diffeomorphism on a neighborhood of 0, with $\psi(0) = 0$ and derivative $L \in GL^+(2, \mathbb{R})$. It has a global bi-Lipschitz extension Ψ agreeing with ψ on a smaller neighborhood and fixing ∞, such that*

$$\sup_{z \in \widehat{\mathbb{C}}} d_{\mathrm{sph}}\big(\lambda \Psi(z), \lambda L z\big) \longrightarrow 0 \quad (\lambda \to \infty). \tag{WP3}$$

Consequently the same upper bound controls $d_{\mathcal{C}}(\lambda \Psi Y, \lambda L Y)$ for every curve Y.

Proof. Write $\psi(z) = Lz + r(z)$, where $r(0) = 0$ and $Dr(z) \to 0$. Choose a smooth radial cutoff χ_δ, equal to one in $B(0, \delta)$ and zero outside $B(0, 2\delta)$, and set $\Psi(z) = Lz + \chi_\delta(z) r(z)$, with the perturbation zero outside the chart. For sufficiently small δ the Lipschitz norm of the perturbation is less than half the smallest singular value of L. Hence Ψ is globally bi-Lipschitz. Surjectivity follows, for example, from the contraction mapping applied to $z = L^{-1}(w - \chi_\delta(z) r(z))$ for each w.

There is $c > 0$ with $|\Psi(z)| \geq c|z|$ and $|Lz| \geq c|z|$. Given $\eta > 0$, choose $a > 0$ such that $|\Psi(z) - Lz| \leq \eta |z|$ for $|z| \leq a$. The chordal formula (our convention for d_{sph})

$$d_{\mathrm{sph}}(u, v) = \frac{|u - v|}{\sqrt{1 + |u|^2}\sqrt{1 + |v|^2}}$$

then bounds the distance in (WP3), for $|z| \leq a$, by

$$\frac{\eta \lambda |z|}{1 + c^2 \lambda^2 |z|^2} \leq \frac{\eta}{2c}.$$

For $|z| \geq a$, the triangle inequality through ∞ bounds the distance by $2/(c\lambda a)$. First let $\lambda \to \infty$ and then $\eta \downarrow 0$. Pairing identical curve parameters gives the Fréchet bound. □

Lemma 15.4 (Changing a distant tail is invisible after magnification)**.** *Suppose two proper source-avoiding curves share a nonconstant initial ordered segment. Then*

$$d_{\mathcal{C}}(\lambda Y, \lambda \widetilde{Y}) \longrightarrow 0 \quad (\lambda \to \infty). \tag{WP4}$$

Here source avoidance means that neither curve visits 0 after its initial departure.

Proof. Synchronize parameters along a common initial segment ending at a nonzero point. Each remaining tail is a compact subset of the sphere disjoint from 0; its finite planar points therefore have modulus at least some positive b. After multiplication by λ, both tails lie within chordal distance $1/(\lambda b)$ of ∞. Match their remaining parameters arbitrarily. The common prefixes agree exactly and the two tails have mutual distance at most $2/(\lambda b)$. □

Theorem 15.5 (A robust affine decoder on initial ordered germs)**.** *Let μ be a nonconstant rotation- and dilation-invariant law on $\mathcal{C}$, supported on proper curves, and suppose dilation by one factor $\rho > 1$ is ergodic. There is one Borel tensor-valued decoder D, measurable from every positive-radius initial prefix on the proper source-avoiding curve set, with the following simultaneous property. There is a single μ-conull set $\mathcal{G}$ such that, for every $Y \in \mathcal{G}$, every $A \in GL^+(2,\mathbb{R})$, and every local orientation-preserving C^1 diffeomorphism ψ fixing 0, with derivative L,*

$$D(\psi(AY)) = [((LA)^{-1})^{\mathsf{T}}(LA)^{-1}]. \tag{WP5}$$

The argument of D can be any proper source-avoiding completion of the indicated local curve segment; the value is independent of the completion. In particular A and ψ may depend arbitrarily on Y and on additional randomness, provided the displayed local regularity holds for each realization.

Proof. Let T act by multiplication by ρ, and form empirical orbit measures in the Polish probability-measure space $\mathcal{P}(\mathcal{C})$:

$$\Pi_N(Y) = \frac{1}{N}\sum_{n=1}^{N} \delta_{T^n Y}.$$

The ergodic theorem applied to a countable bounded continuous convergence-determining family gives a single conull Borel set $\mathcal{G}$ on which $\Pi_N(Y) \Rightarrow \mu$. Intersect with the proper source-avoiding set, using Lemma 15.2.

For every $Y \in \mathcal{G}$ and every fixed linear B, commutation of B with T and continuity of its action give

$$\Pi_N(BY) = B_*\Pi_N(Y) \Rightarrow B_*\mu. \tag{WP6}$$

This is a deterministic consequence of weak convergence for the given Y, so it holds simultaneously for all B; it does not intersect uncountably many probability-one events.

Extend ψ to Ψ as in Lemma 15.3. Applying that lemma to $\Psi \circ A$ gives $d_{\mathcal{C}}(T^n \Psi AY, T^n LAY) \to 0$. For every bounded Lipschitz test function, the difference of its empirical averages consequently tends to zero by Cesàro averaging. Together with (WP6), this gives

$$\Pi_N(\Psi AY) \Rightarrow (LA)_*\mu. \tag{WP7}$$

Again this is a pathwise assertion for every admissible A, ψ.

To decode the limiting law, parameterize projective positive definite tensors by determinant-one matrices Q and set $A_Q = Q^{-1/2}$. The map

$$Q \longmapsto (A_Q)_*\mu \in \mathcal{P}(\mathcal{C})$$

is continuous by the continuous affine action and is injective by Lemma 15.1. Its image is Borel and its inverse is Borel by Lusin–Souslin. Apply that inverse to the weak limit of Π_N when the limit exists and belongs to this image; otherwise assign a cemetery value. This defines a Borel decoder. Rotation and scalar invariance identify the tensor of the law on the right side of (WP7) as the class in (WP5).

It remains to check germ measurability. For a proper source-avoiding curve Y, replace the tail after its first radius-r exit by the radial ray to ∞, obtaining $E_r Y$. This is a Borel function of the initial prefix: use Borel stopping followed by continuous concatenation with the radial ray in the spherical curve coding. Lemma 15.4 gives $d_{\mathcal{C}}(T^n Y, T^n E_r Y) \to 0$. Their empirical measures therefore have weak limits simultaneously, with the same limit when it exists. More precisely, their bounded-Lipschitz distance is at most $N^{-1}\sum_{n\le N} d_{\mathcal{C}}(T^n Y, T^n E_r Y) \to 0$. Hence $D(Y) = D(E_r Y)$ for every $r > 0$, including when the decoder has the cemetery value. The same argument makes the decoder independent of the completion of a local curve segment and completes the proof. □

Scope for random tip charts. If L is conformal, the decoded tensor changes by the usual covariant rule $Q \mapsto [L^{-\mathsf{T}} Q L^{-1}]$. If a random physical local curve is obtained by first applying a random conformal local map to Y and then a fixed affine map A, its derivative is $A(aR)$ and (WP5) decodes $[(A^{-1})^{\mathsf{T}} A^{-1}]$, independently of the random similarity aR. Thus dependence of the local map on the sampled curve is not by itself an obstruction once this proper, dilation-ergodic whole-plane model and a C^1 local chart are known. This theorem still supplies an ordered-germ decoder. It does not select a canonical point of an unlabelled frontier, prove an intrinsic typical-tip theorem, establish compatibility at arbitrary exploration stopping times, or derive the required whole-plane law and dilation ergodicity from the three exploration axioms.

15.2 Full scaling ergodicity from the stationary cylinder driver

Zhan's construction in Section 3 of [11] gives more than an ergodic one-dimensional angular statistic once the full cylinder process is retained. The following argument supplies the full discrete-dilation ergodicity premise; it uses the classical Hörmander hypoellipticity theorem for the smoothing step.

Proposition 15.6 (Every stationary cylinder skeleton is ergodic)**.** *Let $\kappa > 0$, $\varrho \geq \kappa/2 - 2$, and $a = \varrho/2 + 1 > 0$. Let $Z_t \in (0, 2\pi)$ be the two-sided stationary diffusion*

$$dZ_t = \sqrt{\kappa}\, dB_t + a \cot(Z_t/2)\, dt, \qquad \pi_Z(dz) = c \sin(z/2)^{4a/\kappa}\, dz. \tag{WP8}$$

Let q_0 be uniform modulo 2π, independent of the entire stationary path Z, and set

$$q_t = q_0 - \int_0^t \cot(Z_s/2)\, ds \pmod{2\pi}. \tag{WP9}$$

Then the cylinder process (q_t, Z_t) is stationary, and its two-sided stationary path law is ergodic under every fixed time shift $h \neq 0$. Consequently any Borel whole-plane Loewner trace constructed from $e^{i\lambda_t}$, $\lambda_t = q_t + Z_t$, is ergodic under multiplication by every fixed scalar $e^h \neq 1$ as an unparameterized curve, on a conull invariant set where the trace is defined.

Proof. The dimension of the radial Bessel diffusion is $1 + 4a/\kappa \geq 2$, so its endpoints are not hit; this is the stationary construction in [11, Section 3 and Appendix B]. Uniform independent q_0 makes q_t uniform conditional on the entire path Z, proving stationarity and giving the invariant probability $\pi = (dq/(2\pi))\pi_Z(dz)$, with full support.

The Markov generator on the open cylinder is

$$\mathcal{L} = \frac{\kappa}{2}\partial_z^2 + a \cot(z/2)\partial_z - \cot(z/2)\partial_q. \tag{WP10}$$

Write $V_1 = \sqrt{\kappa}\, \partial_z$ and $V_0 = a \cot(z/2)\partial_z - \cot(z/2)\partial_q$. Then

$$[V_1, V_0] = \frac{\sqrt{\kappa}}{2} \csc^2(z/2)(-a\partial_z + \partial_q). \tag{WP11}$$

Thus V_1 and $[V_1, V_0]$ span the spatial tangent space at every interior point. The parabolic Hörmander theorem applied to $\mathcal{L} - \partial_t$ gives the strong Feller property at every positive time. Here is the localization needed for the singular endpoints. On a relatively compact subcylinder, the killed backward semigroup applied to a bounded Borel function solves the homogeneous backward Kolmogorov equation in the interior at positive times, in the distributional sense. Hörmander's theorem [17, Theorem 1.1] makes this solution smooth there. For any initial interior point and finite horizon, endpoint nonattainment gives a compact subcylinder containing the entire path with

probability at least $1-\varepsilon$. Choose a slightly larger subcylinder. Coupling nearby starting points with the same Brownian motion, local Lipschitz continuity and Gronwall's inequality show that their paths remain in the larger subcylinder on this same event, for all sufficiently close starting points. The full semigroup thus differs from that killed semigroup by at most $\|f\|_\infty\varepsilon$, uniformly near the given point. Let $\varepsilon \downarrow 0$ to obtain continuity for every bounded Borel f. Hörmander smoothing is an external analytic input, not a consequence of stationarity alone.

For clarity, topological irreducibility at every fixed time $h > 0$ can be checked directly, without a general support theorem. Given two cylinder points (q, z) and (q', z'), choose a smooth controlled path $z(s)$, $0 \le s \le h$, from z to z', remaining in a compact subinterval of $(0, 2\pi)$. It can be chosen so that

$$-\int_0^h \cot(z(s)/2)\, ds = q' - q \pmod{2\pi}.$$

Indeed let the path move to a variable interior level, remain there for a positive time, and return to z'. As that level ranges from 0 to 2π, the integral varies continuously from $-\infty$ to $+\infty$. The required deterministic Brownian control is

$$w(s) = \kappa^{-1/2}\left(z(s) - z - a\int_0^s \cot(z(u)/2)\, du\right).$$

It is a Cameron–Martin path. Brownian motion has positive probability of lying in every uniform neighborhood of w. Local Lipschitz dependence of the additive-noise SDE and its integrated q coordinate shows that the uncontrolled process ends in any prescribed neighborhood of (q', z') with positive probability. Hence each skeleton kernel P_h is topologically irreducible.

A strong Feller irreducible Markov kernel has at most one invariant probability, and its stationary probability is an extreme invariant probability. Equivalently its two-sided stationary Markov path law is shift-ergodic. Apply this standard Markov-kernel fact to P_h for every fixed $h > 0$. The same conclusion holds on the full continuous path space under time shift h. Indeed consider the Markov chain of path segments $(X_{nh+s})_{0\le s\le h}$, where $X = (q, Z)$. Its transition law depends only on the previous endpoint. The endpoint law of any invariant segment law must be invariant for P_h, hence must equal π. The segment law is then necessarily the law obtained by starting the diffusion from π and recording its next length-h segment. Thus the segment chain has a unique invariant law and is ergodic; concatenation intertwines its shift with the full continuous-path shift by h. Negative shifts have the same invariant events.

Finally the deterministic whole-plane Loewner correspondence intertwines shifting λ_t to λ_{t+h} with mapping the capacity-parameterized trace $\gamma(t)$ to $e^{-h}\gamma(t+h)$. After forgetting the parameter this is scalar multiplication by e^{-h}. A Borel factor of an ergodic transformation is ergodic, proving the final assertion. □

Application and remaining premise. For $0 < \kappa < 4$ and $\varrho = \kappa + 2$, the stationary whole-plane SLE$(\kappa; \kappa+2)$ driver of [11, Section 3] therefore has full dilation ergodicity, not only the ergodicity of its coordinate Z used in Section 6 of that paper. Applying Theorem 15.5 still requires its stated Borel spherical curve coding and proper continuous trace from 0 to ∞; this proposition supplies the ergodicity input but does not prove those geometric inputs.

Corollary 15.7 (An affine tensor at a deterministic simple-SLE tip)**.** *Fix $0 < \kappa < 4$ and assume the proper spherical curve coding premises of Theorem 15.5 for the whole-plane SLE$(\kappa; \kappa+2)$ law. Let β be a chordal SLE$_\kappa$ trace, let $t_0 > 0$ be a deterministic half-plane capacity time, and let $A \in GL^+(2, \mathbb{R})$ be fixed. The same decoder applied to the reversed ordered germ of $A\beta$ at $A\beta(t_0)$, translated to start at 0, almost surely returns*

$$[(A^{-1})^{\mathsf{T}} A^{-1}]. \tag{WP12}$$

For a general simply connected domain, use a conformal parameter chart and its corresponding deterministic chordal capacity time.

Proof. Theorem 5.3 of [11] maps the reversed past of β at t_0 by a random conformal map to an initial segment of a whole-plane SLE$(\kappa;\kappa+2)$ trace. Its local inverse at 0 has a random nonzero conformal derivative aR. Proposition 15.6 supplies the whole-plane discrete-dilation ergodicity. Apply Theorem 15.5 to this random inverse chart followed by A. The derivative is $A(aR)$, whose tensor class is exactly (WP12). The decoder needs only the initial ordered germ, so the whole-plane segment can be completed arbitrarily. □

This corollary holds simultaneously at any prescribed countable set of deterministic capacity times. It does not assert correctness at every tip, at a tip chosen by an arbitrary intrinsic rule, or at a stopping time selected by the trace.

15.3 An intrinsic measure class without affine Minkowski convergence

There is a way to address the measure-selection part of the frontier problem without first proving existence of Minkowski content after every affine deformation. This does not supply a typical-tip theorem or a Borel choice of the relevant local frontier arc.

Fix $0 < d < 2$. For a nonempty compact planar set E and $\epsilon > 0$ put

$$m_{E,\epsilon}(dx) = \epsilon^{d-2}\mathbf{1}_{\{\operatorname{dist}(x,E)<\epsilon\}}\,dx. \tag{IM1}$$

Call E content regular if these finite measures converge weakly, as $\epsilon \downarrow 0$, to a finite nonzero measure μ_E. This is an explicit local Minkowski-content hypothesis. The theorem below is deterministic; no SLE or exploration assumption enters its proof.

Theorem 15.8 (A Euclidean canonical affine-stable measure class)**.** *There is a Borel map $E \mapsto \nu_E$ from compact planar sets to finite positive measures, supported on E, with the following properties. It has exact d-dimensional similarity covariance: for $F(x) = sRx + t$, $s > 0$ and R orthogonal, $\nu_{FE} = s^d F_*\nu_E$. For every content-regular E, $\nu_E = \mu_E$. If A is invertible real-linear, with smallest and largest singular values $a, b > 0$, then*

$$|\det A|\, b^{d-2} A_*\mu_E \;\le\; \nu_{AE} \;\le\; |\det A|\, a^{d-2} A_*\mu_E \tag{IM2}$$

as measures. In particular, ν_{AE} and $A_*\mu_E$ have precisely the same null sets. No convergence of the tube measures of AE is assumed. Exact affine covariance of the selected measure is not claimed.

Proof. Consider the Borel set of compact sets

$$\mathcal{E}_d = \left\{ E : 0 < \liminf_{\epsilon\downarrow 0} m_{E,\epsilon}(\mathbb{R}^2) \le \limsup_{\epsilon\downarrow 0} m_{E,\epsilon}(\mathbb{R}^2) < \infty \right\}.$$

The tube-measure map in (IM1) is Borel in the Hausdorff hyperspace of compact sets: the distance is jointly continuous in (E, x), so integrals against bounded continuous test functions are Borel by measurable integration. The tube measure also depends continuously on positive ϵ. Indeed every positive distance level has zero area: if $\operatorname{dist}(x, E) = r > 0$, a nearest point of E gives, in each sufficiently small ball about x, a ball of one quarter its radius where the distance is strictly below r. Thus the level set has no Lebesgue density points. Dominated convergence now proves the asserted continuity. All small-radius limit operations may consequently be coded using rational radii, proving that $\mathcal{E}_d$ is Borel. For $E \in \mathcal{E}_d$, the sufficiently small-radius tube measures have uniformly bounded

masses and a common compact support, namely the closed unit neighborhood of E. Their weak cluster set $K(E)$ as $\epsilon \downarrow 0$ is therefore nonempty compact. Every cluster measure is supported on E, and its mass lies between the displayed positive liminf and finite limsup. Let $C(E)$ be the closed convex hull of $K(E)$; it is compact, since it lies in the compact set of measures supported on E with the same uniform mass bounds.

For clarity, these compact-valued constructions admit Borel coding. Choose a complete metric for weak convergence of finite positive measures. For every open set O in that measure space,

$$K(E) \cap O \neq \varnothing \quad \Longleftrightarrow \quad \begin{array}{c} \text{for some } k, \text{ for every } m, \text{ some } \epsilon \in \mathbb{Q} \cap (0, 1/m) \\ \text{has } \operatorname{dist}(m_{E,\epsilon}, O^c) > 1/k. \end{array} \tag{IM3}$$

Precompactness proves both implications. The right side is Borel. Using a countable basis of balls and (IM3), choose a point of $K(E)$ in each basis open set which meets it: successively take the first smaller basis ball meeting $K(E)$ whose closure lies in the preceding open ball, with radii tending to zero. The centers form a Cauchy sequence; its limit lies in $K(E)$ and in the original basis open set. All choices are least-integer choices of Borel predicates. This gives a Borel countable dense family in $K(E)$, and its rational convex combinations give a Borel countable dense family in $C(E)$.

Select ν_E to minimize the Gaussian energy

$$J_E(\nu) = \iint e^{-|x-y|^2/\operatorname{diam}(E)^2}\, \nu(dx)\nu(dy) \tag{IM4}$$

over $C(E)$. The diameter is positive on $\mathcal{E}_d$, since a singleton has tube mass of order $\epsilon^d \to 0$. The energy is continuous for weak convergence of finite positive measures, and its dependence on E is Borel. It is strictly convex: for a nonzero finite signed measure η, the Gaussian Fourier identity gives

$$\iint e^{-|x-y|^2/\operatorname{diam}(E)^2}\, \eta(dx)\eta(dy) = c_E \int e^{-\operatorname{diam}(E)^2|\xi|^2/4} |\widehat{\eta}(\xi)|^2 d\xi > 0,$$

where $c_E > 0$. The last inequality uses continuity and uniqueness of the Fourier transform of a finite measure. Thus the minimizer exists and is unique. It is Borel in E: the infimum over the countable dense family in $C(E)$ is Borel; choose its first member within $1/j$ of that infimum. Compactness and uniqueness force these approximate minimizers to converge to ν_E. Set $\nu_E = 0$ outside $\mathcal{E}_d$, also for the empty set.

For a similarity $F(x) = sRx + t$, the tube change of variables gives $m_{FE,\epsilon} = s^d F_* m_{E,\epsilon/s}$. The full-radius cluster construction therefore gives $C(FE) = s^d F_* C(E)$. Also $J_{FE}(s^d F_* \nu) = s^{2d} J_E(\nu)$. Uniqueness gives the stated exact similarity covariance. If E is content regular, $K(E) = C(E) = \{\mu_E\}$.

For the affine comparison, the singular-value bounds give tube inclusions

$$A(E_{\epsilon/b}) \subset (AE)_\epsilon \subset A(E_{\epsilon/a}),$$

where E_r is the open radius-r neighborhood. Hence for every bounded nonnegative continuous f,

$$|\det A|\, b^{d-2} m_{E,\epsilon/b}(f \circ A) \leq m_{AE,\epsilon}(f) \leq |\det A|\, a^{d-2} m_{E,\epsilon/a}(f \circ A). \tag{IM5}$$

Full tube convergence for E shows that every weak cluster measure of AE satisfies (IM2). The total masses are bounded above and away from zero, so $AE \in \mathcal{E}_d$. The measure inequalities are preserved by convex combinations and weak limits. They therefore hold for every element of $C(AE)$, in particular its selected minimizer. Testing against nonnegative continuous functions proves the inequalities as Borel measures and finishes the proof. □

Corollary 15.9 (Removing the randomness of an intrinsic-point decoder)**.** *Suppose a compact trace is an affine image $F = AE$ of a content-regular trace, and $\Theta(F, x)$ is a Borel local-arc tensor observation, with a cemetery value permitted. If for one tensor Q,*

$$\Theta(F, x) = Q \quad \text{for } A_*\mu_E\text{-almost every } x, \tag{IM6}$$

then the same equality holds for ν_F-almost every x. Moreover Q is recoverable as a Borel function of F alone whenever (IM6) holds, without choosing a capacity time or a random point.

Proof. The null-set assertion is (IM2). On the Borel set where $\nu_F(\mathbb{R}^2) > 0$, push $\nu_F/\nu_F(F)$ forward by $x \mapsto \Theta(F, x)$; outside it return the cemetery value. This is a Borel probability kernel in F and, under (IM6), equals δ_Q. The Dirac embedding of the Polish tensor space into its probability-measure space is Borel injective and has a Borel inverse on its image. Apply that inverse if the pushed measure is Dirac, and otherwise return the cemetery value. This constructs the claimed deterministic Borel decoder. □

Known natural-length sampling and the gap that remains. Lawler–Rezaei [12] proves existence of SLE Minkowski content and identifies natural parametrization. Zhan's decomposition theorem [13], Theorem 4.1 and Corollaries 4.2–4.3, identifies the chordal-SLE curve–point measure obtained from Minkowski-content sampling with the integral of two-sided radial SLE through z against its Green-function density. Thus a natural-length Palm decomposition is available, and is different from the deterministic-capacity-tip theorem in [11]. To use (IM6) one still needs a Borel extraction of the appropriate ordered frontier arc and a decoder valid at its natural-length typical point. A capacity-almost-everywhere statement alone supplies neither. The robust random-chart decoder reduces the remaining model question to its explicit properness, isotropy, scale-ergodicity, and local-chart premises for that intrinsic-point law. No such frontier identification or all-stopping-time compatibility is asserted here.

15.4 The tensor at a natural-length typical point of a simple SLE

The natural-length Palm law uses a different whole-plane model from the deterministic-capacity tip law. The stationary cylinder argument applies also to this model, with force parameter $\varrho = 2$. We restrict to $0 < \kappa < 4$ so that the relevant traces are simple. Let μ_2 be the whole-plane SLE$(\kappa; 2)$ law from 0 to ∞. Continuous proper traces for this explicit SLE model are established in [14, Theorem 1.12 and Proposition 2.5]. Retain the spherical curve coding and Borel local stopping/completion premises of Theorem 15.5. Proposition 15.6, with $\varrho = 2$, supplies its full discrete-dilation ergodicity.

Call a source-avoiding ordered germ at 0 μ_2-generic if a proper source-avoiding completion Y has

$$\frac{1}{N}\sum_{n=1}^{N} \delta_{\rho^n Y} \implies \mu_2 \quad \text{for one fixed } \rho > 1. \tag{NP1}$$

By tail replacement this does not depend on the completion and is a Borel germ property. It is stronger than merely assigning a tensor value to an arbitrary decoder. The spherical linearization argument shows pathwise that this property is preserved by every local conformal map fixing 0 with nonzero derivative: the derivative is a similarity, and μ_2 is rotation and dilation invariant. More generally applying any fixed affine A makes the limit $A_*\mu_2$. These assertions hold simultaneously for sample-dependent local conformal maps, without another exceptional-set intersection.

Proposition 15.10 (The terminal germ of radial SLE$(\kappa; 2)$)**.** *For every $\theta \in (0, 2\pi)$, take radial SLE$(\kappa; 2)$ in $\mathbb{D}$ from $e^{i\theta}$ to 0, with boundary force point 1. Its reversed terminal ordered germ at 0 is*

μ_2-generic almost surely. The same assertion holds in any conformally equivalent marked domain, using the physical Euclidean coordinate at its interior target.

Proof. We use whole-plane SLE$(\kappa;2)$ reversibility, proved in [14, Theorem 1.20], and its conformal Markov property; the relation with radial SLE$(\kappa;2)$ is recalled explicitly in [16, Section 2.1] and [15, Section 2.2]. These are external SLE inputs.

First construct a full stationary whole-plane SLE$(\kappa;2)$ trace γ from 0 to ∞, parametrized by whole-plane capacity, with Loewner maps g_t and cylinder coordinates λ_t, q_t. Let $Z_t = \lambda_t - q_t \in (0, 2\pi)$. Under inversion $J(z) = 1/z$, $\Gamma = J\gamma$ is a whole-plane SLE$(\kappa;2)$ from ∞ to 0. At time 0, conditional on its explored past, the remaining path is radial SLE$(\kappa;2)$ in the remaining domain, from its current tip to 0, with force point at the prime end corresponding to ∞. The conformal chart

$$f(z) = \frac{e^{iq_0}}{g_0(1/z)} \tag{NP2}$$

maps that domain to the unit disk, fixes 0, takes the force point to 1, and takes the tip to e^{-iZ_0}. It is analytic with nonzero derivative at 0. The normalization at capacity zero gives $f'(0) = e^{iq_0}$.

Reversibility makes the full reversed Γ a whole-plane SLE$(\kappa;2)$ from 0 to ∞. Its initial germ is therefore μ_2-generic almost surely. Applying the sample-dependent chart f preserves (NP1). The resulting reversed terminal germ belongs, conditionally on the past, to the radial law in the statement with angle $\theta = -Z_0$ modulo 2π. The stationary gap density is

$$\pi_Z(dz) = c \sin(z/2)^{8/\kappa}\, dz, \tag{NP3}$$

which is positive on $(0, 2\pi)$. Disintegration therefore proves the proposition for Lebesgue-almost every starting angle θ.

To obtain every angle, start the radial law at an arbitrary $\theta_0 \in (0, 2\pi)$ and run it for a deterministic radial time $s > 0$. After mapping by its radial Loewner map and rotating to put the current force point at 1, its future is a radial law with gap X_s, where

$$dX_t = \sqrt{\kappa}\, dB_t + 2\cot(X_t/2)\, dt, \qquad X_0 = \theta_0. \tag{NP4}$$

The endpoint Bessel dimension is $1 + 8/\kappa > 2$, so neither endpoint is reached. The distribution of X_s is absolutely continuous with respect to Lebesgue measure: on each compact subinterval of $(0, 2\pi)$, the stopped drift is bounded, and Girsanov compares its endpoint law with Brownian motion on the event of no exit. These no-exit events increase to probability one as the compact intervals exhaust $(0, 2\pi)$. Thus X_s lies in the already proved good-angle set almost surely. Conditional DMP gives (NP1) for the future terminal germ in its new chart. The inverse radial chart is analytic near 0 with nonzero conformal derivative, so pathwise conformal invariance of (NP1) transfers the property back to the original terminal germ. For a general marked domain use an interior conformal chart at the target and the same pathwise invariance. No continuity of the good event as a function of θ is assumed. □

Theorem 15.11 (A natural-length almost-everywhere affine tensor decoder)**.** *Let β be chordal SLE$_\kappa$, $0 < \kappa < 4$, in a simply connected domain, with its Minkowski-content measure μ_β. Let $A \in GL^+(2, \mathbb{R})$ be fixed. Use the decoder of Theorem 15.5 constructed from the model μ_2. Almost surely, for μ_β-almost every interior point $z = \beta(t)$, the reversed incoming ordered germ of $A\beta$ at Az, translated to start at 0, has decoded tensor*

$$[(A^{-1})^{\mathsf{T}} A^{-1}]. \tag{NP5}$$

The same holds for the outgoing germ. This is a statement about the intrinsic length measure, not capacity-almost every time.

Proof. Zhan's natural-length Palm decomposition [13, Theorem 4.1 and Corollaries 4.2–4.3] identifies chordal SLE with a point sampled using its Minkowski content with two-sided radial SLE through that point, integrated against the SLE Green-function density. For every marked interior point z, the incoming arm of the two-sided radial curve is radial SLE$(\kappa;2)$ from the source to z, with force point at the remote boundary target. Proposition 15.10 makes its reversed terminal germ μ_2-generic. Hence the affine-image empirical law is $A_*\mu_2$, which the robust decoder identifies as (NP5). The bad-point indicator is jointly Borel in (β, z): on a simple representative the point z has a unique hit (up to pauses), whose first-hit cut is Borel; restriction, reversal, translation, and the germ decoder with its Borel completion are Borel. One may first cut a branch inside $B(z, \operatorname{dist}(z, \partial D)/2)$, so rough source or target boundaries do not enter this observation. Integrating the Palm identity over compact interior sets, followed by a countable exhaustion, proves that the expected Minkowski-content mass of bad points is zero. That mass is nonnegative, so it vanishes almost surely. Reversibility of two-sided radial SLE, recalled in [15, Section 2.2], gives the outgoing assertion. □

What this removes and what remains. For compact simple-SLE pieces satisfying the content-regularity premise of Theorem 15.8, (NP5) supplies (IM6). Consequently a tensor can be recovered from their unlabelled trace by the intrinsic-measure construction, provided the local incoming or outgoing branch is Borel extracted from that trace. A simple compact arc has two possible orientations; (NP5) works for both, so choosing an orientation does not change its almost-everywhere tensor. This does not yet identify the laws of all accessible frontier arcs of an exploration, establish the content premise on a chosen frontier exhaustion, or prove compatibility at every random slit state and every exploration stopping time. It does not derive an affine-SLE representation from the three axioms.

15.5 A tensor from an unlabelled compact simple-SLE arc

The natural-point theorem and the intrinsic measure can be combined without leaving an unproved choice of orientation for a compact simple arc. We give the coding and local-content details explicitly. This result concerns arcs already known to be simple SLE; it does not identify arbitrary accessible boundary pieces of an unknown exploration.

Lemma 15.12 (Borel orientation and cutting of simple compact arcs)**.** *Use the space $\mathcal{C}_c$ of arbitrary compact planar oriented curves, with pauses identified and the Fréchet metric. This is a separate space from the fixed-endpoint spherical 0-to-∞ space. Use its continuous quotient map from $C([0,1],\mathbb{R}^2)$. The nonconstant simple-arc subspace is Borel. Its trace map to the Hausdorff hyperspace of compact planar sets has exactly two elements in each nonempty fiber. Consequently its image is Borel and there is a Borel choice of an orientation for every compact simple arc. Given an oriented simple arc and a point in its trace, cutting there and reversing either resulting branch are jointly Borel operations.*

Proof. The compact planar curve space is Polish: images of rational polygonal paths give separability. For completeness, align the representatives of a Cauchy subsequence successively by increasing homeomorphisms so that their successive uniform distances are summable. Their uniform limit is continuous and represents the Fréchet limit; the original Cauchy sequence has the same limit. The quotient map is continuous.

For continuous representatives, failure of simplicity modulo pauses is the existence of $r \le s \le t$ with

$$\gamma(r) = \gamma(t), \qquad |\gamma(s) - \gamma(r)| \ge 1/n \tag{UA1}$$

for some integer $n \geq 1$. For fixed n this is a closed condition: the three times range over a compact set and subsequential limits preserve both relations. These conditions are invariant under the identification of pauses. Their images and their complements under the continuous Polish quotient are analytic, hence Borel. Excluding their countable union and the constant curves gives the claimed Borel subspace. The absence of a nonconstant closed subpath is equivalent to being an embedded arc after its constant intervals are collapsed.

The trace map is continuous for Fréchet and Hausdorff metrics. An embedded compact arc has exactly two orientations, so its restriction to this Borel subspace is two-to-one. The countable-fiber Borel uniformization theorem gives both a Borel image and a Borel orientation selector. For cutting, the unique point hit (or its constant interval in a representative with pauses) is a Borel first hitting time. Indeed its sublevel events are given by the minimum distance on a compact initial interval being zero. Restriction and reversal are Borel on representatives, invariant after quotienting, and therefore descend by the same analytic/complement argument. Endpoints or a constant branch may be assigned a cemetery observation. □

Lemma 15.13 (Segment content implies weak local tube convergence)**.** *Let $y : [s,t] \to \mathbb{R}^2$ be a simple continuous curve, and let Θ be continuous and strictly increasing. Suppose for every $s \leq u < v \leq t$ the d-dimensional total Minkowski content of $y([u,v])$ exists and equals $\Theta(v) - \Theta(u)$. Then $E = y([s,t])$ is content regular in the sense of (IM1), with*

$$\mu_E = y_*(d\Theta|_{[s,t]}). \tag{UA2}$$

In particular the simultaneous segment-content theorem of Lawler–Rezaei [12, Theorem 1.1] supplies this property for every compact nontrivial interior time segment of a simple chordal SLE in $\mathbb{H}$, on one probability-one set.

Proof. For a finite partition $s = t_0 < \cdots < t_m = t$, set $E_j = y([t_{j-1}, t_j])$. The tube measures satisfy $m_{E,\epsilon} \leq \sum_j m_{E_j,\epsilon}$. The difference is nonnegative, and its total mass tends to

$$\sum_j \{\Theta(t_j) - \Theta(t_{j-1})\} - \{\Theta(t) - \Theta(s)\} = 0.$$

Thus replacing $m_{E,\epsilon}$ by the sum costs $o(1)$ against any bounded continuous test f. Choose the partition fine enough that each E_j has diameter at most δ. On its radius-ϵ tube, f differs from $f(y(t_{j-1}))$ by at most its modulus of continuity at $\delta + 2\epsilon$ on a fixed compact neighborhood of E. The tube masses converge to the corresponding increments of Θ. Riemann–Stieltjes sums give (UA2) by first taking $\epsilon \downarrow 0$ and then $\delta \downarrow 0$. □

Theorem 15.14 (Simultaneous affine decoding from every compact simple-SLE subarc)**.** *Fix $0 < \kappa < 4$, put $d = 1 + \kappa/8$, and retain the stated Polish spherical curve and Borel completion coding. There is a Borel map $\mathcal{T}_\kappa$ from compact planar sets to projective positive definite tensors, with a cemetery value, such that the following holds. If β is chordal SLE_κ in $\mathbb{H}$, then on one probability-one set, simultaneously for every $A \in GL^+(2,\mathbb{R})$ and every $0 < s < t < \infty$,*

$$\mathcal{T}_\kappa(A\beta([s,t])) = [(A^{-1})^{\mathsf{T}} A^{-1}]. \tag{UA3}$$

The input is only the unlabelled compact trace. Neither its orientation, capacity time, source, nor target is supplied. In particular A, s, t may be selected as functions of the sample on this conull set.

Proof. The proof of Theorem 15.11 gives more than correctness for a single fixed affine map: on one conull set, at content-almost every point of β, both local ordered branches are μ_2-generic before affine deformation. For each such germ, (NP1) and commutation with linear maps give the affine empirical law $A_*\mu_2$ simultaneously for all A. No uncountable intersection of probability-one events is needed. Intersect with the simultaneous segment-content event of Lawler–Rezaei.

For any compact simple arc F, choose its orientation by Lemma 15.12. At each interior point x, apply the μ_2 tensor decoder to its reversed incoming branch translated to start at 0, using a Borel radial completion. Call this observation $\Theta(F, x)$. A branch can be cut inside a small ball about x before completing it, so only its germ enters. This is a Borel observation by the lemma and the previously established Borel germ decoder.

Set $E = \beta([s, t])$ and $F = AE$ on the conull set above. Lemma 15.13 makes E content regular, with positive finite atomless measure μ_E. By simplicity and the natural parametrization identity, μ_E is exactly the restriction of the full natural measure μ_β to E. At $A_*\mu_E$-almost every point of F, the chosen incoming branch is one of the two already good branches of the full curve. Its observation is therefore the right side of (UA3), regardless of the sample-dependent orientation choice. Theorem 15.8 makes ν_F equivalent to $A_*\mu_E$. Apply the Dirac-pushforward construction of Corollary 15.9 to $\Theta(F, \cdot)$. Return the cemetery value off the Borel simple-arc set or when the selected measure vanishes or the pushforward is not a single non-cemetery Dirac mass. This defines the required Borel map and proves (UA3).

All comparisons after the two conull events were deterministic, and their statements hold for every affine map and every subinterval. This proves the simultaneous assertion. □

Limit of this advance. This provides an intrinsic measure and orientation-independent decoder on known affine simple-SLE arcs, including sample-dependent subarcs. It does not assert that an arbitrary accessible frontier decomposes into such arcs with Borel extraction, identify those arc laws for the unknown exploration family, or supply continuation kernels at every random slit state. No classification follows without those steps.

15.6 A hereditary Borel arc certificate from comparable tube clusters

A Borel tensor decoder on known simple-SLE arcs need not have hereditary fibers on arbitrary arcs. Uniform comparison of the entire tube-cluster class repairs this issue without a new two-arm Palm theorem or a continuity assumption on the decoder. Retain the full-radius cluster set $K(X)$, its closed convex hull $C(X)$, and the canonical measure $\nu_X \in C(X)$ of Section 15.3. Thus $X \in \mathcal{E}_d$ means that its d-dimensional total tube masses have finite limsup and strictly positive liminf as the radius tends to zero. The canonical measure is Borel in X and has exact similarity covariance. Take $1 < d < 2$.

For a nonconstant compact simple arc X, use either Borel orientation from Lemma 15.12. At an interior point x, cut both branches, translate them to start at 0, and complete each from its other endpoint by the outward radial ray to infinity. Apply a fixed Borel initial-germ tensor decoder D to obtain $\Theta^-(X, x)$ and $\Theta^+(X, x)$. We require the already proved Borelness and germ locality of D, including for arbitrary proper source-avoiding completions. Give the endpoints cemetery observations. Both observations are jointly Borel; their unordered pair does not depend on the chosen orientation.

Definition 15.15 (Comparable-cluster certificate)**.** For a fixed tensor Q, a compact set X has certificate C_Q if:

1. X is a nonconstant compact simple arc in $\mathcal{E}_d$;

2. ν_X is atomless and has full support X;

3. for some integer $L \geq 1$, every $\lambda \in K(X)$ satisfies

$$L^{-1}\nu_X \leq \lambda \leq L\nu_X; \tag{HC1}$$

4. both branch observations equal Q at ν_X-almost every point:

$$\Theta^-(X,x) = \Theta^+(X,x) = Q \quad \text{for } \nu_X\text{-almost every } x. \tag{HC2}$$

Full support prevents a nontrivial subarc from being invisible to all cluster measures. Uniform comparison of every cluster in (HC1), rather than equivalence of only one selected measure, is used below.

Theorem 15.16 (A hereditary Borel known-model certificate)**.** *The class C_Q is Borel, translation invariant, and hereditary under taking every nonconstant compact subarc. It contains no nonconstant finite-length C^1 arc.*

For $0 < \kappa < 4$, take $d = 1 + \kappa/8$ and the decoder constructed from the whole-plane SLE$(\kappa; 2)$ natural-point law. If β is chordal SLE$_\kappa$ in $\mathbb{H}$, then on one probability-one event, simultaneously for every $A \in GL^+(2,\mathbb{R})$ and every nonconstant compact interior subarc E of β,

$$AE \in \mathsf{C}_{[(A^{-1})^{\mathsf{T}}A^{-1}]}. \tag{HC3}$$

It belongs to no certificate class with a different tensor.

Proof. First check Borelness. The intrinsic-tube-measure theorem constructs the Borel set $\mathcal{E}_d$, the compact-valued multifunction $K(X)$ on it, Borel dense selections $\lambda_n(X)$ in $K(X)$, and ν_X. For fixed L, (HC1) is equivalent to the inequalities for every $\lambda_n(X)$: measure order is preserved under weak convergence. Each inequality is tested against a fixed countable nonnegative continuous compactly supported family determining measure order. One may use nonnegative rational piecewise-linear functions on rational square meshes. Thus the existence of an integer L is a Borel predicate.

For a finite compactly supported measure, atomlessness is equivalent to the supremum of its masses on balls of radius 2^{-j} tending to zero. Rational centers give the same supremum for open balls, or one can use slightly enlarged rational radii. Full support X is equivalent to positive measure in every rational open ball meeting X. These are Borel tests on (X, ν_X). The integral against ν_X of the jointly Borel indicator of the failure of (HC2) is Borel. Together with the Borel simple-arc and cutting lemma, this proves Borelness. Translation translates the full cluster set and its canonical measure, and leaves each translated-to-zero branch unchanged, proving translation invariance.

For heredity, let Y be any nonconstant compact subarc of $X \in \mathsf{C}_Q$. Along any sequence $\epsilon_n \downarrow 0$,

$$m_{Y,\epsilon_n} \leq m_{X,\epsilon_n}. \tag{HC4}$$

The right side has bounded mass and common compact support. Pass to a subsequence where both sides converge weakly to η and $\lambda \in K(X)$. Then $\eta \leq \lambda$. Since $\lambda \leq L\nu_X$ and ν_X is atomless, both limits are atomless, and η is supported on Y.

Every point in the relative interior of Y has a physical ball whose closure is separated from the complementary tails of X. On a smaller such ball the X- and Y-tube densities are identical for all sufficiently small radii. The two limit measures therefore coincide on the relative interior of Y. Neither charges its endpoints, so

$$\eta = \lambda|_Y, \qquad L^{-1}\nu_X|_Y \leq \eta \leq L\nu_X|_Y. \tag{HC5}$$

The restricted measure $\nu_X|_Y$ is nonzero, atomless, and fully supported on Y. Indeed any ball meeting Y contains an interior point of Y and a smaller ball around it separated from the complementary tails; full support of ν_X gives positive mass.

Because the original radius sequence was arbitrary, (HC5) proves that the total Y-tube masses are bounded above and away from zero. Thus $Y \in \mathcal{E}_d$, and every cluster measure of Y obeys the two bounds in (HC5). They pass to convex combinations and weak limits, hence to $\nu_Y \in C(Y)$. Consequently

$$L^{-2}\nu_Y \le \eta \le L^2\nu_Y \quad \text{for every } \eta \in K(Y). \tag{HC6}$$

The same comparisons make ν_Y atomless, fully supported on Y, and equivalent to $\nu_X|_Y$. At an interior point of Y, the two ordered branch germs are exactly the corresponding germs of X, possibly with their orientations interchanged. Germ locality of D and (HC2), together with $\nu_Y \ll \nu_X|_Y$, establish (HC2) for Y. This proves heredity for arbitrary, including sample-dependent, subarc endpoints. The Borel definition did not use an uncountable quantifier over subarcs.

A finite-length C^1 arc has tube area $O(\epsilon)$: cover its arclength parameter by $O(1 + \mathrm{length}(X)/\epsilon)$ balls of radius 2ϵ. Its total d-dimensional tube mass is $O(\epsilon^{d-1}) \to 0$, so it does not belong to $\mathcal{E}_d$.

For the SLE assertion, take the simultaneous natural-point and segment-content event in Theorem 15.14. On every compact nonconstant interior subarc E, the natural measure μ_E is finite, positive, atomless, and fully supported. By (IM5), every $\lambda \in K(AE)$ lies between fixed positive finite multiples $c_A A_*\mu_E$ and $C_A A_*\mu_E$. The canonical $\nu_{AE} \in C(AE)$ obeys the same bounds. Thus (HC1) holds with any sufficiently large integer $L \ge C_A/c_A$, and ν_{AE} is atomless and fully supported. The both-branch natural-point theorem gives (HC2) at $A_*\mu_E$-almost every point, hence at ν_{AE}-almost every point. Distinct certificate tensors cannot both be almost-everywhere constant values of the same observations under a nonzero measure. All comparisons after the common SLE conull event are deterministic, giving simultaneity for every A and every subarc. □

Lemma 15.17 (Pathwise stability under local conformal maps)**.** *Use specifically the empirical-law decoder of Theorem 15.5, built from the isotropic whole-plane law μ, so that $D = I$ means that the branch empirical law is exactly μ. If $X \in \mathsf{C}_I$ and ψ is conformal and injective on an open neighborhood of X, then for every $A \in GL^+(2,\mathbb{R})$,*

$$A\psi(X) \in \mathsf{C}_{[(A^{-1})^{\mathsf{T}}A^{-1}]}. \tag{HC7}$$

This is a deterministic assertion for every such A, ψ after X is fixed. In particular the map and the affine transformation may depend on an underlying random sample.

Proof. Write $\Psi = A\psi$. On a sufficiently small relatively compact neighborhood of X, this is a C^1 diffeomorphism with bi-Lipschitz constants $0 < a \le b < \infty$ and Jacobian bounds $0 < j_- \le |\det D\Psi| \le j_+ < \infty$. The lower Lipschitz bound follows near the diagonal from the nonvanishing derivative, and away from it from injectivity and compactness. The image neighborhood contains all sufficiently small tubes around $\Psi(X)$. For small ϵ,

$$\Psi(X_{\epsilon/b}) \subset (\Psi X)_\epsilon \subset \Psi(X_{\epsilon/a}). \tag{HC8}$$

For any nonnegative continuous test function f, change of variables therefore gives

$$j_- b^{d-2} m_{X,\epsilon/b}(f \circ \Psi) \le m_{\Psi X,\epsilon}(f) \le j_+ a^{d-2} m_{X,\epsilon/a}(f \circ \Psi). \tag{HC9}$$

The compositions may be extended continuously away from a fixed small neighborhood of X; this does not affect the small tubes.

Along any target-radius sequence, extract jointly convergent source tube measures at radii ϵ/a and ϵ/b. Every source cluster lies between $L^{-1}\nu_X$ and $L\nu_X$. Thus every target cluster η satisfies, with fixed constants $0 < c \le C$,

$$c\Psi_*\nu_X \le \eta \le C\Psi_*\nu_X. \tag{HC10}$$

The same argument bounds target masses above and away from zero, so $\Psi X \in \mathcal{E}_d$. Its canonical measure satisfies (HC10) as well. Hence it is atomless and fully supported and all its clusters are uniformly comparable to it.

At ν_X-almost every x, both branch empirical laws are exactly μ. Centering at x and at $\Psi(x)$, the map between the two local branches has derivative $A\psi'(x) = A(s_x R_x)$, with $s_x > 0$ and R_x a rotation. The pathwise spherical linearization and tail-replacement proof of Theorem 15.5 sends each empirical law to $A_*\mu$, because μ is invariant under similarities. Their decoded tensor is therefore the tensor in (HC7). The canonical target measure is equivalent to $\Psi_*\nu_X$ by (HC10), establishing its almost-everywhere branch condition. All conclusions are deterministic for the fixed generic branches, so no exceptional-set intersection over A, ψ is involved. □

Scope. These are hereditary certificates for compact arcs, with a known-model simple-SLE application. They do not by themselves identify accessible SLE_6 frontier laws, prove certified-arc supply at every random slit state, or construct a Borel family with a consistent one-time change of affine metric. Those model and restart questions remain separate.

16 Restart states and direct driving bounds for nonsimple traces

Scope. The exploration itself is not assumed to contain any embedded time interval. It is an ordered, possibly densely self-touching curve. A decoder theorem for simple SLE arcs is a separate statement about different objects. At a finite capacity time τ, its state is the remaining target component with its inherited colored boundary and its distinguished source and target *prime ends*, not just the two Euclidean endpoint locations. If $f : \mathbb{H} \to D$ is the initial inverse chart, its restarted inverse chart is

$$f_\tau(z) = f\big(g_\tau^{-1}(z + W_\tau)\big). \tag{NS1}$$

The centered future driver is $W_{\tau+s} - W_\tau$. This identity uses Loewner composition and does not assert conformal covariance of the unknown physical family. The marked-state space is required to be Borel, and genuine strong DMP is a hypothesis on this state space. The chart is coordinate bookkeeping, not an extra physical state variable: the law in the physical domain does not depend on its choice.

Uniform approximation of a deterministic ordered curve by simple interior paths does not by itself assert strict growth of its filled hulls. The topological characterization in Yuan [7], Theorem 1.1 and Remark 1.4, additionally requires strictly increasing filled hulls. For the results below continuous Loewner generation and the usual capacity clock are standing admissibility hypotheses. No stochastic semimartingale property is included in admissibility. Initially the capacity lifetime may be $\zeta \le \infty$; strong DMP and all restart assertions are imposed on $\{\tau < \zeta\}$ for stopping times τ, allowing $\tau = \infty$ elsewhere. The full curve tends to its target prime end, so the charted trace tends to infinity as capacity tends to ζ.

A continuum crossing assumption. Fix $M > 0$ and $0 < q < 1$. At every admissible marked state, suppose that every nonseparating boundary quadrilateral of modulus at least M has exploration crossing probability at most q. Here the two side arcs of the quadrilateral lie on the domain boundary,

the two crossed arcs lie in the domain, and nonseparating means that the source and target can be connected while avoiding the quadrilateral. All crossing events, admissible pairs of states and quads, and evaluation kernels are assumed jointly Borel. The constants are uniform over the state class, including its random slit states. This is a precise upper unforced-crossing input; it is not a substitute for a two-sided configuration-level RSW hypothesis. We use a boundary-robust crossing convention: the joining subcurve may lie in the closed quadrilateral and may touch its two boundary side arcs between its crosscut hits. Closures at boundary sides are interpreted with their inherited prime-end sides. In particular, a nonsimple trace which joins the two semicircles of a half-annulus while touching the real line is a crossing. An estimate stated only for subcurves strictly inside the open quad is not silently substituted for this assumption.

Lemma 16.1 (Time-zero crossing bounds become conditional bounds)**.** *Suppose a Borel family satisfies genuine strong DMP on the stated restart class and the preceding time-zero crossing assumption at every state. For every stopping time τ, on $\{\tau < \zeta\}$, and every $\mathcal{F}_\tau$-measurable admissible nonseparating quad Q_τ of modulus at least M in the remaining state,*

$$\mathbb{P}\{\textit{future exploration crosses } Q_\tau \mid \mathcal{F}_\tau\} \le q \quad \textit{almost surely.} \tag{NS2}$$

The same bound holds in the restarted half-plane chart, with the same constants.

Proof. Let X_τ be the marked state and let P_x be the exploration kernel. Strong DMP identifies the conditional crossing probability with $P_{X_\tau}(\text{cross } Q_\tau)$. The numerical time-zero bound holds for every admissible pair (x, Q), so evaluation at the random pair gives (NS2). This does not intersect an uncountable collection of almost-sure statements. Conformal maps preserve the quad modulus, its separation status, and the crossing event. Consequently its pullback to the physical state has exactly the same bound. We have transported a geometric inequality, not identified the law of a conformal image with a law in another physical domain. □

Lemma 16.2 (A deterministic driver localization bound)**.** *Let a continuous chordal Loewner trace have filled hulls K_s and driver W_s, with* $\operatorname{hcap}(K_s) = 2s$*. If $K_t \subset \{z : |z| \le R\}$, then*

$$\sup_{0\le s\le t} |W_s| \le 3R. \tag{NS3}$$

Also $\sup\{\operatorname{Im} z : z \in K_t\} \le 2\sqrt{t}$*. These facts do not require a simple trace.*

Proof. For fixed $s \le t$, Schwarz-reflect the hydrodynamic map g_s across the real line outside $[-R, R]$. On $|z| > R$ it is univalent and has expansion $g_s(z) = z + \sum_{n\ge1} b_n z^{-n}$. The area theorem applied after rescaling by R gives

$$\sum_{n\ge1} \frac{n|b_n|^2}{R^{2n+2}} \le 1.$$

Consequently Cauchy–Schwarz gives, at $z = \pm 2R$,

$$|g_s(z) - z| \le R\left(\sum_{n\ge1} \frac{1}{n4^n}\right)^{1/2} < R.$$

The real interval which is the image of the two sides of the hull, and in particular its current tip image W_s, lies between $g_s(-2R)$ and $g_s(2R)$ by boundary order. This proves (NS3). For the height bound, while z is unswallowed its image height Y_s satisfies

$$\partial_s Y_s = -\frac{2Y_s}{|g_s(z) - W_s|^2}, \qquad \partial_s Y_s^2 \ge -4.$$

A point of original height greater than $2\sqrt{t}$ therefore cannot be swallowed by time t. The closed-hull bound follows by closure. □

Proposition 16.3 (Conditional Brownian-size driving bounds without simplicity)**.** *For a continuous-driver family satisfying the hypotheses of Lemma 16.1, whose full traces tend to the target prime end (and hence to infinity in their half-plane chart), the capacity lifetime is infinite almost surely. There are $c, C > 0$, depending only on M, q, such that, for every finite stopping time τ, $h > 0$, and $x \geq 0$,*

$$\mathbb{P}\left\{ \sup_{0\leq s\leq h} |W_{\tau+s} - W_\tau| > x\sqrt{h} \mid \mathcal{F}_\tau \right\} \leq Ce^{-cx}. \tag{NS4}$$

In particular, for some $a, C' > 0$,

$$\mathbb{E}\left[\exp\left\{ a \sup_{0\leq s\leq h} |W_{\tau+s} - W_\tau|/\sqrt{h} \right\} \middle| \mathcal{F}_\tau \right] \leq C'. \tag{NS5}$$

For every $p > 0$ this gives a constant C_p with conditional p-th moment at most $C_p h^{p/2}$. The driver is almost surely locally α-Hölder for every $\alpha < 1/2$.

Proof. Work first at time zero in the half-plane chart, before the possible capacity lifetime. Fix $u > 0$ and choose $d = 4e^{\pi M}$. To bound horizontal travel before reaching height u, use the successive first hitting times of the lines $\operatorname{Re} z = kdu$, $k = 1, 2, \ldots$, before upper exit. At the first hit of line kdu, all the earlier trace and its filled hull lie to its left. The half-annulus centered at $(k+1)du$, of inner radius $2u$ and outer radius $du/2$, lies entirely to the right of that line and is untouched. As a boundary quad its modulus for crossing between the semicircles is

$$\pi^{-1} \log \frac{du/2}{2u} = M.$$

It does not separate the current source prime end from infinity. Indeed, a source access arc eventually lies near the tip, to the left of this quad. Any path from such an access point to infinity first enters the empty right half-plane to the left of the quad, where it can be rerouted around its outer semicircle. This verifies the prime-end separation statement without equating prime ends which have the same physical impression. Reaching the next line at height at most u requires crossing this quad from its outer to its inner semicircle. Its conditional probability is therefore at most q by (NS2). The same construction starts at $k = 0$ with the source at zero.

Iterating gives probability at most q^n for hitting line ndu before upper exit. Reflection of this geometric construction treats left exit, without assuming reflection invariance of the laws. If the capacity lifetime were at most a fixed $T < \infty$, the height bound would confine the full trace to height $2\sqrt{T}$, while convergence to infinity would require unbounded horizontal extent. Use the preceding estimate with $u = 3\sqrt{T}$, so that no upper exit is possible on this event. The iterated crossing estimate makes that event have probability zero. A countable union over integer T proves infinite lifetime.

Now put $u = 2\sqrt{h}$. While the trace stays in $[-L, L] \times [0, u]$, each filled hull lies in the disk of radius $\sqrt{L^2 + u^2}$. Lemma 16.2 implies that, for $L \geq u$, a driver supremum greater than $5L$ requires a horizontal trace exit before time h; the height bound excludes upper exit. The probability of reaching either side $\pm L$ before upper exit is at most $2q^{\lfloor L/(du)\rfloor}$. Taking $L = x\sqrt{h}/5$ when $x \geq 10$, and enlarging the prefactor for $x < 10$, proves (NS4) at zero. The proof only used target-visible Loewner hulls and first hits of vertical lines, so dense self-touching does not affect it.

Apply the identical argument to the centered future driver in (NS1). Lemma 16.1 supplies the same crossing constants after conditioning at τ. This proves (NS4) for all finite stopping

times. Tail integration gives (NS5) for $0 < a < c$ and the moment bounds. Kolmogorov continuity, using arbitrarily large p, also gives pathwise Hölder exponents $\alpha < 1/2$ on every compact capacity interval. □

Relation to the existing literature and limits of the estimate. This is a direct version of the rectangle argument underlying Kemppainen–Smirnov [4], Section 3.3, Propositions 3.8–3.9. Their Theorem 1.5 starts with families of simple curves and obtains nonsimple limits. One cannot apply that formulation merely by choosing deterministic simple approximations of each sample: the approximating *laws* need not retain the conditional crossing bound. The direct argument above instead assumes continuous-driver admissibility for the continuum family and proves its quantitative estimate there.

These estimates control unsigned increments. They do not bound $|\mathbb{E}[W_{t+h} - W_t \mid \mathcal{F}_t]|$ by $O(h)$, establish semimartingality, or identify an affine conformal structure. Uniform quad bounds are stable under a fixed quasiconformal change with altered constants; Brownian-size bounds therefore also hold in coordinates which need not be the correct affine coordinates for stochastic driving rigidity.

17 Unsigned regularity and the missing continuum signed budget

17.1 Unsigned Brownian-scale regularity does not supply a semimartingale

The distinction between topological Loewner admissibility and stochastic regularity is substantive even after strong quantitative estimates are available. The following process example isolates the missing signed cancellation. It is not asserted to obey source locality, domain Markov for an undecorated slit domain, or a curve-level unforced-crossing estimate.

Proposition 17.1 (A driver-level obstruction to an unsigned regularity argument)**.** *For every $\kappa > 0$, $H \in (1/2, 1)$, and $a \neq 0$, there is a continuous process X on $[0,1]$ with all the following properties in its usual natural filtration:*

1. *Its paths are locally α-Hölder for every $\alpha < 1/2$.*
2. *For every $c > 0$ there is $C_c < \infty$ such that, for every stopping time τ and deterministic $h > 0$ with $\tau + h \leq 1$ almost surely,*

$$\mathbb{E}\left[\exp\left(c\,\frac{\sup_{0\leq u\leq h}|X_{\tau+u} - X_\tau|}{\sqrt{h}}\right)\,\middle|\,\mathcal{F}_\tau\right] \leq C_c. \tag{UR1}$$

3. *Its sums of squared increments along dyadic partitions have the same limit as Brownian motion of variance rate κ.*
4. *It is not a semimartingale.*

Proof. Let $\varphi(x) = \operatorname{dist}(x, \mathbb{Z})$ and define

$$g_H(t) = \sum_{n=0}^{\infty} 2^{-nH}\varphi(2^n t), \qquad X_t = \sqrt{\kappa}B_t + a g_H(t), \tag{UR2}$$

where B is standard Brownian motion. Uniform convergence defines a continuous deterministic function. The bounds $|\varphi(x) - \varphi(y)| \leq \min\{|x - y|, 1/2\}$ show

$$|g_H(t) - g_H(s)| \leq C_H|t - s|^H. \tag{UR3}$$

Indeed split the series where $2^n|t-s|$ first exceeds one, and sum the finite geometric series of Lipschitz bounds and the remaining geometric tail separately. Both sums are bounded by a constant times $|t-s|^H$. The pathwise Hölder assertion follows.

The natural filtrations of X and B are identical, since the shift ag_H is deterministic. Brownian strong Markov and (UR3) give, for $h \le 1$, the bound in (UR1) by

$$e^{c|a|C_H}\,\mathbb{E}\exp\left(c\sqrt{\kappa}\sup_{0\le u\le 1}|B_u|\right)<\infty.$$

Reflection bounds for Brownian maxima make the last expectation finite for every $c>0$.

Here is an explicit verification of the variation obstruction. Set $q=2^{1-H}>1$, and use the mesh $t_j=j2^{-m}$, $0\le j\le 2^m$. All terms with $n\ge m$ vanish at the mesh points. On each mesh interval all the earlier tent functions are linear, so

$$\Delta_j g_H = 2^{-m}\sum_{n=0}^{m-1} q^n r_{n+1}(j), \tag{UR4}$$

where the sign vectors r_{n+1} are the dyadic Rademacher vectors. Their orthogonality on this mesh gives

$$Q_m := \sum_j (\Delta_j g_H)^2 = 2^{-m}\frac{q^{2m}-1}{q^2-1}, \qquad \max_j |\Delta_j g_H| \le 2^{-m}\frac{q^m-1}{q-1}. \tag{UR5}$$

Consequently

$$\sum_j |\Delta_j g_H| \ge \frac{Q_m}{\max_j |\Delta_j g_H|} \ge \frac{q^m+1}{q+1} \longrightarrow \infty. \tag{UR6}$$

Thus g_H has infinite variation. On the other hand $Q_m\to 0$ because $H>1/2$. More generally (UR3) bounds the sums of squared increments on any subinterval of the dyadic mesh by $C_H^2 2^{-m(2H-1)}$, so they tend to zero uniformly over mesh endpoints. Cauchy–Schwarz bounds the mixed Brownian/g_H sums by the square root of the product of their squared increment sums. The standard Brownian dyadic quadratic-variation limit therefore proves the asserted Brownian quadratic variation for X.

For completeness, the failure of semimartingality can also be seen without relying only on the deterministic infinite-variation criterion. The bounded deterministic elementary predictable strategies $L_t^m = \operatorname{sign}(a\Delta_j g_H)$ on $(t_j, t_{j+1}]$ give the elementary sums

$$\sum_j L^m_{t_{j+1}}(X_{t_{j+1}}-X_{t_j}) = |a|\sum_j |\Delta_j g_H| + Z_m, \tag{UR7}$$

where Z_m is centered Gaussian with variance at most κ. They diverge to $+\infty$ in probability by (UR6). For a continuous semimartingale, its elementary integrals with predictable integrands bounded by one are bounded in probability on a fixed finite horizon: localize the martingale bracket and finite-variation total variation, then use the martingale L^2 inequality. This contradicts (UR7). □

What RSW regularity can legitimately supply. Kemppainen–Smirnov [4], Proposition 2.6, Theorem 1.5, and Section 3.3, give conformally stable conditional unforced-crossing criteria, continuous driving with every Hölder exponent below 1/2, and Brownian-scale exponential driving bounds in their stated setup. Their Section 1.4 explicitly permits nonsimple curves already having continuous Loewner drivers; it does not alone give that property for an arbitrary law of uniformly approximable nonsimple curves. These are unsigned estimates; they do not assert semimartingality. The example above proves that even the listed stronger process-level conclusions, including conditional estimates at stopping times and Brownian quadratic variation, cannot replace the missing signed cancellation argument. It is not a counterexample to the full curve axioms.

17.2 A continuum signed budget on visited slit states

A useful first target is weaker than the uniform linear conditional-bias bound (BI1). It asks for finite aggregate predictable variation, not for a pointwise drift density. Its probabilistic content is classical quasimartingale theory; the formulation below makes the exact continuum restart observable and its stopping rule explicit.

Assume only the Borel continuous-law and restart-identity premises of (KR1), not the semimartingale premises of Proposition 4.1. Let $\{P_f\}$ be this family, with drivers starting at zero and defined on $[0,\infty)$. Fix $L, T > 0$, set

$$\sigma_L = \inf\{t \ge 0 : |W_t| \ge L\}, \qquad X_t = W_{t\wedge\sigma_L}, \quad 0 \le t \le T, \tag{QV1}$$

and use the usual natural filtration of the original W. For $\ell < 0 < r$, define the bounded restarted mean

$$m_h(f;\ell,r) = \mathbb{E}_{P_f} W_{h\wedge\inf\{u:W_u\notin(\ell,r)\}}. \tag{QV2}$$

This is a Borel function of its arguments: the stopped evaluation of a continuous path at its first exit from an open interval is Borel and is bounded between the two endpoints. No claim of continuity in f is made.

Proposition 17.2 (An exact signed restart budget)**.** *Suppose (KR1) holds in the inherited centered capacity charts. For every deterministic partition $\pi = (0 = t_0 < \cdots < t_n = T)$,*

$$\mathbb{E}[X_{t_{j+1}} - X_{t_j} \mid \mathcal{F}_{t_j}] = \mathbf{1}_{\{t_j<\sigma_L\}} m_{t_{j+1}-t_j}(f_{t_j}; -L - W_{t_j}, L - W_{t_j}). \tag{QV3}$$

Consequently the aggregate budget

$$\mathcal{V}_{L,T}(f) = \sup_{\pi} \sum_{j=0}^{n-1} \mathbb{E}_{P_f}\left[\mathbf{1}_{\{t_j<\sigma_L\}} |m_{t_{j+1}-t_j}(f_{t_j}; -L - W_{t_j}, L - W_{t_j})|\right] \tag{QV4}$$

is exactly the conditional variation of X. It is finite if and only if $X = M + A$, $A_0 = 0$, has a continuous canonical decomposition with M a uniformly integrable martingale on $[0,T]$ and $\mathbb{E}\operatorname{Var}_{[0,T]} A < \infty$. In that case

$$\mathcal{V}_{L,T}(f) = \mathbb{E}\operatorname{Var}_{[0,T]} A. \tag{QV5}$$

If it is finite for every L, T, every driver is a continuous semimartingale locally. No absolute continuity of A, initial tangent, or Brownian bracket follows just from this finiteness.

Proof. On $\{t_j \ge \sigma_L\}$ the stopped increment vanishes. Otherwise freeze the past at t_j. The conditional future driver is $P_{f_{t_j}}$ by (KR1). The original stopping boundary $(-L, L)$ becomes exactly $(-L - W_{t_j}, L - W_{t_j})$ for that centered future driver. This proves (QV3), hence the conditional-variation identity (QV4).

Since X is bounded and continuous, the quasimartingale decomposition theorem gives precisely the stated continuous canonical decomposition when its conditional variation is finite; see Rao [5]. Conversely, given this decomposition, martingale increments have conditional mean zero, so each partition sum is at most $\mathbb{E}\operatorname{Var} A$. For the reverse inequality, bounded predictable elementary processes on deterministic time partitions are dense among bounded predictable integrands for the finite measure $H \mapsto \mathbb{E}\int_0^T H_s\, d|A|_s$. Approximate the predictable sign of the signed measure dA by such processes, all bounded by one. For an elementary process with value $H_j \in \mathcal{F}_{t_j}$ on $(t_j, t_{j+1}]$,

$$\mathbb{E}\int_0^T H_s\, dA_s = \sum_j \mathbb{E}[H_j(X_{t_{j+1}} - X_{t_j})] \le \sum_j \mathbb{E}|\mathbb{E}[X_{t_{j+1}} - X_{t_j} \mid \mathcal{F}_{t_j}]|.$$

The approximation gives (QV5). The local conclusion follows by taking $L \uparrow \infty$ and $T \uparrow \infty$, using continuity and uniqueness of continuous canonical decompositions. □

Why this is a weaker target, and why it is still missing. The bound (BI1) gives $\mathcal{V}_{L,T} \le KT$ and thus suffices. Finiteness of (QV4) alone also permits singular finite-variation drift. For example, Brownian motion plus a deterministic Cantor function has finite conditional variation on compact intervals but its canonical drift is not absolutely continuous. This example illustrates the distinction between the two probabilistic conditions, not a proposed exploration family. An unsigned estimate of Brownian order gives only a bound of order $\sum_j \sqrt{t_{j+1} - t_j}$, which diverges on finer equal partitions. The budget must be proved from interactions of locality and DMP, in a suitable common affine structure; none of those additional cancellation estimates has been established here. One cannot replace the offsets in (QV3) by the fresh symmetric interval $(-L, L)$: the localization boundary is inherited from the past.

17.3 The shrinking-slit anomaly fits inside Brownian-size bounds

Proposition 17.3 (Uniform unsigned bounds do not control initial bias rates)**.** *Let $b_n \neq 0$ with $|b_n| \to \infty$, and let $\kappa > 0$. Put*

$$a_n(t) = b_n(t \wedge b_n^{-2}), \qquad X_t^n = \sqrt{\kappa} B_t + a_n(t), \quad t \ge 0. \tag{SP1}$$

Then $\sup_t |X_t^n - \sqrt{\kappa} B_t| \le |b_n|^{-1} \to 0$ under the displayed coupling. Uniformly in n, the processes satisfy the conditional Brownian-scale exponential bound (UR1), at every bounded stopping time and every deterministic $h > 0$. Every X^n is a semimartingale with bracket κt, but its initial drift is b_n. The optimal constant in its unstopped linear conditional-bias bound is $K_n = |b_n|$.

Proof. The deterministic shifts obey

$$|a_n(t) - a_n(s)| \le \min\{|b_n||t - s|, |b_n|^{-1}\} \le \sqrt{|t - s|}. \tag{SP2}$$

Brownian strong Markov therefore gives (UR1), uniformly in n, by adding one inside the normalized supremum. The uniform-convergence and bracket claims are immediate. For $0 < h < b_n^{-2}$ the initial mean increment is $b_n h$, proving the optimal lower bound on K_n; the Lipschitz bound $|b_n|$ proves the upper bound. □

For the mirrored slit derivatives in (DR9), a variance anomaly produces

$$b_\epsilon^\pm = (\kappa/2 - 3)\frac{e_\epsilon^\pm}{d_\epsilon^2}, \qquad \frac{e_\epsilon^\pm}{d_\epsilon^2} = \mp \frac{4}{\epsilon^3 \sqrt{\epsilon^2 + 4}}. \tag{SP3}$$

For $\kappa \neq 6$ these are of order ϵ^{-3}. Substituting either sequence into (SP1) gives pulses of duration $\asymp \epsilon^6$ and height $\asymp \epsilon^3$. Thus even uniform convergence to Brownian motion, exact Brownian bracket, and uniform conditional Brownian-size bounds allow precisely the inverse-cubic initial-bias scale which the slit test needs to exclude. Choosing $b_\epsilon = b/d_\epsilon$ similarly realizes the inverse-linear drift scale. The sharp corner at the end of a pulse can be smoothed without changing these orders.

These processes are not claimed to be the locality-related drivers of one exploration family. The point is narrower and rigorous: neither uniform unsigned driving estimates nor weak continuity of the driver laws implies the rate $\epsilon K_\epsilon \to 0$ required in (BI5). A successful shrinking-slit proof must use additional structure of the locality comparison, not only the size and continuity conclusions of RSW.

18 What the continuum crossing estimates give once the signed budget is finite

The finite aggregate signed budget supplies a continuous semimartingale, but by itself it supplies no regularity of its canonical characteristics. The unsigned continuum estimate nevertheless supplies one additional conclusion at that point: the bracket is absolutely continuous with a bounded density. This does not require a tangent of the drift, or any regularity of its total variation.

Proposition 18.1 (Brownian-order conditional moments dominate the bracket)**.** *Let X be a continuous semimartingale in a usual filtration, with canonical decomposition $X = X_0 + M + A$, where M is a continuous local martingale starting at zero and A is continuous predictable finite variation. Suppose that for every $0 \le s < t \le T$,*

$$\mathbb{E}\big[(X_t - X_s)^2 \mid \mathcal{F}_s\big] \le K(t-s) \quad \textit{almost surely} \tag{BR1}$$

with one deterministic finite K. Then, pathwise,

$$d\langle M\rangle_t \le K\, dt. \tag{BR2}$$

Consequently M is square integrable on $[0,T]$ and its bracket has a predictable density v with $0 \le v \le K$ almost everywhere. No bound on the variation or initial tangent of A is asserted.

Proof. Write $C = \langle M\rangle$. For fixed deterministic $s < t$, the quadratic increment sums along refining deterministic partitions of $[s,t]$ converge in probability to $C_t - C_s$. For any $E \in \mathcal{F}_s$, the tower property and (BR1) give

$$\mathbb{E}\left[\mathbf{1}_E \sum_i (X_{t_{i+1}} - X_{t_i})^2\right] \le K(t-s)\mathbb{P}(E).$$

Choose an almost-surely converging subsequence of the nonnegative sums and apply Fatou. Thus

$$\mathbb{E}[\mathbf{1}_E(C_t - C_s)] \le K(t-s)\mathbb{P}(E). \tag{BR3}$$

In particular $\mathbb{E}C_T \le KT$, and $C_t - Kt$ is a continuous supermartingale of class D: all of its stopped absolute values are dominated by $C_T + KT$.

A continuous predictable finite-variation supermartingale of class D is nonincreasing: its existing special-semimartingale decomposition has zero martingale part, and uniqueness identifies its drift with the negative increasing Doob–Meyer compensator. Apply this fact to $C_t - Kt$ to obtain (BR2). Finite expected bracket makes M square integrable, and the predictable Radon–Nikodym density of dC gives the final assertion. □

Application to the continuum classification problem. The conditional exponential estimate already proved in the nonsimple curve class implies (BR1) with a uniform numerical constant depending only on the crossing constants. If the aggregate signed restart budget is finite for a bounded stopped driver, then the quasimartingale criterion supplies its canonical decomposition. Such a stopped driver retains (BR1): on $\{\sigma_L > s\}$ its increment is bounded by the original driver supremum over $[s,t]$, and otherwise it vanishes. The proposition then gives bounded absolutely continuous bracket automatically. Thus the first signed-budget target, if established, leaves drift regularity as the characteristic bottleneck; it does not leave both bracket regularity and drift regularity independent.

Corollary 18.2 (The remaining singular drift is transported without an anomaly)**.** *Assume the proposition for all relevant chart drivers. Before contact with a locality deletion, let h_t be the evolving deletion map and $u'(t) = h_t'(W_t)^2 > 0$. Write $dA = b_t\,dt + dA_t^{\perp}$ for the decomposition relative to capacity Lebesgue measure. Then the restarted deleted-chart characteristics satisfy*

$$d\widehat{A}_{u(t)} = h_t'(W_t)\,dA_t + \tfrac{1}{2}h_t''(W_t)(v_t - 6)\,dt, \qquad d\widehat{A}^{\perp}_{u(t)} = h_t'(W_t)\,dA_t^{\perp}. \tag{BR4}$$

In particular the deletion cannot create or cancel a singular drift component by the variance-six anomaly. The singular part transforms as a one-form, with its capacity clock changed as stated.

Proof. The deletion Itô identity is

$$d\widehat{W}_{u(t)} = h_t'(W_t)\,dM_t + h_t'(W_t)\,dA_t + \tfrac{1}{2}h_t''(W_t)\,dC_t - 3h_t''(W_t)\,dt.$$

Insert $dC = v\,dt$ and use uniqueness of canonical decomposition. The strictly positive continuous capacity-clock derivative makes u locally bi-Lipschitz before deletion contact, so it preserves the distinction between absolutely continuous and singular measures. All terms except the indicated pushforward of $dA^{\perp}$ are absolutely continuous. This proves (BR4). □

A failed uniform-variation shortcut. An unscaled bound on total drift variation in mirrored shrinking deletion charts does not by itself give the local first-order rates needed by the existing rigidity theorem. The anomaly can be supported on capacity windows much shorter than the slit scale. The previously proved pulse example shows why bounded variation budgets alone cannot be replaced silently by drift-density bounds. Conversely, imposing a uniform $o(\sqrt{t})$ initial variation modulus over *all* positive scalings of every chart would be far too strong: if $A^{(\lambda)}_{\lambda^2 t} = \lambda A_t$, then

$$\lambda\,\mathbb{E}\,\mathrm{Var}(A;[0,T]) \le \omega(\lambda^2 T)\lambda\sqrt{T}, \qquad \omega(r) \longrightarrow 0,$$

and $\lambda \downarrow 0$ already forces $A = 0$, without locality. Any useful added variation premise must therefore specify its chart normalization and its domain dependence rather than hide an all-scalings martingale assumption in a uniform modulus.

Remaining research question. Can source locality, physical translation covariance, and the precise crossing hypothesis control the finite signed budget, or rule out the one-form-valued singular drift in (BR4), in one fixed unknown affine structure? The present proposition does not answer either question. It narrows the required characteristic input once semimartingality has been obtained.

Why an interior-hitting shortcut would be invalid. The boundary-unforced crossing estimate alone does not imply a positive lower driving tail: it includes deterministic SLE_0 among its examples. In the Kemppainen–Smirnov definition the unforced annulus part is empty when the inner circle misses the current domain boundary; see [4, Definition 2.3 and Section 1.4]. Thus an argument treating first entry into an interior ball as an already controlled unforced crossing would add an unproved premise. The next ray-deletion comparison repairs the particular driving-confinement argument by using corrected locality; it does not assert general interior small-ball nonconcentration.

18.1 Source locality and a ray deletion give a lower driving tail

The upper unforced-quad estimate alone does not control first entry into an interior ball. The following argument uses source locality essentially, turning the relevant interior point into a boundary slit tip in a comparison domain. It proves a lower *driving* tail, not a general interior small-ball nonconcentration estimate.

Admissible comparison states. For a physical inverse chart $f : \mathbb{H} \to D$ and $r > 0$, require that the domain

$$D' = f(\mathbb{H} \setminus \{iy : y \geq r\}) \tag{RL1}$$

belongs to the family, with source $f(0)$ and either of the two appropriate target prime ends over the original target infinity. It is enough to use the right-hand target end. A canonical inverse chart for this comparison domain is $f \circ F_r$, where

$$F_r(w) = \frac{r}{2}\left(\sqrt{w+1} - \frac{1}{\sqrt{w+1}}\right), \tag{RL1a}$$

with the principal square root. Indeed the square root maps the half-plane onto a quadrant, and $s \mapsto (r/2)(s - 1/s)$ maps that quadrant onto the ray-deleted half-plane, taking 1 to 0. This also provides a jointly Borel comparison state and target choice. Choose boundary colors so they agree with the original colors on the two real sides near zero. Source locality is assumed to compare the ordered prefixes up to first encounter with the removed ray. For regular distinct source and target points this is a usual source-neighborhood deletion. At rough states, this is an explicit prime-end/source-deletion admissibility requirement, not a consequence of Euclidean neighborhood locality at coincident impressions.

Lemma 18.3 (A lower driving tail from locality and boundary quads)**.** *Assume continuous Loewner generation with capacity $2t$, corrected source locality for (RL1), and the uniform boundary-robust unforced quad bound with constants $M > 0$, $q < 1$ on all these states. Put*

$$\delta = \frac{1}{4}e^{-2\pi M}. \tag{RL2}$$

Then, for every $h > 0$,

$$\mathbb{P}\left\{\sup_{0\leq s\leq h} |W_s| > \delta\sqrt{h}\right\} \geq 1 - q. \tag{RL3}$$

Strong DMP gives the same conditional bound in the inherited centered chart after any finite stopping time.

Proof. In the original half-plane chart set $z = i\sqrt{h}$ and delete the upward ray $A = \{iy : y \geq \sqrt{h}\}$. The comparison domain is $U = \mathbb{H} \setminus A$. It is simply connected. Its two ends at infinity are distinguished prime ends; choosing either one as target allows the boundary colors on the real line to match the original source colors. Corrected source locality couples the original trace and the comparison trace through their first hit of A.

Let E be the event $\sup_{s\leq h} |W_s| \leq \delta\sqrt{h}$. On E, the original trace and hull up to time h lie in the vertical strip $|\operatorname{Re} w| \leq \delta\sqrt{h}$ by the Loewner ODE. If its trace height were at most $\sqrt{h}$, its hull would lie in a disk of radius $\sqrt{(1+\delta^2)h}$. The area theorem would give $2h \leq (1+\delta^2)h$, impossible because $\delta < 1$. Consequently the trace reaches height $\sqrt{h}$ by time h and there enters $B(z, 2\delta\sqrt{h})$.

Entry into the open ball occurs strictly before the first encounter with A. If the first ray encounter before h is outside this ball, its height is greater than $\sqrt{h}$, so an earlier height-$\sqrt{h}$ point was already in the ball. If the first encounter itself lies in the ball, continuity puts an earlier point in that open ball. If there is no ray encounter before h, the asserted visit already precedes it. Thus the coupled comparison trace visits the ball on E, using only the common prefix; E itself need not be a common-prefix event.

In U, consider the annulus centered at z of inner radius $2\delta\sqrt{h}$ and outer radius $\sqrt{h}/2$, slit along the upward ray. It is a boundary quad of opening angle 2π, with both side arcs on the two prime-end

sides of A. Its modulus is

$$\frac{1}{2\pi}\log\frac{\sqrt{h}/2}{2\delta\sqrt{h}} = M.$$

It is unforced: source zero and either target end at infinity can be connected outside its outer circle, on the chosen target side. The comparison trace starts outside its outer circle, so visiting the inner ball requires a crossing. The boundary-robust convention counts side contacts and a visit ending at the slit tip as well. The quad bound gives comparison visit probability at most q. Hence $\mathbb{P}(E) \le q$, proving (RL3).

The same comparison can be made in every inherited restart chart. Genuine strong DMP therefore supplies the conditional version with the same constants. No conformal covariance of the physical family was used. □

What this does and does not establish. Together with the upper bracket theorem and the stochastic lower-tail criterion, this gives uniformly elliptic bracket for a semimartingale driver from the current crossing bound and the stated ray-deletion locality. It does not identify the density as six, remove singular drift, or prove semimartingality. In particular it does not infer a general interior hitting estimate from the boundary-quad condition. Deterministic vertical SLE_0 fails the source-locality comparison used here, so it does not contradict this argument.

18.2 The driving variation index is two before semimartingality

The lower driving tail has a consequence which does not assume a finite signed budget, a bracket, or a semimartingale decomposition. For a continuous real path x on $[a,b]$, write

$$V_r(x;[a,b]) = \sup_{\pi}\sum_{j}|x(t_{j+1})-x(t_j)|^r, \qquad r>0, \tag{VI1}$$

where the supremum is over finite partitions of that interval.

Proposition 18.4 (Uniform diffusive lower tails force variation index two)**.** *Let X be continuous and adapted, and suppose there are fixed $\delta, p > 0$ such that for all deterministic $t \ge 0$ and $h > 0$,*

$$\mathbb{P}\{\sup_{0\le u\le h}|X_{t+u}-X_t| > \delta\sqrt{h} \mid \mathcal{F}_t\} \ge p. \tag{VI2}$$

Then, on one probability-one event, $V_r(X;[a,b]) = \infty$ for every nontrivial interval and every $0 < r < 2$. If its paths are also locally Hölder for every exponent below $1/2$, then on that same event $V_r(X;[a,b]) < \infty$ on every compact interval for every $r > 2$. Thus the variation index is exactly two on every nontrivial interval. No assertion about V_2 or a quadratic-variation limit is made.

Proof. Fix first a deterministic compact interval $[a,b]$, put $n = 2^m$, $h = (b-a)/n$, $t_j = a + jh$, and define

$$I_j = \mathbf{1}_{\{\sup_{0\le u\le h}|X_{t_j+u}-X_{t_j}|>\delta\sqrt{h}\}}.$$

These indicators are measurable at t_{j+1}, and $\mathbb{E}[I_j \mid \mathcal{F}_{t_j}] \ge p$. The martingale differences $D_j = I_j - \mathbb{E}[I_j \mid \mathcal{F}_{t_j}]$ are orthogonal in L^2 and satisfy $\mathbb{E}(\sum_j D_j)^2 \le n$. Therefore

$$\mathbb{P}\{\textstyle\sum_j I_j < pn/2\} \le \frac{4}{p^2 n}. \tag{VI3}$$

This is summable over dyadic n. Borel–Cantelli gives $\sum_j I_j \ge pn/2$ for all sufficiently large m, almost surely. For every successful block, choose a time where its displacement from the left endpoint

exceeds $\delta\sqrt{h}$. The ordered union of these times and the block endpoints is a finite partition, so simultaneously for every $r > 0$,

$$V_r(X;[a,b]) \geq \frac{pn}{2}\delta^r h^{r/2} = \frac{p\delta^r}{2}(b-a)^{r/2}n^{1-r/2}. \tag{VI4}$$

No adapted selection is needed: this is a pathwise lower bound for a supremum over partitions. It diverges for every $r < 2$ on the same event. Intersect over rational intervals, and place a rational subinterval inside any other nontrivial interval.

If $r > 2$, choose $1/r < \alpha < 1/2$. A Hölder-α constant K on a compact interval bounds every partition sum by $K^r \sum_j |t_{j+1} - t_j|^{\alpha r} \leq K^r(b-a)^{\alpha r}$. The simultaneous upper assertion follows from the countable collection of rational Hölder exponents. □

By the ray-locality lemma and the already proved upper driving bounds, the continuum family has this variation index under the stated ray-deletion admissibility. In particular its drivers cannot have finite variation on any nontrivial interval. The lower proof controls block oscillations, not terminal increments of the blocks; replacing it by a claim of positive quadratic variation along deterministic partitions would be unjustified. Nor does variation index two imply semimartingality: the earlier Brownian-plus-Takagi example already shows the latter conclusion needs signed information.

18.3 A bounded bracket becomes uniformly elliptic

Proposition 18.5 (Conditional lower tails make a bounded bracket elliptic)**.** *Let $X = X_0 + M + A$ be a continuous semimartingale with $d\langle M\rangle = v_t dt$, $0 \leq v_t \leq K$, and A continuous finite variation. Suppose the conditional lower-tail bound (RL3), with lower probability $p > 0$, holds with X in place of W, at least at every deterministic $\tau = t$ and for all $h > 0$ in a finite horizon. Then*

$$v_t \geq \frac{p\delta^2}{32} \quad dt \otimes d\mathbb{P}\text{-almost everywhere.} \tag{NC3}$$

No density or initial tangent of A is assumed.

Proof. Choose a bounded predictable version of v. For Lebesgue-almost every deterministic t, Fubini and Lebesgue differentiation give, almost surely,

$$\frac{1}{h}\int_t^{t+h} v_s ds \longrightarrow v_t, \qquad \frac{\mathrm{Var}(A;[t,t+h])}{\sqrt{h}} \longrightarrow 0. \tag{NC4}$$

The second assertion follows because the finite Stieltjes variation measure has a finite one-sided Lebesgue derivative almost everywhere. Fix such a t, set $\eta = p\delta^2/32$, and let $E = \{v_t < \eta\} \in \mathcal{F}_t$. The lower tail assumption implies

$$p\mathbb{P}(E) \leq \mathbb{P}\{E, \sup_{u\leq h} |X_{t+u} - X_t| > \delta\sqrt{h}\}.$$

The corresponding drift supremum is at most $\mathrm{Var}(A;[t,t+h])$, so its probability of exceeding $\delta\sqrt{h}/2$ tends to zero by (NC4). Since $v \leq K$, M is square integrable on finite horizons and Doob's inequality gives

$$\mathbb{P}\{E, \sup_{u\leq h} |M_{t+u} - M_t| > \delta\sqrt{h}/2\} \leq \frac{16}{\delta^2 h}\mathbb{E}[\mathbf{1}_E(C_{t+h} - C_t)].$$

Boundedness of v and (NC4) permit dominated convergence of the last normalized expectation to $\mathbb{E}[\mathbf{1}_E v_t]$. Thus $p\mathbb{P}(E) \leq 16\eta\mathbb{P}(E)/\delta^2 = (p/2)\mathbb{P}(E)$, forcing $\mathbb{P}(E) = 0$. Apply Fubini again. □

Combining this proposition with the upper bracket theorem and the ray-locality lemma gives $c\,dt \le dC \le K\,dt$, with $c = (1-q)\delta^2/32 > 0$, once the finite signed budget supplies semimartingality. The density is not yet identified as six. The lower conclusion uses corrected ray-deletion locality, not the boundary-quad estimate alone.

18.4 Bounded Brownian bracket does not remove a singular drift

The reflected Brownian process $X_t = |B_t|$ is a sharp process-level warning at the next stage. In its usual natural filtration it has canonical decomposition $X = N + \Lambda$, where N is Brownian motion and Λ is the continuous Skorokhod regulator, increasing only on $\{X = 0\}$. That set has zero Lebesgue measure almost surely, so $d\Lambda$ is singular and nonzero. Tanaka's formula gives the decomposition in the Brownian filtration; the regulator is recovered from the occupation local time of X, so both summands are adapted to its smaller natural filtration, where the martingale identity continues to hold by conditioning.

For every stopping time τ of X,

$$\sup_{u\le h}\big||B_{\tau+u}| - |B_\tau|\big| \le \sup_{u\le h}|B_{\tau+u} - B_\tau|. \tag{RB1}$$

Brownian strong Markov and conditioning therefore give uniform conditional Gaussian-size increment bounds for X, even though its finite signed budget is carried by a singular regulator. In particular its bracket density is exactly one. This is not a curve-family counterexample: neither source locality nor the domain-indexed exploration axioms are asserted for this example. It shows why bounded bracket and unsigned driving estimates do not finish the drift argument.

19 Analytic locality and signed characteristic measures

Scope and matched localization. This section uses source locality only to identify two initial ordered trace laws and their natural filtrations. On their common trace, let h_t be the evolving analytic deletion map, recentered at the source, and write

$$p_t = h_t'(W_t) > 0, \quad q_t = h_t''(W_t), \quad u(t) = \int_0^t p_s^2\,ds, \quad Y_t = \widehat{W}_{u(t)} = h_t(W_t) - h_0(0). \tag{LB1}$$

The exact recentering convention has no effect on the derivatives or characteristics. The two capacity clocks and their inverses are stopping-time clocks in the corresponding natural trace filtrations, by geometric reconstruction of a stopped trace. Stop on a strictly positive interval where the deletion has not collided with the trace, its field and inverse are absolutely continuous in time as C^2-valued fields on compact analytic source neighborhoods, with bounded spatial derivatives, and

$$|W| \le L, \quad |Y| \le \widehat{L}, \quad 0 < m \le p \le P, \quad |q| \le Q, \quad \theta \le T. \tag{LB2}$$

All identities below concern the processes stopped at this same physical trace time θ, not two independently selected exit rules. The inherited stop $u(\theta)$ is a stopping time in deleted capacity, and $u(\theta) \le P^2T$. The deletion PDE gives the stated time regularity before collision. In particular, the time dependence of the field is pathwise finite variation; no stochastic regularity of W is assumed in writing (LB1).

Proposition 19.1 (Local semimartingality and finite signed budgets are analytic-locality invariants)**.** *Under this setup, the original driver is locally a continuous semimartingale if and only if the deleted*

driver is, on the matched initial intervals. Suppose they have canonical decompositions $W = M + A$ *and* $\widehat{W} = \widehat{M} + \widehat{A}$*, with brackets* C *and* $\widehat{C}$*. Set* $D_t = \widehat{A}_{u(t)}$*. Then, before the matched stop,*

$$dD = p\,dA + \frac{q}{2}(dC - 6dt), \qquad d\widehat{C}_{u(t)} = p^2\,dC. \tag{LB3}$$

For the bounded stopped processes let $V = \mathbb{E}\operatorname{Var} A$ *and* $\widehat{V} = \mathbb{E}\operatorname{Var} D$*, equivalently their conditional-variation budgets when finite. Then*

$$\widehat{V} \le (P + LQ)V + \tfrac{1}{2}QL^2 + 3QT, \tag{LB4}$$

and

$$V \le \left(m^{-1} + Q\widehat{L}m^{-3}\right)\widehat{V} + \tfrac{1}{2}Q\widehat{L}^2m^{-3} + 3QT/m. \tag{LB5}$$

In particular finite expected canonical variation, hence finite aggregate signed budget, transfers both ways after these matched bounded localizations. Neither estimate supplies a first-order time rate.

Proof. Assume first W is a semimartingale. The finite-variation-field Itô–Wentzell formula applied to $h_t(W_t)$ gives

$$dY_t = p_t\,dM_t + p_t\,dA_t + \tfrac{1}{2}q_t\,dC_t - 3q_t dt.$$

Here the final term is the Loewner deletion field identity $(\partial_t h_t)(W_t) = -3h_t''(W_t)$, which requires no stochastic calculus to derive. The geometric stopping-time clocks transfer the semimartingale and its natural filtration to deleted capacity. Conversely, if the deleted driver is a semimartingale, its pullback Y is one in the common trace filtration. Apply the same finite-variation-field formula to $h_t^{-1}(Y_t + h_0(0))$. The inverse field is adapted, spatially analytic, and of finite variation in time on this interval; its derivatives are bounded under localization. Thus W is a semimartingale. Uniqueness of canonical continuous decompositions now gives (LB3), without assuming any drift density or initial tangent.

We use a useful elementary budget/bracket estimate. If a continuous semimartingale $Z = N + F$ starts at zero, is bounded by K, and has $\mathbb{E}\operatorname{Var} F < \infty$, then

$$\mathbb{E}\langle N\rangle_T \le K^2 + 2K\mathbb{E}\operatorname{Var} F. \tag{LB6}$$

To prove it, stop to bound the bracket and total variation, take expectations in Itô's formula for Z^2, and then use Fatou. The resulting finite expected bracket also makes the martingale square integrable, so the usual unstopped identity is justified. Apply (LB6) first to W and use (LB3) and (LB2):

$$\widehat{V} \le PV + \tfrac{1}{2}Q\mathbb{E}C_\theta + 3QT,$$

which gives (LB4). In the opposite direction put $\widetilde{C}_t = \widehat{C}_{u(t)}$ and solve (LB3):

$$dA = p^{-1}dD - \frac{q}{2p^3}d\widetilde{C} + 3qp^{-1}dt.$$

Apply (LB6) to Y and use (LB2) to obtain (LB5). These estimates are invoked with the source-side budget finite; the corresponding conclusion then guarantees finiteness on the other side. The equality between budgets and expected variation is the earlier exact conditional-variation identity, applied to the matched stopped processes. A budget in the original capacity and one in the deleted capacity must retain their inherited stopping rules. □

Proposition 19.2 (Two deletion directions eliminate singular characteristics)**.** *Take two admissible deletion maps h_t^+, h_t^- on the same initial trace. Suppose their drivers are continuous semimartingales. Define $p^\pm, q^\pm, u_\pm$ as in (LB1), let $D_t^\pm = \widehat{A}^\pm_{u_\pm(t)}$, and suppose*

$$\Delta_t = p_t^+ q_t^- - p_t^- q_t^+ \neq 0 \tag{LB7}$$

on a positive initial interval. Then the original driver is a semimartingale and its characteristic measures satisfy

$$\boxed{\begin{aligned} dA &= \frac{q^-\, dD^+ - q^+\, dD^-}{\Delta}, \\ dC - 6dt &= \frac{2(p^+\, dD^- - p^-\, dD^+)}{\Delta}. \end{aligned}} \tag{LB8}$$

No initial derivatives or drift densities are required. In particular, if both deleted drift measures are absolutely continuous in their capacity clocks, then the original drift and bracket are absolutely continuous on this interval. The original drift need not have been assumed absolutely continuous.

Proof. The preceding proposition transfers semimartingality from either deleted chart to the original one. Apply (LB3) to both, and invert the two-by-two system of signed measures. This gives (LB8). The clocks $du_\pm = (p^\pm)^2 dt$ are locally bi-Lipschitz after localization, so absolute continuity of a deleted drift in its own capacity is exactly absolute continuity of its pullback $D^\pm$ in t. Equation (LB8) then proves the assertion. □

The singular-part alternative is exact. Take pathwise Lebesgue decompositions with respect to dt. Writing superscript $\perp$ for singular Stieltjes parts, (LB3) says

$$dD^\perp = p\, dA^\perp + \tfrac{1}{2} q\, dC^\perp, \qquad d\widetilde{C}^\perp = p^2\, dC^\perp. \tag{LB9}$$

Thus absence of a singular bracket is invariant under analytic local coordinate changes. Once the bracket is absolutely continuous, absence of a singular drift is invariant as well. If the bracket has a singular part, an Itô correction can cancel that part from a particular chart's drift; two distinct directions with (LB7) cannot both cancel it. Formula (LB8) also gives the pathwise measure bounds

$$\begin{aligned} |dA| &\le \frac{|q^-|\,|dD^+| + |q^+|\,|dD^-|}{|\Delta|}, \\ |dC - 6dt| &\le \frac{2p^+|dD^-| + 2p^-|dD^+|}{|\Delta|}. \end{aligned} \tag{LB10}$$

These also hold for singular parts, with the second left side replaced by the positive measure $dC^\perp$.

For the mirrored shrinking slits of (DR6), initially $p_0^\pm = d_\epsilon \asymp \epsilon$ and $q_0^- = -q_0^+ = -e_\epsilon^+$ with $|e_\epsilon^+| \asymp \epsilon^{-1}$. The determinant is nonzero. By stopping when the coefficients depart by a sufficiently small relative amount from these initial values, (LB10) gives, with an absolute numerical K independent of small ϵ,

$$|dA| \le K\epsilon^{-1}(|dD^+| + |dD^-|), \qquad |dC - 6dt| \le K\epsilon(|dD^+| + |dD^-|). \tag{LB11}$$

In particular singular bracket mass costs at least inverse-linear singular drift mass across these two deleted charts. The matched positive stopping interval may shrink with ϵ. Therefore finiteness of each deleted budget does not permit passing to $\epsilon = 0$ in (LB11) on a fixed interval. A quantitative uniform budget estimate is still needed; no regularity or classification is being deduced from bare locality by this calculation.

What this changes in the research target. Finiteness of a localized signed budget is a property of the local physical source law within a fixed complex structure, rather than an artifact of choosing one of its analytic deletion charts. Its numerical size is not invariant: (LB3)–(LB5) exhibit the bracket correction and small first derivative which can inflate it. Analytic coordinate changes are being used throughout; an arbitrary real affine normalization is not such a change. Thus this result neither selects the unknown affine structure nor asserts semimartingality in all affine structures. The new positive regularization is the two-direction elimination of singular characteristics. Obtaining finite budgets and the quantitative scale control required for Brownian rigidity remains open.

19.1 Target-invariant force-point branches still fail source locality

A useful test is the familiar target-invariant branching $\mathrm{SLE}_\kappa(\kappa-6)$ construction. Target invariance holds within one physical domain; it does not assert invariance under deleting remote pieces of that domain. The following calculation excludes the standard one-sided force-point family from the present conjecture, including the singular drift at an initially coincident force point. It does not assume a finite initial drift tangent.

Use the convention

$$dW_t = \sqrt{\kappa}\, dB_t + \frac{\rho}{W_t - V_t}\, dt, \qquad dV_t = \frac{2}{V_t - W_t}\, dt, \qquad (W_0, V_0) = (0, 0^+), \quad \rho > -2. \tag{FP1}$$

The right force point is the boundary-prime-end limit immediately to the right of the source, transported by the Loewner map. The gap $(V - W)/\sqrt{\kappa}$ is a Bessel process of dimension $\delta = 1 + 2(\rho+2)/\kappa > 1$. In particular, its zero set has Lebesgue measure zero, and the force drift in (FP1) is locally integrable. These standard statements follow from the Bessel construction; see Sheffield [3], Section 3.1, Propositions 3.1 and 3.3, and Section 3.3.

Proposition 19.3 (Only the ordinary SLE_6 member is source-local)**.** *Let $\kappa > 0$. Consider the canonical conformally covariant family with one force point immediately on a specified side of its source, defined by (FP1). If it satisfies source-neighborhood locality under remote hull deletions, then $\kappa = 6$ and $\rho = 0$. In particular, for $4 < \kappa < 8$, the target-invariant $\mathrm{SLE}_\kappa(\kappa-6)$ family fails that locality whenever $\kappa \neq 6$.*

Proof. Let A be a fixed half-plane hull away from 0, let $h : \mathbb{H} \setminus A \to \mathbb{H}$ be its hydrodynamic map, and compare the exploration in $\mathbb{H}$ with that in $\mathbb{H} \setminus A$. Center the new source at $h(0)$. Work up to a positive stopping time before the growing hull encounters A, chosen so that the comparison maps are analytic across a fixed interval containing W_t and V_t. Let h_t be the evolving deletion map and $ds = h_t'(W_t)^2 dt$. The centered transformed driver and force point are $h_t(W_t) - h(0)$ and $h_t(V_t) - h(0)$. The force point is transported exactly: the deletion-map PDE and (FP1) give

$$d\{h_t(V_t)\} = \frac{2h_t'(W_t)^2}{h_t(V_t) - h_t(W_t)}\, dt. \tag{FP2}$$

The driving change-of-coordinate calculation yields the Brownian part $\sqrt{\kappa}\, d\widehat{B}_s$ and drift, in the original dt clock,

$$\frac{\rho h_t'(W_t)}{W_t - V_t} + \left(\frac{\kappa}{2} - 3\right) h_t''(W_t).$$

For the force-point law in the deleted domain, the required drift in that same clock is

$$\frac{\rho h_t'(W_t)^2}{h_t(W_t) - h_t(V_t)}.$$

Therefore locality would require

$$R_t = \rho \left\{ \frac{h_t'(W_t)}{W_t - V_t} - \frac{h_t'(W_t)^2}{h_t(W_t) - h_t(V_t)} \right\} + \frac{\kappa - 6}{2} h_t''(W_t) = 0 \tag{FP3}$$

for Lebesgue-almost every sufficiently small t. This follows from uniqueness of the canonical semimartingale drift: the force point is the canonical one-sided source-boundary Loewner flow, with its instant-reflecting continuation at collisions. It is adapted to the driver and is transported as the specified boundary side, not an independently reassignable mark. The capacity change is invertible on the stopped comparison interval. All drift terms are integrable since $\delta > 1$.

For an analytic H with $H'(w) > 0$, put $d = w - v$. Taylor expansion gives

$$\frac{H'(w)}{d} - \frac{H'(w)^2}{H(w) - H(v)} = -\frac{1}{2} H''(w) + \frac{1}{6} H'(w) \mathcal{S}H(w)\, d + O(d^2), \tag{FP4}$$

where $\mathcal{S}H = H'''/H' - \frac{3}{2}(H''/H')^2$ is the Schwarzian. The remainder is uniform in a compact analytic neighborhood on which H' is bounded away from zero. Consequently

$$R_t = \frac{\kappa - 6 - \rho}{2} h_t''(W_t) + \frac{\rho}{6} h_t'(W_t) \mathcal{S}h_t(W_t)(W_t - V_t) + O((W_t - V_t)^2). \tag{FP5}$$

Choose, for example, the vertical slit $A = \{x + iy : 0 \le y \le a\}$ with $x, a > 0$. Its map is $h(z) = x + \sqrt{(z-x)^2 + a^2}$, with the hydrodynamic branch. At 0 both $h''(0) \neq 0$ and $\mathcal{S}h(0) \neq 0$. More explicitly, writing $u = z - x$, the Schwarzian on the real analytic source interval is

$$\mathcal{S}h(z) = -\frac{3a^2(u^2 + a^2/2)}{u^2(u^2 + a^2)^2} < 0. \tag{FP6}$$

As $t \downarrow 0$, $W_t, V_t \to 0$ and the analytic maps h_t converge locally to h, with their derivatives. If $\kappa - 6 - \rho \neq 0$, (FP5) therefore makes R_t bounded away from zero on a positive initial interval, contradicting (FP3). Thus $\rho = \kappa - 6$.

If this common value is nonzero, the second term of (FP5), divided by $W_t - V_t$, tends to $\rho h'(0) \mathcal{S}h(0)/6 \neq 0$. After decreasing the positive comparison time, $R_t \neq 0$ whenever $W_t \neq V_t$. The latter holds for Lebesgue-almost every time by the Bessel zero-set statement, again contradicting (FP3). Hence $\rho = 0$ and $\kappa = 6$.

Although written in half-plane coordinates, the test applies to bounded physical domains: map the half-plane and the remote slit into a disk by one fixed Möbius map. The resulting physical domains agree near their common source. The same calculation in their chordal charts then contradicts source locality there. □

Why target invariance does not contradict this exclusion. For a Möbius H one has the exact identity

$$\frac{H'(w)}{w - v} - \frac{H'(w)^2}{H(w) - H(v)} = -\frac{1}{2} H''(w). \tag{FP7}$$

Thus the force-point drift with $\rho = \kappa - 6$ cancels the entire coordinate anomaly for Möbius changes of target. A remote domain hull deletion instead has nonzero Schwarzian, leaving the second term of (FP5). Schramm–Wilson [2], Theorem 3, establishes the Möbius coordinate-change theorem with sum of force weights $\kappa - 6$; it does not assert remote-domain locality: https://nyjm.albany.edu/nyjm/j/2005/11-31.pdf. Sheffield's Proposition 3.14 and Section 4.2 use target invariance to construct branching radial explorations within one fixed domain. Those statements are consistent with the calculation above.

There is also an extra-state issue for strong DMP. A force point started at 0^+ is usually separated from the tip at positive times. Conditional continuation then carries its current boundary-prime-end image V_t, rather than restarting with a fresh force point at W_t^+. The standard force-point Markov property is therefore a property of the domain with this additional mark (and, in side-swapping variants, the appropriate excursion state). It is not by itself the required strong-DMP identity on a domain with only the two colored boundary arcs and marked source/target. A proposed intrinsic recovery of that mark from the colored domain would require its own proof. This issue is independent of the locality failure already proved.

Consequence for the present research direction. Even if the desired RSW estimates were granted to these branches, they would not be counterexamples to the current conjecture: the locality test excludes them before RSW enters. They are nevertheless a useful singular-drift test. Their Bessel force drift is integrable, so merely obtaining semimartingality or finite aggregate signed variation would not eliminate them. The exact locality comparison supplies information beyond initial drift tangents; in this explicit ansatz it is the Schwarzian remainder that distinguishes target independence from full source locality.

The failure of source locality persists after any one fixed affine change applied to the entire family: such a change preserves common source neighborhoods and transports their stopped-law equality.

20 Near-critical germ singularity and the energy of a driving drift

A stronger external obstruction. Aumann [9], Theorem 2.1 and Corollary 2.2, proves mutual singularity of distinct near-critical triangular-percolation interface limits already on the raw initial ordered-germ sigma-field. The theorem assumes a common subsequence with the stated quad-crossing convergence; for comparison with the critical parameter zero, that condition follows from Cardy's formula. Corollary 5.1 gives singularity under nontrivial scalar dilations at a fixed nonzero parameter, again with its convergence hypotheses. Thus a curve-level dilation obstruction is available, without identifying random slit-domain continuation kernels. It does not construct the coherent Borel strong-DMP family still needed for a counterexample to the corrected three-axiom conjecture.

The following consequence concerns a proposed driving representation, not a claim of semimartingality of a near-critical interface.

Proposition 20.1 (Finite driving energy preserves the Brownian initial germ)**.** *Let $X_0 = 0$ be continuous on $[0,T]$, and let B be Brownian motion in a usual filtration to which X is adapted. Suppose $\kappa > 0$, $\rho > 0$ is a stopping time almost surely, and, for $t \le \rho \wedge T$,*

$$X_t = \sqrt{\kappa} B_t + \int_0^t b_s \, ds, \qquad \int_0^{\rho \wedge T} b_s^2 \, ds < \infty \quad \text{almost surely,} \tag{GE1}$$

where b is progressively measurable. On the canonical continuous-path space put $\mathcal{H}_{0+} = \bigcap_{t>0} \sigma(x_s : 0 \le s \le t)$, using the raw sigma-fields. Then, for every $E \in \mathcal{H}_{0+}$,

$$\mathbb{P}\{X \in E\} = \mathbb{W}_\kappa(E) \in \{0,1\}, \tag{GE2}$$

where $\mathbb{W}_\kappa$ is Wiener measure of variance rate κ.

Proof. Set

$$\sigma = \rho \wedge T \wedge \inf\left\{ t : \int_0^{t \wedge \rho} b_s^2 ds \ge \kappa \right\}.$$

The finite energy in (GE1) implies $\sigma > 0$ almost surely. Define $\theta_s = \kappa^{-1/2} b_s \mathbf{1}_{\{s \le \sigma\}}$. Its integrated square is at most one. Novikov's criterion and Girsanov's theorem (for example [10], Theorems 1–2) therefore make

$$Z_T = \exp\left\{ -\int_0^T \theta_s \, dB_s - \frac{1}{2}\int_0^T \theta_s^2 ds \right\}$$

a strictly positive density of an equivalent probability Q. Under Q,

$$\widetilde{X}_t = \sqrt{\kappa} B_t + \int_0^{t\wedge\sigma} b_s ds$$

is Brownian motion of variance rate κ on $[0, T]$. The paths X and $\widetilde{X}$ agree on the strictly positive interval $[0, \sigma]$. Every raw germ event has the same membership for two paths agreeing on some positive initial interval: choose any deterministic $1/n$ inside that interval and use measurability in $\sigma(x_s : s \le 1/n)$. Consequently $Q\{X \in E\} = \mathbb{W}_\kappa(E)$. Blumenthal's zero–one law makes the latter probability zero or one; equivalence of P and Q gives (GE2). There is no global Novikov assumption on the original drift. □

Corollary 20.2 (A near-critical limit cannot be a finite-energy drift of SLE)**.** *Consider a nonzero near-critical interface limit covered by Aumann's comparison with the critical limit. Assume it has a continuous Loewner-generating parametrization, with an adapted Borel correspondence between driver prefixes and ordered trace prefixes. Its driver in a capacity chart where the critical comparison has law $\sqrt{6}B$ cannot satisfy (GE1) with $\kappa = 6$.*

Proof. A raw ordered trace-germ event is unchanged when two nonconstant Loewner traces agree for a positive capacity interval. In the preceding proof the original and the Q-Brownian drivers have precisely such an interval in common. The corresponding traces therefore have the same raw germ-event membership. Under Q the comparison trace is SLE_6; its raw initial germ is zero–one (equivalently use the Brownian germ and the adapted correspondence). Equivalence then gives the same germ probabilities under the original law. This contradicts the germ event separating the nonzero near-critical and critical laws in Aumann's Corollary 2.2. □

The energy must diverge immediately if such an AC representation exists. More precisely, suppose $X = \sqrt{\kappa}B + \int_0^{\cdot} b_s ds$ on $[0, T]$, with b progressively measurable and integrable in absolute value, and suppose a raw germ event E has probability one for X and zero under $\mathbb{W}_\kappa$. Then

$$\int_0^t b_s^2 ds = \infty \quad \text{for every } 0 < t \le T, \quad \text{almost surely.} \tag{GE3}$$

Indeed set $\sigma = T \wedge \inf\{t : \int_0^t b_s^2 ds \ge \kappa\}$, allowing $\sigma = 0$ on instantaneous infinite energy. This is a stopping time in the usual filtration. In all cases $\int_0^\sigma b_s^2 ds \le \kappa$, by monotone convergence from times strictly below σ when $\sigma > 0$. The same bounded-energy change of measure as above makes $\widetilde{X} = \sqrt{\kappa}B + \int_0^{\cdot\wedge\sigma} b_s ds$ Brownian under an equivalent probability. Thus $E(\widetilde{X})$ has probability zero under both probabilities. On $\{\sigma > 0\}$, however, X and $\widetilde{X}$ have the same germ, so $E(X) = 1$ forces $\mathbb{P}(\sigma > 0) = 0$. Finite energy on any positive initial interval would imply $\sigma > 0$, proving (GE3). The identical argument uses the trace-germ separator for the near-critical comparison.

Why square integrability cannot be weakened to absolute integrability. For $0 < \alpha < 1/2$, the process $X_t = \sqrt{\kappa}B_t + t^\alpha$ has a continuous absolutely continuous finite-variation drift, with density $\alpha t^{\alpha-1}$ integrable at zero but with infinite square integral on every positive interval. Brownian

small-time Hölder regularity gives $B_t/t^\alpha \to 0$ almost surely. Hence $X_t/t^\alpha \to 1$ almost surely, while the corresponding Wiener limit is zero. This explicitly separates their raw initial germs and shows why the energy hypothesis in Proposition 20.1 is substantive.

Scope and compatibility with source shielding. This excludes Brownian driving with a locally bounded, or more generally locally square-integrable, absolutely continuous drift. It does not prove that a near-critical driver is not a semimartingale: a singular finite-variation part, infinite drift energy at every positive initial time, or a different martingale bracket is not excluded. Neither the bracket-six premise nor Loewner generation is supplied by the singularity theorem. Raw germ singularity between two laws is compatible with an individual germ zero–one law for each: different laws may assign different deterministic values to the same germ event. Source shielding alone therefore cannot identify the universal germ law.

21 Affine deformation without assuming regular driving characteristics

Scope. The statements in this section are deterministic statements about bounded half-plane hulls and continuously generated Loewner chains. No simple-curve, zero-area, or semimartingale hypothesis is made. Boundary values of conformal maps are interpreted at the distinguished tip prime end. A real-linear map of the half-plane preserves filled hulls, and the transformed chain is assumed to have its continuous capacity parametrization, as supplied by the affine admissibility results for the model under consideration.

Let $g_t : \mathbb{H} \setminus K_t \to \mathbb{H}$ be hydrodynamically normalized, $g_t(z) = z + 2t/z + O(z^{-2})$, with continuous driver W_t, and put $f_t = g_t^{-1}$. Let $L(z) = az + b\bar{z}$ be an orientation-preserving real-linear self-map of $\mathbb{H}$. Write

$$s(t) = \tfrac{1}{2}\,\mathrm{hcap}(LK_t), \qquad F_t = \widehat{g}_{s(t)} \circ L \circ f_t, \qquad U_t = F_t(W_t), \tag{AB1}$$

where $\widehat{g}$ uniformizes the transformed hulls. Then

$$\mu_{F_t}(z) = \frac{b}{a}\,\frac{\overline{f_t'(z)}}{f_t'(z)}. \tag{AB2}$$

For a model in which $L\gamma$ is SLE_6, $U_t = \sqrt{6}B_{s(t)}$. This is an identity for a known model, not a classification consequence of the exploration axioms.

Proposition 21.1 (An interior Stieltjes evolution)**.** *At every $z \in \mathbb{H}$, locally in finite time,*

$$dF_t(z) = \frac{2\,ds(t)}{F_t(z) - U_t} - \left\{ \frac{2(F_t)_z(z)}{z - W_t} + \frac{2(F_t)_{\bar{z}}(z)}{\bar{z} - W_t} \right\} dt. \tag{AB3}$$

The equation is an equality of finite-variation measures, locally uniformly on compact subsets of $\mathbb{H}$. It does not require that s be absolutely continuous. The field $\mu_t = \mu_{F_t}$ satisfies, in the interior,

$$\partial_t \mu_t = -v_t \partial_z \mu_t - \overline{v_t}\partial_{\bar{z}}\mu_t + (v_t' - \overline{v_t'})\mu_t, \qquad v_t(z) = \frac{2}{z - W_t}. \tag{AB4}$$

Proof. Inverse Loewner evolution gives $\partial_t f_t = -v_t f_t'$. Apply the ordinary chain rule to (AB1), using the Loewner equation for $\widehat{g}$ in its clock s and the real-linear derivative of L. The two contributions from $\partial_t f_t$ are exactly the two dt terms in (AB3). On any compact spatial set and finite localized time interval, the images stay a positive distance from the real line; the spatial derivatives and

denominators are bounded. Since s is continuous increasing, the displayed terms have finite total variation. For (AB4), differentiate $\partial_t f'_t = -v_t f''_t - v'_t f'_t$ and its conjugate in (AB2). This calculation is entirely in the interior. □

Interior finite variation of F does not justify applying Itô's formula to $F_t(W_t)$. Neither boundary differentiability of F_t at W_t nor the necessary uniform boundary time regularity has been proved.

21.1 A normalization-safe first variation at an arbitrary hull

We use the boundary-fixed affine perturbation

$$L_\epsilon(w) = w + \epsilon\nu(\bar{w} - w), \qquad \nu \in \mathbb{C}, \quad \epsilon \in \mathbb{R}. \tag{AV1}$$

For small ϵ it maps $\mathbb{H}$ onto itself and fixes every real boundary point. These perturbations span the two real anisotropy directions modulo similarities.

Fix a bounded hull $K = K_t$ contained in $\{|w| \le R\}$ with $R > 0$. Take a real smooth radial cutoff χ_R equal to one on $|w| \le 2R$ and zero on $|w| \ge 4R$, with derivatives of order j bounded by $C_j R^{-j}$. Set

$$V(w) = \chi_R(w)\nu(\bar{w} - w), \qquad \Phi_\epsilon(w) = w + \epsilon V(w), \qquad f = g_t^{-1}, \quad W = W_t. \tag{AV2}$$

For sufficiently small $|\epsilon|$ this is a plane diffeomorphism, fixes $\mathbb{R}$ pointwise, preserves $\mathbb{H}$, and equals L_ϵ on K. Define the compactly supported bounded coefficient on $\mathbb{H}$

$$\eta(z) = V_{\bar{w}}(f(z))\,\frac{\overline{f'(z)}}{f'(z)}. \tag{AV3}$$

Extend it to the lower half-plane by $\eta(\bar{z}) = \overline{\eta(z)}$.

Proposition 21.2 (First variation indexed by the original hull prefix)**.** *Let*

$$s_\epsilon = \tfrac{1}{2}\,\mathrm{hcap}(L_\epsilon K), \qquad U_\epsilon = \widehat{g}^{\,\epsilon}_{s_\epsilon}(L_\epsilon\gamma(t)), \tag{AV4}$$

where the second expression is interpreted by the tip prime end. Both quantities are differentiable at $\epsilon = 0$, and

$$\boxed{\begin{aligned} \partial_\epsilon U_\epsilon|_0 &= -\frac{2}{\pi}\,\mathrm{Re}\int_{\mathbb{H}} \frac{\eta(z)}{z - W}\,d^2z, \\ \partial_\epsilon s_\epsilon|_0 &= \frac{1}{\pi}\,\mathrm{Re}\int_{\mathbb{H}} \eta(z)\,d^2z. \end{aligned}} \tag{AV5}$$

The integrals are absolutely convergent, including at the tip. Their values are independent of the choice of the stated cutoff. There are absolute cutoff-dependent constants such that

$$|\partial_\epsilon U_\epsilon|_0 \le C|\nu|R, \qquad |\partial_\epsilon s_\epsilon|_0 \le C|\nu|R^2. \tag{AV6}$$

Proof. The derivative of Φ_ϵ is uniformly close to the identity, so this map is a global diffeomorphism for sufficiently small ϵ. Since $\mathrm{Im}\,\Phi_\epsilon(x+iy) = y(1 - 2\epsilon\,\mathrm{Re}(\nu)\chi_R(x+iy))$, it preserves both half-planes. Its image of K agrees with $L_\epsilon K$.

Put $F_\epsilon = \widehat{g}^{\,\epsilon}_{s_\epsilon} \circ \Phi_\epsilon \circ f$ and reflect it across $\mathbb{R}$. It is a plane quasiconformal map with principal normalization

$$F_\epsilon(z) = z + \frac{2(s_\epsilon - t)}{z} + O(z^{-2}) \quad (z \to \infty). \tag{AV7}$$

Indeed Φ_ϵ is the identity outside a fixed disk, so the composition there is the composition of the two hydrodynamic conformal maps. Its Beltrami coefficient is, in the upper half-plane,

$$m_\epsilon(z) = \frac{\epsilon V_{\bar{w}}(f(z))}{1+\epsilon V_w(f(z))}\frac{\overline{f'(z)}}{f'(z)} = \epsilon\eta(z) + O(\epsilon^2), \tag{AV8}$$

with the reflected coefficient in the lower half-plane. The remainder bound is in L^∞, supported on one bounded disk independent of ϵ. Boundedness of the support follows also directly from hydrodynamic univalence: the image under g_t of the portion of $\mathbb{H}\setminus K$ in a fixed disk is bounded. With the above cutoff scales, the support is contained in a disk of radius CR.

Here is an explicit version of the required differentiability argument. Let

$$\mathcal{C}h(z) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{h(\zeta)}{z-\zeta}\, d^2\zeta, \qquad \mathcal{S} = \partial_z\mathcal{C}.$$

For fixed $p>2$, the Beurling transform is bounded on L^p. For sufficiently small ϵ the principal solution has

$$F_\epsilon = z + \mathcal{C}h_\epsilon, \qquad h_\epsilon = (I - m_\epsilon\mathcal{S})^{-1}m_\epsilon. \tag{AV9}$$

The Neumann series and (AV8) imply $h_\epsilon/\epsilon \to \eta$ in L^p, and also in L^1 since all these functions have the same bounded support. On functions supported in this disk, the Cauchy transform maps L^p, $p>2$, continuously to bounded continuous functions, by Hölder's inequality at its integrable singularity and the elementary far-field bound. Consequently

$$\frac{F_\epsilon(z)-z}{\epsilon} \longrightarrow \mathcal{C}\eta(z) \tag{AV10}$$

uniformly in z, including $z=W\in\mathbb{R}$. The boundary identity $U_\epsilon = F_\epsilon(W)$ gives the first formula (AV5); reflection combines the two half-plane integrals. The coefficient of $1/z$ in (AV9) is $\pi^{-1}\int h_\epsilon$, so (AV7) and L^1 convergence give the second formula. Neither step differentiates the physical tip.

All cutoffs produce exactly the same transformed hull $L_\epsilon K$ and distinguished prime end, so their driver and capacity variations coincide. Finally $\|\eta\|_\infty \le C|\nu|$, its support lies in $|z|\le CR$, and $|W|\le 3R$. Integrating $1/|z-W|$ over this disk gives the first bound (AV6); its area gives the second. □

Corollary 21.3 (An explicit constant-anisotropy response)**.** *For the boundary-fixed affine deformation (AV1), the driver response can also be written without a cutoff:*

$$\boxed{\partial_\epsilon U_\epsilon|_0 = -\frac{2}{\pi}\,\mathrm{Re}\int_{\mathbb{H}} \frac{\nu}{z-W}\left(\frac{\overline{f'(z)}}{f'(z)} - 1\right) d^2z.} \tag{AV11}$$

The integral is absolutely convergent. This formula remains indexed by the original hull prefix.

Proof. Use the actual affine map in place of the cutoff and write $F_\epsilon = \widehat{g}^{\epsilon}_{s_\epsilon}\circ L_\epsilon\circ f$ and $Q_\epsilon = F_\epsilon - L_\epsilon$. Set $\phi = \overline{f'}/f'$ and $\mu_\epsilon = \epsilon\nu\phi/(1-\epsilon\nu)$. The Beltrami equation gives exactly

$$(Q_\epsilon)_{\bar{z}} - \mu_\epsilon(Q_\epsilon)_z = \epsilon\nu(\phi-1). \tag{AV12}$$

The map Q_ϵ has real boundary values since L_ϵ fixes $\mathbb{R}$, so reflect Q_ϵ and the two coefficients across the real line. Its normalization at infinity is $Q_\epsilon = O(|z|^{-1})$. The right side divided by ϵ, denoted m, is bounded and satisfies $m(z) = O(|z|^{-2})$ at infinity, because $f'(z) = 1 + O(z^{-2})$. In particular $m\in L^p\cap L^q$ for fixed $p>2>q>1$.

Writing $\xi_\epsilon = (Q_\epsilon)_{\bar z}$, the normalized solution of (AV12) is

$$\xi_\epsilon = \epsilon(I - \mu_\epsilon \mathcal{S})^{-1} m, \qquad Q_\epsilon = \mathcal{C}\xi_\epsilon. \tag{AV13}$$

For small ϵ, the two inverse operators are bounded on both L^p and L^q, so $\xi_\epsilon/\epsilon \to m$ in their intersection. Near the Cauchy kernel singularity use $p > 2$; at infinity use $q < 2$. This proves uniform convergence $Q_\epsilon/\epsilon \to \mathcal{C}m$, including boundary points. The normalized solution in (AV13) agrees with the geometrically defined Q_ϵ: their difference solves the homogeneous Beltrami equation and vanishes at infinity, so factorization through the normalized quasiconformal homeomorphism reduces it to a bounded entire function vanishing at infinity. Finally $L_\epsilon(W) = W$, and reflection yields (AV11). □

Proposition 21.4 (Uniform Taylor expansion along matched prefixes)**.** *Suppose $K_t \subset \{|z| \le R\}$ for $0 \le t \le T$, and use the same cutoff V from (AV2) for every prefix. Write D_t and E_t for the two first derivatives in (AV5). There are absolute constants $c, C > 0$, depending only on the fixed cutoff profile, such that for $|\epsilon\nu| \le c$,*

$$\begin{aligned} \sup_{t\le T} |U_\epsilon(t) - W_t - \epsilon D_t| &\le C\epsilon^2|\nu|^2 R, \\ \sup_{t\le T} |s_\epsilon(t) - t - \epsilon E_t| &\le C\epsilon^2|\nu|^2 R^2, \\ \sup_{t\le T} |D_t| \le C|\nu|R, \qquad \sup_{t\le T} |E_t| &\le C|\nu|R^2. \end{aligned} \tag{AV14}$$

In particular the matched-prefix pair is differentiable in the uniform path norm at $\epsilon = 0$, with continuous first derivative paths D, E. This remains a statement in the original prefix clock.

Proof. For every $t \le T$, the coefficients $m_{\epsilon,t}$ in (AV8) are supported in the same disk of radius CR. The bounds $\|\eta_t\|_\infty \le C|\nu|$ and $\|m_{\epsilon,t} - \epsilon\eta_t\|_\infty \le C\epsilon^2|\nu|^2$ hold with constants independent of t, the hulls, and R. Fix one $p > 2$. Choosing c so that the Neumann-series inverses on L^p have norm at most two gives

$$\|h_{\epsilon,t} - \epsilon\eta_t\|_p \le C\epsilon^2|\nu|^2 R^{2/p}. \tag{AV15}$$

The difference is supported in the same disk. For a function k supported in that disk, Hölder's inequality gives $\|\mathcal{C}k\|_\infty \le CR^{1-2/p}\|k\|_p$. Thus (AV15) gives the first remainder in (AV14), even before evaluating at the moving real point W_t. The coefficient at infinity is estimated in L^1: $\|h_{\epsilon,t} - \epsilon\eta_t\|_1 \le C\epsilon^2|\nu|^2R^2$, proving the second remainder. The two first-derivative bounds follow by the same estimates. Continuity of $U_\epsilon(t)$ and $s_\epsilon(t)$, together with uniform convergence of the difference quotients, gives continuity of D_t and E_t. □

Corollary 21.5 (Temporal one-half Hölder regularity of the linear clock response)**.** *Under the preceding proposition,*

$$|E_t - E_s| \le C|\nu|R\sqrt{t-s}, \qquad 0 \le s < t \le T. \tag{AV17}$$

This is a deterministic statement in the original capacity clock. It does not assert absolute continuity of E or of s_ϵ.

Proof. Write $\Delta = t - s$. Monotonicity of transformed hull capacity gives $s_\epsilon(t) - s_\epsilon(s) \ge 0$ for both signs of ϵ. Choose its sign opposite to that of $E_t - E_s$ and apply the two Taylor remainder bounds. For $0 < a \le c/|\nu|$ this gives

$$|E_t - E_s| \le \frac{\Delta}{a} + 2Ca|\nu|^2R^2.$$

For Δ below a sufficiently small constant times R^2, choose a proportional to $\sqrt{\Delta}/(|\nu|R)$. For larger Δ, use $|E_t - E_s| \le C|\nu|R^2$ from (AV14). The case $\nu = 0$ is immediate. □

Uniform stochastic parameter regularity. For a continuous Loewner driver on $[0,T]$, the strip and height bounds give the deterministic hull-radius estimate

$$R_T \le \sup_{t\le T}|W_t| + 2\sqrt{T}. \tag{AV16}$$

Consequently the conditional driving tail (NS4), applied in a restarted centered half-plane chart, implies conditional exponential tails for $\sup_{t\le h}|D_t|/(|\nu|\sqrt{h})$ and for $\sqrt{\sup_{t\le h}|E_t|/(|\nu|h)}$, with changed numerical constants. The two uniform Taylor remainders have respectively the same Brownian and quadratic size bounds multiplied by $\epsilon^2|\nu|^2$. These conclusions hold for the unknown crossing-controlled driver as well as the affine-SLE model. They give regularity in the *deformation parameter*. Apart from the separate one-half Hölder bound on E, they give no temporal regularity estimate for D and no finite signed variation of W or D.

21.2 Temporal regularity of the first affine driver response

Use the fixed-cutoff matched-prefix setup above, with $K_t \subset B_R$ for $t \le T$. Let $D_t = \partial_\epsilon U_\epsilon(t)|_0$ in affine direction ν. For $0 \le s < t \le T$ set

$$r_{s,t} = \sup_{s\le u\le t}|W_u - W_s| + 2\sqrt{t-s}.$$

Proposition 21.6 (The linear affine driver response retains temporal roughness bounds)**.** *For every integer $N \ge 1$, there is C_N depending only on N and the fixed cutoff profile such that*

$$|D_t - D_s| \le C_N|\nu|R^{1/(N+1)}r_{s,t}^{N/(N+1)}. \tag{TD1}$$

Consequently if W is Hölder for every exponent below $1/2$, then so is its matched-prefix first affine response D.

Proof. Fix $p \ge 2(N+1)$, taking $p > 2$ in the case $N = 1$. The principal Beltrami solution at prefix s is $F_s^\epsilon = \mathrm{id} + \mathcal{C}h_{\epsilon,s}$, with coefficient support in B_{CR} and

$$\|h_{\epsilon,s}\|_p \le C_p|\epsilon\nu|R^{2/p}, \qquad |\epsilon\nu| \le c_p.$$

The Cauchy transform maps $L^p(\mathbb{C})$ to the homogeneous Hölder class with exponent $1-2/p$, by splitting its difference kernel at the separation of the two evaluation points. Thus

$$|F_s^\epsilon(z) - F_s^\epsilon(w)| \le |z-w| + C_p|\epsilon\nu|R^{2/p}|z-w|^{1-2/p}. \tag{TD2}$$

This holds on the closed half-plane, since the Cauchy representation is continuous there.

The old-chart increment hull between s and t is contained in $B(W_s, r_{s,t})$: its centered driving oscillation bounds horizontal extent and its height is at most $2\sqrt{t-s}$. The comparison map F_s^ϵ maps it to the increment hull in the deformed chart. The map agrees on the actual physical hull with the affine deformation, since the fixed cutoff is one on all prefixes. Equation (TD2) and the deterministic driver localization estimate applied to this increment give

$$|U_\epsilon(t) - U_\epsilon(s)| \le 3r_{s,t} + C_p|\epsilon\nu|R^{2/p}r_{s,t}^{1-2/p}. \tag{TD3}$$

No stochastic Markov property is used here.

The rational expression for $m_{\epsilon,t}$ and the uniformly convergent Beurling Neumann series are analytic in the real parameter ϵ on one interval $|\epsilon\nu| < c_p$, uniformly over prefixes in B_R. More

explicitly, their complex analytic extension is obtained by extending the upper and lower reflected rational coefficient formulas with the same complex parameter; reflection symmetry is needed only for real parameters. Their L^p Taylor coefficients are bounded by $R^{2/p}(C_p|\nu|)^j$. The Cauchy transform estimate gives a uniform order-N remainder $C_{N,p}|\nu|^{N+1}|\epsilon|^{N+1}R$ for the matched-prefix driving value.

Choose finite-difference weights $a_1, \dots, a_N$ with $\sum_{k=1}^N a_k k^j = \mathbf{1}_{\{j=1\}}$ for $1 \le j \le N$. They give

$$D_t = \frac{1}{\epsilon}\sum_{k=1}^{N} a_k(U_{k\epsilon}(t) - W_t) + O_N(|\epsilon|^N|\nu|^{N+1}R), \tag{TD4}$$

uniformly in t. Subtract this formula at s and apply (TD3), obtaining

$$|D_t - D_s| \le C_N\left\{\frac{r_{s,t}}{|\epsilon|} + |\nu|R^{2/p}r_{s,t}^{1-2/p} + |\epsilon|^N|\nu|^{N+1}R\right\}. \tag{TD5}$$

For $r_{s,t}/R$ below a sufficiently small constant, choose $|\epsilon\nu| = c_N(r_{s,t}/R)^{1/(N+1)}$ so all the finite-difference parameters remain within the analytic interval. The first and third terms give (TD1); the second is no larger because $2/p \le 1/(N+1)$. For larger $r_{s,t}/R$, (TD1) follows from $\sup|D| \le C|\nu|R$. The case $\nu = 0$ is immediate.

If W is Hölder-α on a fixed horizon with $\alpha < 1/2$, then $r_{s,t} \le C(\omega)(t-s)^\alpha$. Equation (TD1) gives exponent $\alpha N/(N+1)$ for D. Given any $\beta < 1/2$, first choose $\beta < \alpha < 1/2$ and then N sufficiently large. Countable choices of α, N give the simultaneous conclusion. □

Corollary 21.7 (The linear clock response has every temporal exponent below one)**.** *For every $0 < \theta < 2$, the same bounded-hull setup gives*

$$|E_t - E_s| \le C_\theta|\nu|R^{2-\theta}r_{s,t}^\theta. \tag{TD6}$$

If W is Hölder for every exponent below $1/2$, then E is Hölder for every exponent below 1. In particular it has finite q-variation for every $q > 1$, its sums of squared increments tend to zero along every sequence of partitions with mesh tending to zero, and the same is true of mixed sums $\sum \Delta W \Delta E$. Neither bounded variation nor absolute continuity is asserted.

Proof. Choose arbitrarily large $p > 2$, and restrict real parameters to a small interval $|\epsilon\nu| \le \delta_p$ on which the preceding L^p bounds hold. Put $\alpha = 1 - 2/p$. For $r = r_{s,t} \le R$, (TD2) bounds the radius of the transformed incremental hull by $C_pR(r/R)^\alpha$. Its half-plane capacity is at most the square of this radius. Consequently, uniformly in those real parameters,

$$0 \le s_\epsilon(t) - s_\epsilon(s) \le C_pR^2(r/R)^{2\alpha}.$$

For capacity, the holomorphic parameter argument bounds the uniform order-N Taylor remainder by

$$C_{N,p}|\epsilon|^{N+1}|\nu|^{N+1}R^2.$$

Apply the finite-difference formula for its first parameter derivative at both s and t. For $a = |\epsilon\nu|$ small enough to accommodate all sample points this yields

$$|E_t - E_s| \le C_{N,p}|\nu|R^2\{(r/R)^{2\alpha}/a + a^N\}.$$

Optimize a proportional to $(r/R)^{2\alpha/(N+1)}$, capping it at the allowed parameter radius. Small r/R gives exponent $2\alpha N/(N+1)$; larger r/R uses the uniform bound on E. Choose p, N so this exponent

exceeds the prescribed θ. This proves (TD6). The temporal conclusion follows exactly as for D, using $r_{s,t} \le C(\omega)(t-s)^\gamma$ for each $\gamma < 1/2$.

For the final assertions choose a temporal Hölder exponent for E greater than $1/q$ to obtain finite q-variation, or greater than $1/2$ to obtain vanishing squared-increment sums. For mixed sums, choose exponents $\gamma < 1/2$ for W and $\beta < 1$ for E with $\gamma + \beta > 1$; the absolute mixed sum is bounded by a constant times $T\,\mathrm{mesh}(\pi)^{\gamma+\beta-1}$. □

This is temporal path regularity of the linear affine response under the original-prefix clock. It proves neither finite signed variation nor semimartingality, and does not justify differentiating the own-capacity wrong-coordinate driver with respect to anisotropy.

A useful check. For the deterministic vertical slit driven by $W_t = 0$, the pure-shear direction $\nu = i$ gives the response $16\sqrt{t}/\pi$; this agrees with the derivative of the standard square-root driver of a slanted straight ray. Indeed the straight-ray driver $c\sqrt{s}$ has ray angle $\theta(c) = \frac{\pi}{2}(1 - c/\sqrt{c^2+16})$, so $\theta'(0) = -\pi/8$, whereas the shear changes the vertical ray angle by $-2\epsilon + O(\epsilon^2)$. This checks the sign and constant geometrically. For a pure vertical stretch, symmetry gives zero driver response and the geometric capacity derivative is $-4t$.

Indexing warning. The derivative in (AV5) compares the same original ordered prefix. It is not the derivative at fixed numerical capacity in the transformed curve. Inverting $s_\epsilon(t)$ changes the time argument of the original rough driver. Even a differentiable clock perturbation can destroy differentiability in ϵ for a Brownian path. No fixed-capacity pathwise first variation is claimed.

A model-independent consequence of the crossing radius bound. For a continuum family satisfying the conditional crossing assumptions used earlier, the filled hull during a restarted capacity interval of length h lies in a disk of radius R_h with $\mathbb{P}(R_h > x\sqrt{h} \mid \mathcal{F}_\tau) \le Ce^{-cx}$, after changing constants. Apply (AV6) to the restarted half-plane hull. The affine response of its endpoint driver has conditional moments $\mathbb{E}(|\delta U_h|^p \mid \mathcal{F}_\tau) \le C_p|\nu|^p h^{p/2}$; the capacity response has moments bounded by $C_p|\nu|^p h^p$. This concerns affine perturbation of the *restarted chart hull*, not a fixed physical affine perturbation of a rough slit domain. The latter is encoded by (AB2), rather than a constant chart coefficient. Neither estimate establishes temporal semimartingality of the driver or of its first variation.

Literature boundary. Sung–Wang [21, Theorem 1.1], treats a simple chord and deformation supported away from it. The proof above, using the principal-solution method of Ahlfors–Bers [20], instead keeps the bounded filled hull fixed, pulls a compactly supported smooth physical deformation into its uniformizing half-plane, and uses the principal Beltrami solution. It does not use a theorem about simple approximants or omit swallowed positive-area sets from a physical-domain change of variables.

22 Mean affine response through a reverse Loewner flow

Let $W_t = \sqrt{\kappa}B_t$, $\kappa > 0$, be the driver of ordinary chordal SLE, and let $f_t = g_t^{-1}$. The matched-prefix affine driver response in direction ν is

$$D_t = -\frac{2}{\pi}\,\mathrm{Re}\,\nu I_t, \qquad I_t = \int_{\mathbb{H}} \frac{\overline{f_t'(W_t+z)}/f_t'(W_t+z) - 1}{z}\,dA(z).$$

The same original hull prefix, rather than the same deformed capacity, is held fixed.

Absolute integrability and Fubini. For $K_t \subset B_R$, the inverse map is hydrodynamically univalent and $f_t'(z) = 1 + O(R^2/|z|^2)$ uniformly for $|z| > CR$. Since $|W_t| \le 3R$,

$$\int_{\mathbb{H}} \frac{|\overline{f_t'(W_t+z)}/f_t'(W_t+z) - 1|}{|z|}\, dA(z) \le CR.$$

Indeed the near part is bounded by $2/|z|$ over a disk of radius CR, and the far part by $CR^2/|z|^3$. For Brownian driving we can take $R = \sup_{s\le t}|W_s| + 2\sqrt{t}$, so $\mathbb{E}R \le C_\kappa\sqrt{t}$. Thus expectation and the displayed integral may be interchanged. This uses neither a simple trace nor a zero-area hull.

Reverse-flow representation. For fixed t and $0 \le r \le t$, put

$$H_r(z) = g_{t-r}(f_t(z + W_t)) - W_{t-r}.$$

Then $H_0(z) = z$, $H_t(z) = f_t(z + W_t)$, and pathwise the reverse Loewner equation is driven by $W_t - W_{t-r}$. This reversed increment process has Brownian law. Hence the derivative field at time t has the same law as the spatial derivative of

$$dh_r(z) = -\frac{2}{h_r(z)}dr + \sqrt{\kappa}\, d\widetilde{B}_r, \qquad h_0(z) = z.$$

The sign of the Brownian term does not affect its law. Its spatial derivative satisfies $\partial_r h_r' = 2h_r'/h_r^2$. Therefore

$$\frac{\overline{h_t'(z)}}{h_t'(z)} = \exp\left\{\int_0^t 2\left(\frac{1}{\overline{h_r(z)}^2} - \frac{1}{h_r(z)^2}\right) dr\right\}.$$

For fixed initial height $y > 0$, the reverse image height increases, so all the coefficients are bounded away from their boundary singularity on finite time intervals.

Define

$$u(t,z) = \mathbb{E}\frac{\overline{h_t'(z)}}{h_t'(z)}.$$

It is the bounded Feynman–Kac solution, in the stochastic semigroup sense, of

$$\partial_t u = \mathcal{L}u + qu, \quad u(0,z) = 1,$$

where

$$\mathcal{L} = \frac{\kappa}{2}\partial_x^2 - \frac{2x}{x^2+y^2}\partial_x + \frac{2y}{x^2+y^2}\partial_y, \qquad q(z) = \frac{8ixy}{(x^2+y^2)^2}.$$

No boundary condition at $y = 0$ is inserted: the stochastic representation from each interior starting point specifies the solution. Uniqueness among bounded classical solutions, or within the stated stochastic semigroup class, follows by stopped Itô/Feynman–Kac on $y \ge y_0 > 0$ and localization in x, since the phase potential is purely imaginary. Consequently

$$\mathbb{E}D_t = -\frac{2}{\pi}\,\mathrm{Re}\left\{\nu\int_{\mathbb{H}} \frac{u(t,z)-1}{z}\, dA(z)\right\}.$$

The integral is absolutely convergent by the preceding Fubini bound.

Symmetry and scaling. Reflection of Brownian driving gives $u(t,-\bar{z}) = \overline{u(t,z)}$, so the last integral is purely imaginary. Brownian scaling makes it $iJ_\kappa\sqrt{t}$ for a real constant J_κ. Thus

$$\mathbb{E}D_t = c_\kappa \operatorname{Im}(\nu)\sqrt{t}, \qquad c_\kappa = 2J_\kappa/\pi.$$

The first-order mean vertical-stretch response vanishes, whereas a shear response is allowed by symmetry. Determining whether c_6 is zero, and more generally computing c_κ, is a concrete next model task. This calculation does not determine temporal semimartingality of D or of the finite-affine wrong-coordinate driver.

An interior cancellation, not a boundary conclusion. Let $\mathcal{L}^*$ denote the formal transpose under the bilinear integral pairing, so the multiplication potential remains q, not its complex conjugate. Direct differentiation gives the pointwise identity

$$(\mathcal{L}^* + q)\frac{1}{z} = \frac{\kappa - 6}{z^3}, \qquad z \in \mathbb{H}.$$

At $\kappa = 6$ the interior term vanishes. This does not justify integrating by parts over the full half-plane: the test $1/z$ is singular at the boundary and has a noncompact tail. Boundary and infinity fluxes have not been controlled. In particular no vanishing or nonvanishing of c_6 is deduced. Studying those fluxes is a precise next question for the affine-SLE_6 model.

23 Coordinate-compatible exit clocks before semimartingality

The upper and lower conditional oscillation bounds already proved for the continuum family have a stronger clock consequence. It is compatible with working in a wrong affine structure: it assumes neither a signed budget nor a canonical semimartingale decomposition. All constants may depend on the fixed affine distortion through the crossing constants.

Let X be a continuous adapted process on $[0,\infty)$ in a usual filtration. Suppose constants $C, c, \delta, p > 0$, with $0 < p < 1$, satisfy, for every finite stopping time τ, every $h > 0$, and every $x \geq 0$,

$$\mathbb{P}\{\sup_{u \leq h} |X_{\tau+u} - X_\tau| > x\sqrt{h} \mid \mathcal{F}_\tau\} \leq Ce^{-cx}, \tag{3}$$

$$\mathbb{P}\{\sup_{u \leq h} |X_{\tau+u} - X_\tau| > \delta\sqrt{h} \mid \mathcal{F}_\tau\} \geq p. \tag{4}$$

For the current exploration assumptions, these follow from the upper crossing estimate, corrected ray-deletion locality, and strong DMP. The rough-state admissibility qualifications of those results remain in force. Physical translation covariance is not needed for this deduction.

Proposition 23.1 (Uniform Brownian-order exit-time control)**.** *For $r > 0$, put*

$$\xi_r(\tau) = \inf\{u \geq 0 : |X_{\tau+u} - X_\tau| \geq r\}.$$

There are constants $a_i > 0$, depending only on the input constants, such that, conditionally on $\mathcal{F}_\tau$,

$$\mathbb{P}\{\xi_r(\tau) \leq h \mid \mathcal{F}_\tau\} \leq C\exp(-cr/\sqrt{h}), \tag{5}$$

$$\mathbb{P}\{\xi_r(\tau) > h \mid \mathcal{F}_\tau\} \leq (1-p)^{\lfloor \delta^2 h/(4r^2)\rfloor}, \tag{6}$$

$$a_1 r^2 \leq \mathbb{E}[\xi_r(\tau) \mid \mathcal{F}_\tau] \leq a_2 r^2, \qquad \mathbb{E}[\exp(a_3\xi_r(\tau)/r^2) \mid \mathcal{F}_\tau] \leq a_4. \tag{7}$$

In particular, exits occur in finite time almost surely.

Proof. For the fast bound, the exit event implies that the supremum on $[0,h]$ is at least r. Apply (3) first at thresholds $r-\varepsilon$ and then let $\varepsilon \downarrow 0$. For the slow bound, split time into blocks of length dr^2, where $d = 4/\delta^2$. If the exit has not occurred by the end of a block, both the path and its value at the start of that block lie in the original interval $[X_\tau - r, X_\tau + r]$. Their displacement throughout the block is at most $2r = \delta\sqrt{dr^2}$. The conditional lower bound at the block's starting stopping time therefore bounds the conditional survival probability by $1-p$. Iteration proves (6). Integrating that tail gives the upper mean and an exponential moment. Choose $a > 0$ small enough that $Ce^{-c/\sqrt{a}} \le 1/2$. The fast bound gives $\mathbb{P}\{\xi_r(\tau) > ar^2 \mid \mathcal{F}_\tau\} \ge 1/2$, hence the lower mean bound $ar^2/2$. □

Proposition 23.2 (Adaptive quadratic activity is nondegenerate)**.** *Define $\sigma_0^r = 0$ and successive exits by*

$$\sigma_{k+1}^r = \inf\{t \ge \sigma_k^r : |X_t - X_{\sigma_k^r}| \ge r\}, \qquad \nu_r = r^2 \sum_{k\ge 1} \delta_{\sigma_k^r}.$$

The exit times do not accumulate on a finite interval. There are fixed $0 < a < b < \infty$ and $A, \alpha > 0$ such that for every deterministic interval $I = [s,t]$, $h = t - s > 0$,

$$\mathbb{P}\{\nu_r(I) < ah \text{ or } \nu_r(I) > bh\} \le A\exp(-\alpha h/r^2).$$

Consequently, on one probability-one event, the measures $\nu_{2^{-m}}$ are relatively compact on every finite interval, and every subsequential weak limit ν satisfies

$$a\,dt \le d\nu \le b\,dt$$

(after decreasing a and increasing b if necessary). These measures are realized squared increments on the corresponding exit partitions, not a claimed canonical quadratic variation.

Proof. Finite-time nonaccumulation follows from continuity: infinitely many oscillations of size r on a compact interval contradict uniform continuity. Every exit time is finite by the preceding proposition, and every completed exit increment has square exactly r^2.

Write $D_k = (\sigma_{k+1}^r - \sigma_k^r)/r^2$. For some $\lambda > 0$ and $K < \infty$, the preceding proposition gives

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{\sigma_k^r}] \le K.$$

Also its fast-tail proof supplies $a_0 > 0$ with $\mathbb{P}\{D_k > a_0 \mid \mathcal{F}_{\sigma_k^r}\} \ge 1/2$. Thus, for every $\eta > 0$,

$$\mathbb{E}[e^{-\eta D_k} \mid \mathcal{F}_{\sigma_k^r}] \le \rho := \tfrac{1}{2}(1 + e^{-\eta a_0}) < 1.$$

Iteration and exponential Markov inequalities give, for sums of n successive full exit durations, the bounds

$$\mathbb{P}\{\textstyle\sum_{j=1}^n D_j > H \mid \text{initial past}\} \le K^n e^{-\lambda H}, \qquad \mathbb{P}\{\sum_{j=1}^n D_j \le H \mid \text{initial past}\} \le \rho^n e^{\eta H}.$$

No independence of the durations is used.

To count exits in a fixed interval starting at s, first wait for the first global exit $\theta \ge s$. The residual time $R = \theta - s$ also has a uniform conditional exponential moment at scale r^2. Indeed, before θ, the process remains within r of the previous exit value, hence within $2r$ of X_s. Repeating the slow-tail proof with that larger confinement band gives $\mathbb{P}\{R > u \mid \mathcal{F}_s\} \le (1-p)^{\lfloor \delta^2 u/(16r^2)\rfloor}$. Consequently for a sufficiently small common λ, $\mathbb{E}[e^{\lambda R/r^2} \mid \mathcal{F}_s] \le K_0$. If an interval has at most n

exits, its residual duration plus n full durations is at least h. If it has at least $n+1$ exits, n full durations after its first exit fit inside h. Thus

$$\mathbb{P}\{\nu_r(I)/r^2 \le n\} \le K_0 K^n e^{-\lambda h/r^2}, \qquad \mathbb{P}\{\nu_r(I)/r^2 \ge n+1\} \le \rho^n e^{\eta h/r^2}.$$

Choose n respectively as a sufficiently small and sufficiently large multiple of h/r^2. Rounding and bounded values of h/r^2 are absorbed in A. This proves (23.2).

For dyadic r, the right side of that estimate is summable for every fixed rational interval. Borel–Cantelli and a countable intersection give eventual upper and lower mass bounds on every rational interval. The upper bounds give relative compactness of finite measures on compact intervals. For any weak subsequential limit, use rational open intervals for the upper mass bound and rational closed intervals for the lower bound, applying the Portmanteau inequalities. Approximation by rational endpoints gives (23.2); in particular the limits have no atoms. Finite endpoint conventions in the counts do not affect the conclusion, since each atom of ν_r has mass r^2. □

What is gained and what is still missing. The continuum axioms already supply nonzero finite quadratic activity along observable stopping-time partitions, with subsequential activity clocks comparable to capacity. This is stronger information than merely stating a variation index. It does not establish uniqueness of the clock, convergence along deterministic partitions, semimartingality, or variance six. Even uniqueness of a quadratic-activity clock could not by itself establish semimartingality: the earlier Brownian-plus-infinite-variation Hölder example remains a warning about missing signed cancellation.

At exit times the displacement is $\varepsilon_k r$, with $\varepsilon_k \in \{-1, 1\}$. The remaining observable signed quantity is

$$r \sum_{k<N} \left| \mathbb{E}[\varepsilon_k \mid \mathcal{F}_{\sigma_k^r}] \right|.$$

The new clock estimate controls the number of summands by order T/r^2; it does not improve the resulting crude bound of order T/r on this signed budget. A useful next question is how the predictable exit-sign bias changes under a candidate constant affine normalization, or whether locality yields cancellations for a corrected observable depending on the transported Beltrami field. Finite signed variation of the raw driver in every affine coordinate system is not being assumed or recommended.

24 Affine capacity clocks and the limits of scaling arguments

24.1 Global affine capacity comparison does not control clock increments

The affine clock at least has a deterministic global comparison. This uses only the geometric capacity theorem of Lalley–Lawler–Narayanan [19, Theorem 1], not any regularity of the trace or driver.

Proposition 24.1 (Capacity distortion by a fixed affine map)**.** *Let $L(x+iy) = \alpha x + \beta y + i\delta y$, where $\alpha, \delta > 0$ and $\beta \in \mathbb{R}$. There are constants $0 < c_L \le C_L < \infty$ such that every bounded half-plane hull K satisfies*

$$c_L \operatorname{hcap}(K) \le \operatorname{hcap}(LK) \le C_L \operatorname{hcap}(K). \tag{GC1}$$

Consequently, if $\operatorname{hcap}(K_t) = 2t$, its affine clock $s(t) = \operatorname{hcap}(LK_t)/2$ satisfies $c_L t \le s(t) \le C_L t$. The constants can be chosen uniformly when L ranges over a compact set of invertible half-plane-preserving linear maps.

Proof. For a family of disks $\{B_j\}$ with bounded radii, the elementary Vitali selection argument gives, for every $\lambda \geq 1$,

$$\left|\bigcup_j \lambda B_j\right| \leq (2\lambda+3)^2 \left|\bigcup_j B_j\right|. \tag{GC2}$$

Indeed one can select disjoint disks such that each original disk meets a selected disk of at least half its radius. Its λ-dilate then lies in the $(2\lambda+3)$-dilate of that selected disk. Sum their areas; arbitrary families reduce to countable subfamilies of the open union.

Let m, M be the smallest and largest singular values of L, and $d = \det L$. Put

$$H(K) = \bigcup_{z\in K} B(z, \operatorname{Im} z), \qquad \operatorname{hsiz}(K) = |H(K)|.$$

For $z = x + iy$, the ellipse $LB(z, y)$ contains $B(Lz, my)$ and is contained in $B(Lz, My)$, whereas the corresponding disk in $H(LK)$ is $B(Lz, \delta y)$. Since $m \leq \delta \leq M$, (GC2) yields

$$\frac{d}{(2M/\delta+3)^2} \operatorname{hsiz}(K) \leq \operatorname{hsiz}(LK) \leq d(2\delta/m+3)^2 \operatorname{hsiz}(K). \tag{GC3}$$

The cited theorem gives universal positive a, b with $a \operatorname{hsiz}(K) \leq \operatorname{hcap}(K) \leq b \operatorname{hsiz}(K)$. Apply it to both K and LK. This proves (GC1), the clock assertion, and local uniformity in L. □

This is a comparison from the original source time zero, not an increment estimate. After mapping out K_t, the transported affine map is quasiconformal with a history-dependent field; it is no longer the original linear map. Applying (GC1) to that increment would be invalid. Even for increasing continuous functions, $s(t) \asymp t$ does not imply absolute continuity: $s(t) = t + \epsilon C(t)$, with C a Cantor staircase supported on $[1, 2]$, has a singular part and still satisfies such a comparison. Determining whether the actual affine-SLE clock has a singular component remains open here.

24.2 A necessary clock regularity condition for wrong-coordinate semimartingality

Let η_s be chordal SLE_6 in $\mathbb{H}$, parametrized by $\operatorname{hcap}(K_s) = 2s$, and let B_s be its ordinary Brownian driver divided by $\sqrt{6}$. Let L be an orientation-preserving real-linear self-map of $\mathbb{H}$. Put

$$\tau(s) = \tfrac{1}{2} \operatorname{hcap}(LK_s), \qquad \gamma_{\tau(s)} = L\eta_s,$$

and denote the ordinary Loewner driver of γ by W_t. We assume continuous Loewner generation, strictly growing hulls, and infinite lifetimes for both parametrizations. Thus τ and its inverse are continuous increasing clocks. Completed natural filtrations are used.

Proposition 24.2 (Semimartingality forces absolute continuity of the affine clock)**.** *Suppose $W = M + A$ is a continuous semimartingale in its natural filtration and its martingale bracket $C = \langle M\rangle$ satisfies*

$$c\,dt \leq dC_t \leq C_0\,dt \tag{AC1}$$

on the relevant localized intervals, with deterministic constants $0 < c \leq C_0 < \infty$. Then τ is absolutely continuous in s. There exists a Brownian-predictable H such that, locally,

$$M_{\tau(s)} = \int_0^s H_u\,dB_u, \qquad c\,d\tau(s) \leq H_s^2 ds \leq C_0\,d\tau(s). \tag{AC2}$$

In particular, a nonzero singular component of $d\tau$ would exclude semimartingality of W, whenever the crossing/locality hypotheses needed for (AC1) hold.

Proof. An initial segment of W reconstructs the corresponding ordered Loewner trace and hull. Applying L^{-1} reconstructs the original SLE segment, its original capacity, and its driver. Conversely the Brownian driver up to s reconstructs the transformed segment and its driver up to $\tau(s)$. Thus, with the usual completions,

$$\mathcal{F}^W_{\tau(s)} = \mathcal{F}^B_s. \tag{AC3}$$

This assertion retains the ordered trace and distinguished tip; an unlabelled hull or frontier alone is not substituted for that state.

For every fixed s, $\tau(s)$ is a stopping time in the W filtration: at observed time t one can reconstruct the original capacity $\tau^{-1}(t)$, and $\{\tau(s) \le t\} = \{\tau^{-1}(t) \ge s\}$. Consequently $N_s = M_{\tau(s)}$ is a continuous local martingale in the time-changed filtration, which is the Brownian filtration by (AC3). Localize both clocks and the martingale in the usual way. The Brownian martingale representation theorem gives $N_s = \int_0^s H_u\, dB_u$, and therefore $C_{\tau(s)} = \int_0^s H_u^2 du$.

The measure inequality (AC1), applied pathwise to each clock interval, gives

$$c\, d\tau \le d(C \circ \tau) \le C_0\, d\tau.$$

Its middle measure equals $H_s^2 ds$. The lower inequality forces $d\tau \ll ds$ and proves all assertions. □

The already established upper conditional driving moment bound yields the upper inequality in (AC1) once semimartingality is assumed. The ray-deletion locality lower tail yields the lower inequality. Thus (AC1) uses precisely those explicit continuum crossing and prime-end admissibility hypotheses; it is not a consequence of semimartingality alone. No singularity of the actual affine clock has been proved here. Absolute continuity of τ is only necessary, not sufficient for semimartingality. It also does not imply absolute continuity of its inverse: τ' might vanish on a positive-measure set. Global comparisons $c_L s \le \tau(s) \le C_L s$ do not remove this possibility or give an increment comparison.

24.3 Why scaling and a trivial germ do not regularize the bracket

Here is a process-level obstruction to strengthening the previous scaling lemma without its initial-bracket premise. It is not a proposed exploration family satisfying locality and DMP.

Proposition 24.3 (A self-similar elliptic martingale without an initial bracket tangent)**.** *Let B be standard Brownian motion and define*

$$X_t = \int_0^t \sqrt{2 + \sin(B_s/\sqrt{s})}\, dB_s, \qquad t \ge 0. \tag{AC4}$$

The integrand at zero may be chosen arbitrarily. Then X is a continuous martingale in its own natural filtration, its initial germ is trivial, and it has exact Brownian scaling. Its bracket satisfies

$$dt \le d\langle X\rangle_t \le 3dt. \tag{AC5}$$

Nevertheless $\langle X\rangle_t/t$ has no limit in probability as $t \downarrow 0$. It satisfies conditional Brownian-scale upper oscillation tails and a positive conditional lower oscillation tail uniformly over stopping times.

Proof. The integrand is bounded and progressive, so the integral is a square-integrable martingale on each finite horizon. A martingale remains a martingale in its own smaller natural filtration; its bracket is the pathwise quadratic variation and hence remains

$$Q_t = \int_0^t [2 + \sin(B_s/\sqrt{s})] ds.$$

This proves (AC5). The natural X filtration is contained in the Brownian filtration, so its completed initial germ is trivial. For every $a > 0$, substitution $\widetilde{B}_u = B_{au}/\sqrt{a}$ in the stochastic integral proves $(a^{-1/2}X_{at})_{t\geq 0} \overset{\text{law}}{=} (X_t)_{t\geq 0}$. The same substitution shows Q_t/t has the fixed distribution of Q_1.

This distribution is nondegenerate. Fix small $\epsilon > 0$. On $[\epsilon, 1]$, Brownian paths have positive probability of lying in a sufficiently small uniform neighborhood of either continuous function $s \mapsto \pi\sqrt{s}/2$ or its negative. This follows from Brownian full support after extending each function continuously to zero. On these two events the integral of the sine over $[\epsilon, 1]$ is close respectively to $1 - \epsilon$ and $-(1 - \epsilon)$, whereas its contribution on $[0, \epsilon]$ has absolute value at most ϵ. Thus Q_1 is nonconstant.

If Q_t/t converged in probability at zero, an almost surely converging subsequence would make its limit measurable in the Brownian initial germ. The limit would be deterministic, contradicting the fixed nondegenerate distribution. Hence there is no such limit.

For completeness, the upper bracket bound and the exponential martingale inequality give conditional Gaussian upper tails for $\sup_{u\leq h} |X_{\sigma+u} - X_\sigma|/\sqrt{h}$ at every stopping time σ. For the lower tail, restart the martingale at σ and apply Dambis–Dubins–Schwarz. Its bracket over $[\sigma, \sigma + h]$ is at least h, so its maximum dominates the maximum of a Brownian motion over $[0, h]$. This gives any fixed positive Brownian lower-tail probability, conditionally on the past. Passing to the natural X filtration preserves both bounds by conditional expectation. □

This example rules out the shortcut "exact scaling + Brownian germ triviality + bounded elliptic bracket implies Brownian driving." It does not exclude the possibility that the actual wrong-coordinate affine-SLE driver is a semimartingale whose bracket density oscillates on every logarithmic scale near zero.

25 Finite-mesh locality characteristics before semimartingality

The existing characteristic transformation rules presuppose a semimartingale. The conditional crossing moments already allow a finite-mesh version with an error tending to zero in total expected variation. This identifies the signed quantities to study without first choosing a preferred affine structure or asserting a canonical quadratic variation.

25.1 Compactness of realized squared increments

Proposition 25.1 (Bounded cluster clocks on deterministic meshes)**.** *Let X be continuous and adapted on $[0, T]$ in a usual filtration and suppose, for deterministic $s < t$, that*

$$\mathbb{E}[|X_t - X_s|^2 \mid \mathcal{F}_s] \leq K_2(t - s), \qquad \mathbb{E}[|X_t - X_s|^4 \mid \mathcal{F}_s] \leq K_4(t - s)^2. \tag{FM1}$$

For a deterministic partition $\pi = (t_i)$ write

$$a_i = \mathbb{E}[(X_{t_{i+1}} - X_{t_i})^2 \mid \mathcal{F}_{t_i}].$$

Let Q^π be the continuous piecewise linear function starting at zero with increment a_i on $[t_i, t_{i+1}]$, and let

$$R^\pi(t) = \sum_{t_{i+1}\leq t} (X_{t_{i+1}} - X_{t_i})^2.$$

Every Q^π is increasing and K_2-Lipschitz. There is an absolute constant C such that

$$\mathbb{E} \sup_{t\leq T} |R^\pi(t) - Q^\pi(t)|^2 \leq CK_4T\,|\pi| + CK_2^2|\pi|^2, \tag{FM2}$$

where $|\pi|$ denotes the largest mesh interval. In particular the realized square-sum processes are asymptotically tight with continuous increasing cluster limits whose densities are bounded by K_2. For every fixed mesh sequence with $\sum_n |\pi_n| < \infty$, almost surely their uniform cluster set is nonempty and every cluster path is increasing and K_2-Lipschitz. Neither uniqueness nor a positive lower density is asserted.

Proof. The slope claim follows from $0 \le a_i \le K_2(t_{i+1} - t_i)$. At partition endpoints the difference is the discrete martingale

$$Z_j = \sum_{i<j} \{(X_{t_{i+1}} - X_{t_i})^2 - a_i\}.$$

Conditional centering, orthogonality, and (FM1) give $\mathbb{E} Z_{\#\pi}^2 \le K_4 \sum_i (t_{i+1} - t_i)^2 \le K_4 T |\pi|$. Doob's inequality controls the maximum at grid points. Between grid points R^π is constant and Q^π moves by at most $K_2|\pi|$, which proves (FM2). The Q^π lie in one compact subset of $C([0,T])$ by Arzela–Ascoli. Equation (FM2) proves tightness and the assertion about cluster laws. For summable mesh sizes, Markov's inequality and Borel–Cantelli give $\|R^{\pi_n} - Q^{\pi_n}\|_\infty \to 0$ almost surely, hence the pathwise cluster assertion. A cluster selection need not produce one canonical adapted clock, and no such selection is claimed. □

The conditional exponential bounds from the crossing estimate imply (FM1), also for the bounded stopped drivers used in (QV1). Thus this is a further regularity consequence available before any signed budget is proved. It concerns cluster limits along deterministic meshes, whereas the earlier exit-clock theorem also supplies a lower density for its different, adaptive partitions. These facts do not identify the two kinds of cluster clock.

25.2 The variance-six anomaly has a finite-mesh form

Use a locality deletion map h_t on a common physical trace before contact, as in (LB1). Stop at a matched trace time $\theta \le T$ so that the driver stays in a compact real interval, h_t is analytic on a fixed larger neighborhood of that interval, and its spatial derivatives through order three, and the time derivatives needed below through spatial order two, are bounded by deterministic constants. Also require $0 < m \le h_t'(W_t) \le P$. Such bounds are localization conditions on a deletion map strictly away from contact. Write

$$X_t = W_{t\wedge\theta}, \quad Y_t = h_{t\wedge\theta}(X_t) - h_0(W_0), \quad A_t = t \wedge \theta, \quad U_t = \int_0^{t\wedge\theta} h_s'(W_s)^2 \, ds. \tag{FM3}$$

Here Y is the deleted driver at the inherited clock U, not at deterministic deleted capacity. Suppose the original driving increments have the conditional supremum moment bounds

$$\mathbb{E}[\sup_{0\le u\le h} |X_{t+u} - X_t|^j \mid \mathcal{F}_t] \le C_j h^{j/2}, \qquad j = 1, \dots, 6, \tag{FM4}$$

for $t + h \le T$. These are supplied by the stopped crossing estimate.

For each deterministic partition interval put

$$\begin{aligned} b_i^X &= \mathbb{E}[\Delta_i X \mid \mathcal{F}_{t_i}], & a_i^X &= \mathbb{E}[(\Delta_i X)^2 \mid \mathcal{F}_{t_i}], \\ b_i^Y &= \mathbb{E}[\Delta_i Y \mid \mathcal{F}_{t_i}], & a_i^Y &= \mathbb{E}[(\Delta_i Y)^2 \mid \mathcal{F}_{t_i}], \\ d_i &= \mathbb{E}[\Delta_i A \mid \mathcal{F}_{t_i}], & e_i &= \mathbb{E}[\Delta_i U \mid \mathcal{F}_{t_i}], \\ p_i &= h'_{t_i\wedge\theta}(X_{t_i}), & q_i &= h''_{t_i\wedge\theta}(X_{t_i}). \end{aligned} \tag{FM5}$$

Proposition 25.2 (Conditional characteristic covariance without Ito calculus)**.** *In this setup there is a deterministic C, depending only on the localized map and moment bounds and T, such that, with $\delta_i = t_{i+1} - t_i$,*

$$\begin{aligned} b_i^Y &= p_i b_i^X + \tfrac{1}{2} q_i (a_i^X - 6d_i) + r_i, \\ a_i^Y &= p_i^2 a_i^X + v_i, \qquad e_i = p_i^2 d_i + w_i, \\ |r_i| + |v_i| + |w_i| &\le C\delta_i^{3/2}. \end{aligned} \tag{FM6}$$

All equalities and bounds hold almost surely. In particular

$$a_i^Y - 6e_i = p_i^2 (a_i^X - 6d_i) + O(\delta_i^{3/2}), \tag{FM7}$$

and the sums of absolute errors in (FM6)–(FM7) are bounded by $CT\sqrt{|\pi|}$. No signed-budget, semimartingale, or canonical quadratic-variation assumption is used.

Proof. Put $S_i = \sup_{u \in [t_i, t_{i+1}]} |X_u - X_{t_i}|$ and $\delta = \delta_i$. Spatial Taylor expansion and time integration of the stopped field give

$$\Delta_i Y = p_i \Delta_i X + \tfrac{1}{2} q_i (\Delta_i X)^2 - 3 q_i \Delta_i A + \mathcal{R}_i, \qquad |\mathcal{R}_i| \le C(S_i^3 + \delta S_i + \delta^2). \tag{FM8}$$

For clarity, the time contribution is $\int_{t_i \wedge \theta}^{t_{i+1} \wedge \theta} (\partial_s h_s)(X_{t_{i+1}}) ds$. The Loewner deletion PDE has the removable boundary value $(\partial_s h_s)(W_s) = -3h_s''(W_s)$. The bounded first spatial derivative of $\partial_s h_s$ permits replacing $X_{t_{i+1}}$ by W_s at cost $C\delta S_i$. The bounded time derivative of h_s'' and bounded h_s''' permit replacing $h_s''(W_s)$ by q_i at cost $C\delta(S_i + \delta)$. This proves (FM8), including intervals straddling θ; after θ everything is frozen. Taking conditional expectations and using (FM4) proves the first line of (FM6).

The simpler pathwise bound $|\Delta_i Y - p_i \Delta_i X| \le C(S_i^2 + \delta)$ follows directly from the bounded field derivatives. Squaring and using (FM4) shows

$$\mathbb{E}[|(\Delta_i Y)^2 - p_i^2 (\Delta_i X)^2| \mid \mathcal{F}_{t_i}] \le C\mathbb{E}[S_i^3 + \delta S_i + S_i^4 + \delta^2 \mid \mathcal{F}_{t_i}] \le C\delta^{3/2},$$

after enlarging C on the fixed horizon. Finally $|h_s'(W_s)^2 - p_i^2| \le C(S_i + \delta)$ while the clock is active. Integration and conditional expectation prove the estimate for e_i. Summing $\delta_i^{3/2} \le \sqrt{|\pi|}\delta_i$ gives the last claim. □

What this does and does not bridge. The pair consisting of predictable signed increments and the variance-six defect has a coordinate transformation rule already at finite resolution. Its total error vanishes on fine deterministic meshes. Once a signed budget is finite this recovers the familiar characteristic-measure calculation, but its derivation here does not presuppose that missing regularity. Since $\sum_i |a_i^X - 6d_i| \le (K_2 + 6)T$, it also explains directly why analytic locality changes can add only a bounded amount to an aggregate signed budget under matched bounded localizations. There is still no estimate forcing that budget to be finite.

The inherited times U_{t_i} are stopping times in the deleted natural trace filtration by geometric reconstruction. Their future increments are random, so (FM6) must not be reinterpreted as a fixed-duration transition identity for the deleted driver. Also the maps here are analytic deletion maps. A fixed physical affine deformation of a rough restarted domain carries a nonconstant transported Beltrami coefficient; it is not covered by replacing h_t with a constant affine chart map.

26 Exit signs give exact signed-regularity criteria

The adaptive exits used to measure unsigned activity also suffice to test the missing signed regularity. This gives a concrete grading of the remaining gap, without selecting an affine structure or assuming a semimartingale decomposition. The criteria below are implications and equivalences for processes; no exit-sign estimate is deduced from the curve axioms in this section.

Let W be a continuous adapted process in a usual filtration, with $W_0 = 0$, and suppose its radius-r exits are finite almost surely. Put

$$\sigma_0^r = 0, \qquad \sigma_{k+1}^r = \inf\{t \geq \sigma_k^r : |W_t - W_{\sigma_k^r}| \geq r\}, \qquad G_k^r = \mathcal{F}_{\sigma_k^r}.$$

Write

$$\beta_k^r = \mathbb{E}\left[\frac{W_{\sigma_{k+1}^r} - W_{\sigma_k^r}}{r}\,\middle|\, G_k^r\right], \qquad b_k^r = r\beta_k^r. \tag{EX1}$$

Every displacement divided by r is $+1$ or -1. Fix $L, T > 0$ and

$$\tau = T \wedge \inf\{t : |W_t| \geq L\}, \qquad X_t = W_{t\wedge\tau}, \quad 0 \leq t \leq T.$$

Let $\mathcal{V}(X)$ be the conditional variation over deterministic time partitions, as in (QV4), and define the pure exit-sign budget

$$\mathcal{B}_r = \mathbb{E} \sum_{k:\, \sigma_k^r < \tau} |b_k^r|. \tag{EX2}$$

Proposition 26.1 (One family of exit skeletons tests the whole signed budget)**.** *One has*

$$\mathcal{V}(X) \leq \liminf_{r\downarrow 0} \mathcal{B}_r. \tag{EX3}$$

Thus $\liminf_{r\downarrow 0} \mathcal{B}_r < \infty$ *is equivalent to* $X = M + A$ *with* M *a uniformly integrable continuous martingale and* A *predictable continuous finite variation with* $\mathbb{E}\operatorname{Var} A < \infty$*. In that case*

$$\lim_{r\downarrow 0} \mathcal{B}_r = \mathcal{V}(X) = \mathbb{E}\operatorname{Var}_{[0,T]} A. \tag{EX4}$$

In particular one may test finiteness along any deterministic sequence of radii decreasing to zero. No uniqueness of an unsigned activity clock is required.

Proof. For a deterministic partition $0 = t_0 < \cdots < t_n = T$, let H_j be $\mathcal{F}_{t_j}$-measurable and bounded by one. Set $u_j = t_j \wedge \tau$ and $K_j = \inf\{k : \sigma_k^r \geq u_j\}$. Each K_j is a stopping index for (G_k^r), since $\{K_j \leq k\} = \{u_j \leq \sigma_k^r\}$. The active variable $H_j \mathbf{1}_{\{t_j < \tau\}}$ is measurable in $\mathcal{F}_{u_j}$ and hence in $G_{K_j}^r$. Consequently

$$L_k = \sum_{j=0}^{n-1} H_j \mathbf{1}_{\{t_j<\tau\}} \mathbf{1}_{\{K_j \leq k < K_{j+1}\}}$$

is G_k^r-measurable, bounded by one, and supported on $\sigma_k^r < \tau$. Thus the expected discrete gain is at most $\mathcal{B}_r$:

$$\mathbb{E}\sum_{j=0}^{n-1} H_j \mathbf{1}_{\{t_j<\tau\}} (W_{\sigma_{K_{j+1}}^r} - W_{\sigma_{K_j}^r}) = \mathbb{E}\sum_k L_k b_k^r \leq \mathcal{B}_r. \tag{EX5}$$

For clarity the equality follows first after truncating the sum at a deterministic index. The truncated gain telescopes into at most n differences of driver values prior to the first grid exit after τ; all such values lie in $[-L - r, L + r]$. Dominated convergence applies to the gains, and absolute summability

applies to the predictable terms if $\mathcal{B}_r < \infty$. If that budget is infinite the desired upper bound is automatic. This avoids an unjustified optional sampling at the last completed exit before a deterministic time.

The value at the first grid exit after u_j is within $2r$ of W_{u_j}. Hence (EX5) differs from $\mathbb{E}\sum_j H_j(X_{t_{j+1}} - X_{t_j})$ by at most $4nr$. Taking $r \downarrow 0$ and then the supremum over the predictable variables and partitions proves (EX3).

If the liminf is finite, the bounded continuous quasimartingale theorem and Proposition 17.2 give the asserted decomposition and $\mathcal{V}(X) = \mathbb{E}\operatorname{Var} A$. For the converse set, on $\{\sigma_k^r < \tau\}$,

$$\widetilde{b}_k^r = \mathbb{E}[W_{\sigma_{k+1}^r \wedge \tau} - W_{\sigma_k^r} \mid G_k^r].$$

Optional sampling of the uniformly integrable stopped martingale gives

$$\mathbb{E}\sum_{\sigma_k^r < \tau} |\widetilde{b}_k^r| \le \mathbb{E}\operatorname{Var} A.$$

There is at most one excursion straddling τ, and its displacement between τ and its full exit is at most $2r$. Since $\{\sigma_k^r < \tau\}$ is G_k^r-measurable, conditional expectation and summation give the precise terminal-error bound

$$\mathbb{E}\sum_{\sigma_k^r < \tau} |b_k^r - \widetilde{b}_k^r| \le 2r. \tag{EX6}$$

Therefore $\mathcal{B}_r \le \mathbb{E}\operatorname{Var} A + 2r$. Combining this with (EX3) proves (EX4). □

Additional hypotheses for the energy criterion. Assume now the uniform diffusive exit estimates from Proposition 23.1. In particular, for constants $a_1, a_2 > 0$ independent of k, r,

$$a_1 r^2 \le d_k^r := \mathbb{E}[\sigma_{k+1}^r - \sigma_k^r \mid G_k^r] \le a_2 r^2. \tag{EX7}$$

The same estimates give exponential residual-exit tails after any bounded stopping time: before the next global exit, the process remains within $2r$ of its current position. The block proof of Proposition 23.1 therefore gives

$$\mathbb{E}[\sigma_{K(\theta)}^r - \theta \mid \mathcal{F}_\theta] \le Cr^2, \qquad K(\theta) = \inf\{k : \sigma_k^r \ge \theta\}. \tag{EX8}$$

The original conditional lower oscillation estimate also excludes all constant nontrivial time intervals almost surely: subdivide any fixed rational interval into n pieces, use the conditional bound on each piece, and let $n \to \infty$. Continuity then implies that the maximum length of an exit-grid interval intersecting a fixed compact time interval tends to zero as $r \downarrow 0$. Otherwise converging endpoints of intervals of length bounded below would exhibit a constant interval.

Proposition 26.2 (Exit-sign energy tests square-integrable drift)**.** *Define*

$$\mathcal{E}_r = \mathbb{E}\sum_{\sigma_k^r < \tau} \frac{(b_k^r)^2}{d_k^r}, \qquad \widetilde{\mathcal{E}}_r = \mathbb{E}\sum_{\sigma_k^r < \tau} \frac{(\widetilde{b}_k^r)^2}{d_k^r}. \tag{EX9}$$

The following conditions are equivalent:

1. $\liminf_{r\downarrow 0} \mathcal{E}_r < \infty$;
2. $\liminf_{r\downarrow 0} \widetilde{\mathcal{E}}_r < \infty$;

3. *$X = M + \int_0^{\cdot\wedge\tau} v_s ds$, where M is a uniformly integrable continuous martingale and v predictable, with $\mathbb{E}\int_0^\tau v_s^2 ds < \infty$.*

For the truncated means there is the exact limit

$$\lim_{r\downarrow 0} \widetilde{\mathcal{E}}_r = \mathbb{E}\int_0^\tau v_s^2 ds. \tag{EX10}$$

For the raw pure-sign energy only the boundedness equivalence is asserted; the terminal excursion can cause a uniformly bounded energy error. By (EX7), condition 1 is equivalently boundedness along some radii of $\mathbb{E}\sum_{\sigma_k^r<\tau}(\beta_k^r)^2$.

Proof. For a bounded deterministic-time elementary predictable process $H = \sum_j H_j \mathbf{1}_{(t_j,t_{j+1}]}$, use the indices and grid strategy L_k from the preceding proof, now with the bound $\sup|H|$. Set $L^r(t) = L_k$ on $(\sigma_k^r, \sigma_{k+1}^r]$, and zero beyond the first exit after τ. The vanishing grid mesh and (EX8) imply

$$\mathbb{E}\int_0^\infty |L^r(t)|^2 dt \longrightarrow \mathbb{E}\int_0^\tau |H_t|^2 dt. \tag{EX11}$$

Indeed on $[0,\tau]$ there is bounded pointwise convergence away from the finitely many partition endpoints; the expected time of the excess terminal interval is $O(r^2)$. Predictability gives

$$\mathbb{E}\sum_k L_k^2 d_k^r = \mathbb{E}\int_0^\infty |L^r(t)|^2 dt.$$

Both $\mathbb{E}\sum_k L_k b_k^r$ and $\mathbb{E}\sum_k L_k \widetilde{b}_k^r$ tend to $\ell(H) := \mathbb{E}\sum_j H_j(X_{t_{j+1}} - X_{t_j})$; the first uses the $O(nr\sup|H|)$ discretization error, and the second also uses (EX6). Cauchy–Schwarz, followed by (EX11), yields

$$|\ell(H)| \le \sqrt{\liminf_{r\downarrow 0} \mathcal{E}_r}\ \left(\mathbb{E}\int_0^\tau H_t^2 dt\right)^{1/2},$$

and the same inequality with $\widetilde{\mathcal{E}}_r$. Bounded deterministic-time elementary predictable processes are dense in the predictable $L^2(\mathbb{P}\,dt)$ space. Riesz representation therefore gives a predictable v, zero after τ, such that $\ell(H) = \mathbb{E}\int_0^\tau H_t v_t dt$, and its squared norm is bounded by the corresponding liminf energy. Testing a single interval and every bounded $\mathcal{F}_s$ variable shows that $X - \int_0^{\cdot\wedge\tau} v_t dt$ is a martingale. Its integrable dominating variable is $L + \int_0^\tau |v_t| dt$, so it is uniformly integrable; continuity gives the continuous martingale decomposition.

Conversely, assume condition 3. Conditional Cauchy–Schwarz on each excursion, including its possible truncation at τ, gives

$$(\widetilde{b}_k^r)^2 \le d_k^r\, \mathbb{E}\left[\int_{\sigma_k^r}^{\sigma_{k+1}^r\wedge\tau} v_s^2 ds \,\middle|\, G_k^r\right] \quad \text{on } \{\sigma_k^r < \tau\}.$$

Thus $\widetilde{\mathcal{E}}_r \le \mathbb{E}\int_0^\tau v_s^2 ds$. The lower norm bound from the Riesz argument gives the reverse liminf, proving (EX10). Finally, set $p_k^r = \mathbb{P}\{\sigma_k^r < \tau < \sigma_{k+1}^r \mid G_k^r\}$. Then $|b_k^r - \widetilde{b}_k^r| \le 2rp_k^r$ on active starts, and

$$\mathbb{E}\sum_{\sigma_k^r<\tau} \frac{|b_k^r - \widetilde{b}_k^r|^2}{d_k^r} \le \frac{4}{a_1}\mathbb{E}\sum_k p_k^r \le \frac{4}{a_1}.$$

The triangle inequality in the indexed L^2 space proves equivalence of the two bounded-energy conditions. This also shows why replacing the truncated energy by the raw sign energy need not preserve the numerical limit. □

Corollary 26.3 (Any subquadratic sign energy distinguishes AC drift from singular drift)**.** *Fix* $1 < p < 2$*, and define*

$$\mathcal{E}_{r,p} = \mathbb{E} \sum_{\sigma_k^r < \tau} \frac{|b_k^r|^p}{(d_k^r)^{p-1}}. \tag{EX12}$$

Then $\liminf_{r\downarrow 0} \mathcal{E}_{r,p} < \infty$ *is equivalent to* $X = M + \int_0^{\cdot\wedge\tau} v_s ds$ *with* $\mathbb{E}\int_0^\tau |v_s|^p ds < \infty$*. In that case the* raw *pure-sign energy has the exact limit*

$$\lim_{r\downarrow 0} \mathcal{E}_{r,p} = \mathbb{E}\int_0^\tau |v_s|^p ds. \tag{EX13}$$

By (EX7), boundedness is equivalently boundedness of $r^{2-p}\mathbb{E}\sum_{\sigma_k^r<\tau} |\beta_k^r|^p$*.*

Proof. Repeat the preceding proof with Hölder conjugate $p' = p/(p-1)$. The grid strategy satisfies

$$\mathbb{E}\sum_k |L_k|^{p'} d_k^r \longrightarrow \mathbb{E}\int_0^\tau |H_t|^{p'} dt.$$

Weighted Hölder's inequality and duality between predictable $L^{p'}$ and L^p therefore identify a predictable density v with $\mathbb{E}\int_0^\tau |v_s|^p ds$ bounded by the liminf energy. For the terminal-truncated means, conditional Hölder gives

$$|\widetilde{b}_k^r|^p \le (d_k^r)^{p-1}\mathbb{E}\left[\int_{\sigma_k^r}^{\sigma_{k+1}^r\wedge\tau} |v_s|^p ds \,\middle|\, G_k^r\right]$$

on active starts. Summation supplies the reverse upper bound, so the truncated-mean energy converges to the asserted L^p norm.

For the full-exit means, the terminal correction obeys

$$\mathbb{E}\sum_{\sigma_k^r<\tau} \frac{|b_k^r - \widetilde{b}_k^r|^p}{(d_k^r)^{p-1}} \le \frac{2^p}{a_1^{p-1}} r^{2-p}\mathbb{E}\sum_k (p_k^r)^p \le \frac{2^p}{a_1^{p-1}} r^{2-p} \longrightarrow 0. \tag{EX14}$$

The triangle inequality for the corresponding weighted L^p norms therefore transfers the same numerical limit to the raw sign energy. The exponent $2 - p > 0$ is essential for this last assertion. At $p = 1$ the limiting object is an arbitrary finite-variation drift measure, as in Proposition 26.1; for every $p > 1$ in the stated range, finiteness forces a density with respect to capacity time. □

Corollary 26.4 (Order-r exit-sign bias supplies bounded drift)**.** *If, for all sufficiently small radii and all active starts,* $|\beta_k^r| \le Kr$ *with a deterministic constant* K*, then* $X = M + \int_0^{\cdot\wedge\tau} v_s ds$ *with* $|v_s| \le K/a_1$ *almost everywhere.*

Proof. By (EX7), $|b_k^r| \le Kr^2 \le (K/a_1)d_k^r$. The predictable test argument therefore bounds $|\ell(H)|$ by $(K/a_1)\mathbb{E}\int_0^\tau |H_t| dt$. The L^1–L^∞ dual representation supplies the claimed bounded predictable density and hence the martingale decomposition. □

The classification gap is now quantitatively explicit. Unsigned estimates give $|\beta_k^r| \le 1$ and order T/r^2 exit terms. The exact finite-variation criterion asks for bounded $r\mathbb{E}\sum|\beta_k^r|$. The square-integrable drift criterion asks for bounded $\mathbb{E}\sum(\beta_k^r)^2$. The stronger pointwise bound $|\beta_k^r| \le Kr$ gives the bounded conditional-bias premise used in Theorem 5.3. To apply that theorem one must still control these constants along mirrored shrinking-slit states: with uniform a_1, the required rate is $\epsilon K_\epsilon \to 0$. Pointwise bounded drift in each unrelated state does not supply that rate.

Even bounded drift alone gives no initial tangent without locality's additional comparison: $v(t) = \sin(\log t)$ has bounded magnitude but $t^{-1}\int_0^t v(s)ds = \frac{1}{2}(\sin\log t - \cos\log t)$ has no limit at zero. Brownian motion plus this deterministic integral is only a process illustration, not a proposed family obeying the curve axioms. All these criteria are in a specified coordinate system. Neither finite raw exit-sign budget nor bounded sign energy is asserted in every affine structure, and no preferred affine tensor has been constructed.

27 Source-linear exit energy is sufficient for locality rigidity

The preceding exit criteria can be linked to classification using a weaker hypothesis than pointwise bounded driving drift. The new input is a bound on *expected initial drift energy per unit capacity*, expressible entirely in exit-sign observables. It still requires a quantitative rate along the locality-related shrinking-slit states. Neither that rate nor the energy bound is derived from RSW here.

Fix $1 < p < 2$ and one physical conformal structure. Assume the Borel family, continuous Loewner generation, strong natural-filtration DMP (KR1), and source locality for the mirrored tests (DR6), with the invertible capacity-clock and rough-prime-end requirements used previously. Assume also the uniform upper conditional driving moments and diffusive exit estimates already obtained from the explicit crossing/locality inputs. Thus Proposition 18.1 and the exit-energy criteria apply once their stated signed hypotheses hold.

For each state f, write $\mathcal{E}^f_{r,p}(\theta)$ for the raw energy in (EX12), stopped at a bounded stopping time θ at which the stopped driver is bounded. The proofs of Propositions 26.1–26.2 and Corollary 26.3 apply to every such θ: only boundedness of the stopped driver, the stopping-time property, and the residual-exit estimate are used. In particular θ need not be exactly the first exit from a symmetric driving interval.

Theorem 27.1 (Conditional classification from initial exit energy)**.** *Suppose the following additional signed conditions hold.*

1. *For every deterministic state f and $L, T > 0$, with $\tau_{L,T} = T \wedge \inf\{t : |W_t| \geq L\}$,*

$$\liminf_{r\downarrow 0} \mathcal{E}^f_{r,p}(\tau_{L,T}) < \infty. \tag{ER1}$$

2. *For each deterministic state f, there is a bounded stopping time $\rho_f > 0$ almost surely, with bounded stopped driver, and a finite deterministic constant $K(f)$ such that*

$$\limsup_{h\downarrow 0} \frac{1}{h} \liminf_{r\downarrow 0} \mathcal{E}^f_{r,p}(h \wedge \rho_f) \leq K(f)^p. \tag{ER2}$$

3. *For each state f, along some sequence $\epsilon_n \downarrow 0$, the mirrored deleted states obey*

$$\epsilon_n \max_{\sigma=\pm} K(f^\sigma_{\epsilon_n}) \longrightarrow 0. \tag{ER3}$$

Then every driver is $\sqrt{6}B$ in the specified conformal structure, and the exploration family is chordal SLE_6 in that structure.

Proof. The exit-energy criterion applied to (ER1), and uniqueness on overlapping stopped intervals, supply the global continuous canonical decomposition

$$W = M + A, \qquad A_t = \int_0^t b_s ds,$$

where b is predictable and locally $L^p(\mathbb{P}\,dt)$ after bounded driving localization. The upper conditional moment estimate and Proposition 18.1 give $C = \langle M\rangle = \int_0^t v_s ds$ with a locally bounded predictable density. The exact energy limit (EX13) turns (ER2) into

$$\limsup_{h\downarrow 0} \frac{1}{h}\mathbb{E}_f \int_0^{h\wedge\rho_f} |b_s|^p ds \le K(f)^p. \tag{ER4}$$

No pointwise bound or right limit of b is assumed.

Fix a deterministic state f and one ϵ. Couple its initial trace to each mirrored deleted-domain trace by locality and use their invertible capacity clocks. Put $d = d_\epsilon$, $e = e_\epsilon^+$, and $K_\epsilon = \max_\sigma K(f_\epsilon^\sigma)$. The canonical characteristic identity (BI3) reads, before contact,

$$p_t^\sigma b_t + \tfrac{1}{2} q_t^\sigma (v_t - 6) = (p_t^\sigma)^2 \widehat{b}^\sigma_{u_\sigma(t)}, \qquad du_\sigma(t) = (p_t^\sigma)^2 dt. \tag{ER5}$$

Here $p_0^\sigma = d$ and $q_0^\pm = \pm e$. There is an almost surely positive stopping time λ_ϵ on which the original and both deleted initial localizations remain valid, p_t^σ lies between $d/2$ and $2d$, and each q_t^σ remains within $|e|/2$ of its initial value. The two-by-two system in (ER5) is then uniformly invertible in these relative bounds. With a numerical constant C independent of ϵ,

$$|b_t| \le Cd \sum_{\sigma=\pm} |\widehat{b}^\sigma_{u_\sigma(t)}|, \qquad |v_t - 6| \le C\frac{d^2}{|e|} \sum_{\sigma=\pm} |\widehat{b}^\sigma_{u_\sigma(t)}| \tag{ER6}$$

for almost every $t < \lambda_\epsilon$.

Changing variables in the first inequality, using $u_\sigma(h\wedge\lambda_\epsilon) \le 4d^2 h$ and stopping before $\rho_{f_\epsilon^\sigma}$, gives

$$\mathbb{E}\,\mathrm{Var}(A; [0, h\wedge\lambda_\epsilon]) \le \frac{C}{d} \sum_{\sigma=\pm} \mathbb{E} \int_0^{4d^2 h\wedge\rho_{f_\epsilon^\sigma}} |\widehat{b}^\sigma_u| du.$$

Hölder's inequality on probability times capacity and (ER4) imply, at fixed ϵ,

$$\mathbb{E} \int_0^{4d^2 h\wedge\rho_{f_\epsilon^\sigma}} |\widehat{b}^\sigma_u| du \le 4d^2 h\big(K_\epsilon + o_\epsilon(1)\big), \qquad h \downarrow 0.$$

The second inequality in (ER6) is treated identically. Thus

$$\limsup_{h\downarrow 0} \frac{1}{h}\,\mathbb{E}\,\mathrm{Var}(A; [0, h\wedge\lambda_\epsilon]) \le CdK_\epsilon, \tag{8}$$

$$\limsup_{h\downarrow 0} \frac{1}{h}\,\mathbb{E}\,\mathrm{Var}(C - 6\,\mathrm{id}; [0, h\wedge\lambda_\epsilon]) \le C\frac{d^2}{|e|}K_\epsilon. \tag{9}$$

In applying (ER4), the deleted drift is evaluated in its own canonical filtration and capacity clock. Its law is the deleted-state law by locality; the time change in (ER5) is not replaced by a deterministic capacity equality.

For any $a > 0$, Markov's inequality and $\mathbb{P}\{\lambda_\epsilon < h\} \to 0$ give

$$\limsup_{h\downarrow 0} \mathbb{P}\left\{ h^{-1} \sup_{s\le h} |A_s| > a \right\} \le Ca^{-1} dK_\epsilon,$$

and the corresponding bound for $C_s - 6s$ with dK_ϵ replaced by $d^2K_\epsilon/|e|$. Since $d_\epsilon \asymp \epsilon$ and $|e_\epsilon| \asymp \epsilon^{-1}$, condition (ER3), first taking $h \downarrow 0$ at fixed ϵ_n and then $n \to \infty$, proves

$$h^{-1} \sup_{s\le h} |A_s| \longrightarrow 0, \qquad h^{-1} \sup_{s\le h} |C_s - 6s| \longrightarrow 0 \quad \text{in probability} \tag{ER7}$$

in every deterministic state.

It remains to propagate the initial conclusion without assuming a tangent of the original density. Proposition 4.1 and strong DMP transport the canonical shifted characteristics at each deterministic time t. Along any fixed sequence $h_n \downarrow 0$, for almost every conditioned history their law is the restarted state's canonical law. Statement (ER7) holds in that state. Conditional probabilities are bounded by one, so dominated convergence under the regular conditional law yields

$$\frac{A_{t+h_n} - A_t}{h_n} \longrightarrow 0, \qquad \frac{C_{t+h_n} - C_t}{h_n} \longrightarrow 6 \quad \text{in probability.} \tag{ER8}$$

On the other hand, absolute continuity and Fubini imply that for Lebesgue-almost every deterministic t, these same quotients converge almost surely to b_t and v_t. Uniqueness of limits in probability gives $b_t = 0$ and $v_t = 6$ almost surely for almost every t. Another Fubini argument gives $A = 0$ and $C_t = 6t$ up to indistinguishability. Lévy's characterization identifies $W = \sqrt{6}B$. □

What is sufficient, and what is not proved. This theorem replaces pointwise order-r exit-sign bias by averaged order-r signed energy. Conditions (ER1)–(ER3) are still missing theorems for the unknown family in an appropriate common structure. Finite exit energy alone supplies AC drift and bounded bracket but no rate in the slit comparison. The drift and variance anomaly rates are respectively ϵK_ϵ and $\epsilon^3 K_\epsilon$. All estimates are conditional classification tools, not a construction of the affine normalizer. In arbitrary initial coordinates an adequate regularity statement may need a correction involving an unknown constant physical Beltrami coefficient before these signed-energy hypotheses can be expected to hold.

27.1 A necessary energy blow-up at characteristic Lebesgue states

The shrinking-slit energy condition is quantitatively adapted to the actual characteristic anomalies. Mere finiteness of energy in each state cannot replace it. The following necessity statement applies on the states visited by a driver with locally L^p drift, without any initial energy-density assumption at every deterministic state.

Assume the family and locality premises of Theorem 27.1, and (ER1), with $1 < p < 2$. For a deterministic state g and a bounded stopping time $\rho_g > 0$ with bounded stopped driver, define the extended nonnegative quantity

$$\underline{K}_p(g; \rho_g) = \left(\liminf_{h \downarrow 0} \frac{1}{h} \liminf_{r \downarrow 0} \mathcal{E}^g_{r,p}(h \wedge \rho_g) \right)^{1/p}. \tag{VN1}$$

No finiteness of this quantity is assumed.

Proposition 27.2 (Anomalies force inverse-scale source energy)**.** *Fix an initial law P_f with canonical characteristics $A_t = \int_0^t b_s ds$ and $C_t = \int_0^t v_s ds$. For Lebesgue-almost every deterministic time t, for almost every conditioned history at that time, put $g = f_t$, $b = b_t$, and $v = v_t$. For every fixed admissible mirrored-slit scale ϵ, choose any positive bounded initial localizations $\rho^\pm_\epsilon$ in the two deleted states. With d_ϵ and $e_\epsilon = e^+_\epsilon$ as in (DR9), one has*

$$\underline{K}_p(g^\pm_\epsilon; \rho^\pm_\epsilon) \geq \left| \frac{b}{d_\epsilon} \pm \frac{e_\epsilon}{2d^2_\epsilon}(v - 6) \right|. \tag{VN2}$$

Consequently

$$\max_\pm \underline{K}_p(g^\pm_\epsilon; \rho^\pm_\epsilon) \geq \frac{|b|}{d_\epsilon} + \frac{|e_\epsilon|}{2d^2_\epsilon}|v - 6|. \tag{VN3}$$

The assertion holds simultaneously for every scale in any prescribed countable sequence. Thus a nonzero variance anomaly forces at least order ϵ^{-3} source-energy roots, while nonzero drift forces at least order ϵ^{-1} roots.

Proof. Local absolute continuity and pathwise Lebesgue differentiation, followed by Fubini, give for almost every deterministic t, almost surely,

$$\frac{1}{h}\int_t^{t+h} |b_s - b_t| ds \longrightarrow 0, \qquad \frac{1}{h}\int_t^{t+h} |v_s - v_t| ds \longrightarrow 0. \tag{VN4}$$

Condition on the past at such a time and use strong DMP and canonical restart compatibility. Outside a null set of histories, the restarted canonical drift and bracket satisfy (VN4) at zero with the frozen deterministic constants b_t, v_t. In particular their variation per unit time is bounded near zero, and their integrated initial tangents are respectively b and v.

For a fixed ϵ, couple the restarted trace with each deleted initial trace by locality. The analytic characteristic identity gives

$$\widehat{A}^{\pm}_{u_\pm(h)} = \int_0^h p_s^{\pm} b_s ds + \frac{1}{2}\int_0^h q_s^{\pm}(v_s - 6) ds, \qquad \frac{u_\pm(h)}{h} \longrightarrow d_\epsilon^2.$$

Continuity of $p^{\pm}, q^{\pm}$, their initial values $d_\epsilon, \pm e_\epsilon$, and (VN4) show that the deleted integrated drift has the deterministic initial tangent

$$B_\epsilon^{\pm} = \frac{b}{d_\epsilon} \pm \frac{e_\epsilon}{2d_\epsilon^2}(v - 6). \tag{VN5}$$

These are initial germ statements; the locality coupling is only needed until its strictly positive comparison stopping time.

For any positive bounded localization ρ in a deleted law, Hölder's inequality gives

$$\frac{1}{h}\,\mathbb{E}\int_0^{h\wedge\rho} |\widehat{b}_s|^p ds \geq \mathbb{E}\left|\frac{\widehat{A}_{h\wedge\rho}}{h}\right|^p.$$

The random variables inside the right side converge in probability to $|B_\epsilon^{\pm}|^p$. Fatou's lemma along an almost surely converging subsequence proves the corresponding liminf lower bound. The exact raw exit-energy identity (EX13), applied at $h \wedge \rho$, converts this into (VN2). For real x, y, $\max\{|x+y|, |x-y|\} = |x| + |y|$, proving (VN3). Countable intersections give the simultaneous sequence claim, and the scales follow from $d_\epsilon \asymp \epsilon$, $|e_\epsilon| \asymp \epsilon^{-1}$. □

This is a restriction on possible regularity estimates rather than a new derivation of signed cancellation. At almost every visited state, the rate $\epsilon \max_\pm K(g_\epsilon^{\pm}) \to 0$ can hold only if both characteristic anomalies vanish. Obtaining that rate from the curve axioms therefore remains the substantive classification step.

28 Analytic clock families can acquire singular time change

The affine-prefix results give analytic dependence of the capacity clock on a small deformation parameter in the uniform path norm, and very high temporal regularity of its first parameter derivative. These properties, even supplemented by strict monotonicity and global capacity comparison, do not alone prove absolute continuity of a finite-deformation clock. The following explicit example separates this analytic issue from the unresolved geometry of the actual affine-SLE clock.

Proposition 28.1 (A singular analytic clock family)**.** *There is a family of continuous strictly increasing clocks $s_\epsilon : [0,\infty) \to [0,\infty)$, $|\epsilon| < 1$, with the following properties.*

1. *$s_0(t) = t$, $s_\epsilon(t) = t$ outside $[1,2]$, and $\frac{1}{2}t \le s_\epsilon(t) \le 2t$ for every $t \ge 0$.*
2. *On every compact time interval the family extends holomorphically in ϵ into the Banach space of continuous paths on the disk $|\epsilon| < 4$. In particular it has uniform Taylor remainders of all orders on any smaller parameter disk.*
3. *Every finite parameter derivative at zero is temporally Hölder of every exponent below one. Its quadratic-increment sums vanish along every sequence of partitions of vanishing mesh. Its mixed sums with any path Hölder of some positive exponent also vanish, after choosing an appropriate exponent for the derivative path.*
4. *For every real $0 < |\epsilon| < 1$, ds_ϵ has a nonzero singular continuous component relative to dt. The first parameter derivative $\partial_\epsilon s_\epsilon|_0$ is not of bounded variation.*

This example has no asserted realization by Loewner hulls.

Proof. Work first on the circle $[0, 2\pi]$. For real $|\epsilon| < 1$, set

$$p_{\epsilon,N}(x) = \prod_{n=1}^{N} (1 + \epsilon \cos(5^n x)), \qquad \mu_{\epsilon,N}(dx) = p_{\epsilon,N}(x)\frac{dx}{2\pi}.$$

Every factor is positive. In the Fourier expansion, frequencies are $k = \sum_{n=1}^{N} \sigma_n 5^n$, $\sigma_n \in \{-1, 0, 1\}$, with coefficient $(\epsilon/2)^{\#\{n:\sigma_n \neq 0\}}$. Such representations are unique: the largest nonzero digit in a difference of two representations has magnitude at least 5^m, whereas all lower difference digits have total magnitude at most $2\sum_{n<m} 5^n < 5^m$. In particular the constant Fourier coefficient is one and these are probability measures. Their Fourier coefficients stabilize at every fixed integer frequency, so compactness and uniqueness of Fourier coefficients give a unique weak limit μ_ϵ.

Let

$$C_{\epsilon,N}(x) = \int_0^x p_{\epsilon,N}(u)\, du.$$

The nonconstant term with highest digit m has frequency of magnitude at least $5^m - \sum_{n<m} 5^n \ge (3/4)5^m$. Summing the absolute coefficients of all terms with this highest digit gives $|\epsilon|(1 + |\epsilon|)^{m-1}$. Since $|\int_0^x e^{iku} du| \le 2/|k|$, the largest-digit block contributes at most

$$C 5^{-m} |\epsilon| (1 + |\epsilon|)^{m-1} \tag{SC1}$$

in uniform norm. Therefore $C_{\epsilon,N}$ converges uniformly in x and locally uniformly for complex $|\epsilon| < 4$ to a holomorphic $C([0, 2\pi])$-valued function C_ϵ. For real $|\epsilon| < 1$ it is increasing, $C_\epsilon(0) = 0$, and $C_\epsilon(2\pi) = 2\pi$. The uniform convergence also shows that the weak measure limit has no atoms and $dC_\epsilon = 2\pi\, d\mu_\epsilon$.

Now define

$$s_\epsilon(t) = \begin{cases} t, & t \notin [1,2], \\ 1 + \dfrac{x + C_\epsilon(x)}{4\pi}, & t = 1 + x/(2\pi), \quad 0 \le x \le 2\pi. \end{cases} \tag{SC2}$$

On the middle interval its increment measure has one-half Lebesgue part, so the clock is strictly increasing; all joins are continuous. It maps $[1,2]$ onto $[1,2]$, giving the stated global comparisons. Analyticity and uniform Taylor bounds follow from (SC1).

The first derivative is zero outside $[1,2]$ and inside equals

$$\partial_\epsilon s_\epsilon(t)\big|_0 = \frac{1}{4\pi}\sum_{n\geq 1} 5^{-n}\sin(5^n x). \tag{SC3}$$

For a temporal separation h its increment has magnitude at most $C|h|(1+|\log|h||)$: split the sum at $5^n|h|=1$ and use $|\sin(a+b)-\sin a|\leq \min\{|b|,2\}$. More generally, in the coefficient of ϵ^j the sum of absolute Fourier coefficients in largest-digit block m is $\binom{m-1}{j-1}$, up to constants depending only on j. The same split gives the modulus

$$C_j|h|(1+|\log|h||)^j. \tag{SC4}$$

This holds across the joins as well, since all derivative paths vanish at both endpoints. Thus each derivative is Hölder-α for every $\alpha<1$. An exponent $\alpha>1/2$ gives vanishing quadratic sums. For mixed sums with a Hölder-β path, choose $\alpha<1$ with $\alpha+\beta>1$ and bound their absolute sum by $CT\,\mathrm{mesh}(\pi)^{\alpha+\beta-1}$.

We prove singularity directly. Under μ_ϵ,

$$\mathbb{E}_{\mu_\epsilon}\cos(5^n x)=\epsilon/2, \qquad \mathbb{E}_{\mu_\epsilon}[\cos(5^n x)\cos(5^m x)]=\epsilon^2/4 \quad (n\neq m).$$

These identities are the Fourier coefficients at 5^n and $5^n\pm 5^m$. Hence the centered variables are pairwise orthogonal and bounded. Their Cesaro average converges in $L^2(\mu_\epsilon)$ to $\epsilon/2$, with squared error at most C/N. At $N=k^2$, Chebyshev and Borel–Cantelli give almost sure convergence; boundedness interpolates between consecutive squares. Under normalized Lebesgue measure the same argument gives almost sure limit zero. For $\epsilon\neq 0$ the two full-measure sets are disjoint, proving $\mu_\epsilon\perp dx$. Formula (SC2) therefore has a nonzero singular continuous increment component in $[1,2]$.

Finally put $F(x)=\sum_{n\geq 1}5^{-n}\sin(5^n x)$. If F were BV, its periodic distributional derivative would be a finite signed measure λ with distributional Fourier series $\sum_{n\geq 1}\cos(5^n x)$. Convolution with the normalized Fejer kernel of degree $M=2\cdot 5^N$ would have L^1 norm at most $\|\lambda\|_{\mathrm{TV}}$ and would equal

$$P_N(x)=\sum_{n=1}^{N}\left(1-\frac{5^n}{M+1}\right)\cos(5^n x).$$

All weights are between $1/2$ and one. Orthogonality gives $\|P_N\|_2\geq c\sqrt{N}$. The fourth moment is at most CN^2: in a zero signed sum of four powers of five the largest exponent must cancel with an opposite copy, since three lower powers have magnitude less than the largest. Only pairings remain. Therefore $\|P_N\|_4\leq C\sqrt{N}$. Interpolation gives $\|P_N\|_1\geq \|P_N\|_2^3/\|P_N\|_4^2\geq c\sqrt{N}$, a contradiction. This proves the claimed failure of BV. □

Consequence for the research strategy. Even an all-order affine-parameter Taylor theory whose clock derivatives have every temporal exponent below one does not, by itself, settle absolute continuity of the finite affine capacity clock. One needs an additional estimate that controls the increment measures, or new information from the actual Loewner/SLE geometry. The example does not decide the affine-SLE clock question, and is not a counterexample to the exploration classification conjecture.

28.1 An exact increment-measure target for clock regularity

There is a positive measure-theoretic formulation of the missing estimate. It concerns a martingale in the dyadic *time variable*; it makes no semimartingale assumption about a Loewner driver.

Proposition 28.2 (A dyadic clock criterion)**.** *Let s be a continuous increasing function on $[0,T]$, with $s(0)=0$ and $s(T)=m>0$. For dyadic cells I at level n, put*

$$a_n(t)=\frac{s(\sup I)-s(\inf I)}{|I|},\quad t\in I.$$

Then $ds\ll dt$ if and only if (a_n) is uniformly integrable relative to Lebesgue measure on $[0,T]$. A sufficient condition is

$$\sup_n H_n<\infty,\qquad H_n=\int_0^T a_n\log\frac{a_n}{m/T}\,dt. \tag{SC5}$$

Write $m_I=s(\sup I)-s(\inf I)$ and I_-,I_+ for the children of I. Condition (SC5) is equivalent, up to absolute multiplicative constants, to finiteness of

$$\sum_{I\ dyadic}\frac{(m_{I_-}-m_{I_+})^2}{m_I}, \tag{SC6}$$

where a zero-mass summand is defined to be zero. If the clock is random, a uniform bound on $\mathbb{E}H_n$ is sufficient for almost sure absolute continuity.

Proof. The a_n form a nonnegative martingale on the probability space $([0,T],dt/T)$ with the increasing dyadic sigma-fields. They converge almost everywhere to the density of the absolutely continuous part of ds, by differentiation of measures. Uniform integrability is equivalent to preservation of their common integral m at the limit, which is equivalent to absence of singular mass. If $ds=a\,dt$, then a_n are the dyadic conditional averages of the integrable function a and hence are uniformly integrable.

The entropy condition gives uniform integrability by the elementary $x\log x$ tail bound. Jensen's inequality makes H_n nonnegative and increasing. At a split of a positive-mass cell, put $p=m_{I_-}/m_I$. The entropy increment is exactly

$$m_I\{p\log(2p)+(1-p)\log(2(1-p))\}.$$

For $\delta=2p-1$, its bracket equals $\sum_{k\ge1}\delta^{2k}/[2k(2k-1)]$, and is bounded between $\delta^2/2$ and $(\log 2)\delta^2$. Summing the increments proves the equivalence with (SC6). Finally a uniform expected entropy bound and monotone convergence give finite limiting entropy almost surely, which implies the pathwise criterion. □

This supplies a concrete new model task: control the dyadic mass-split energy of the actual affine-SLE capacity clock. Uniform integrability, rather than merely temporal Hölder regularity of clock tangents, is exactly what is needed for clock absolute continuity. The entropy bound is sufficient, not necessary; no such bound is proved here from the exploration axioms or for affine SLE. Clock AC would remain only a necessary condition in the existing wrong-driver semimartingale test.

29 Selecting an unknown affine tensor through finite signed tests

One need not identify the affine tensor before formulating a useful signed regularity estimate. It can be retained as a compact parameter and selected jointly with the estimates. The following conditional principle makes this precise without extending from a dense collection of domains or assuming a domain Feller property.

Fix $1 < p < \infty$ and $q = p/(p-1)$. On $C([0,T],\mathbb{R})$ let $W_t(w) = w(t)$, and suppose all coordinate variables are integrable under the law P. Let $\mathcal{H}_T$ consist of processes

$$H_t = \sum_{i=0}^{n-1} h_i(W|_{[0,t_i]})\mathbf{1}_{(t_i,t_{i+1}]}(t), \qquad 0 = t_0 < \cdots < t_n = T,$$

where every h_i is bounded and continuous on its past-path space. Put

$$\mathcal{J}_{p,T}(P) = \sup_{H\in\mathcal{H}_T} \left\{ p\,\mathbb{E}_P \sum_i h_i(W)(W_{t_{i+1}} - W_{t_i}) - (p-1)\mathbb{E}_P \int_0^T |H_t|^q dt \right\}. \tag{FS1}$$

Proposition 29.1 (A signed-test characterization of L^p drift)**.** *The functional in (FS1) is finite if and only if, in the completed right-continuous natural filtration, there is a predictable $b \in L^p(P \otimes dt)$ such that*

$$W_t = W_0 + M_t + \int_0^t b_s ds \tag{FS2}$$

with M a continuous martingale starting at zero. In that case

$$\mathcal{J}_{p,T}(P) = \mathbb{E}_P \int_0^T |b_s|^p ds. \tag{FS3}$$

The drift is unique up to $P \otimes dt$-null sets.

Proof. Write $L(H) = \mathbb{E}\sum_i h_i \Delta W_i$ and $\|H\|_q = (\mathbb{E}\int |H|^q dt)^{1/q}$. Scaling H by any real constant in (FS1), and optimizing that scalar, shows

$$|L(H)| \le \mathcal{J}_{p,T}(P)^{1/p} \|H\|_q. \tag{FS4}$$

In particular L respects equivalence modulo $P \otimes dt$. Continuous bounded past-path functions are dense in every finite L^q space of past-measurable variables, since path space is Polish. Consequently $\mathcal{H}_T$ is dense in the predictable L^q space. For the usual augmentation one may first approximate any augmented past variable by raw past variables at times $t + \delta$, then delay the start of its elementary interval by δ; the lost interval has vanishing $P \otimes dt$ norm. This gives the same density result.

Extend L continuously and apply L^q duality on the raw predictable sigma-field (completed for $P \otimes dt$) to obtain a predictable $b \in L^p$, choosing a raw predictable version, with $L(H) = \mathbb{E}\int Hb\,dt$ and $\|b\|_p \le \mathcal{J}_{p,T}(P)^{1/p}$. Testing one interval $(s,t]$ against a bounded continuous functional of $W|_{[0,s]}$ gives zero conditional mean for increments of $W - \int b dt$. The identity extends to every bounded past-measurable variable: it is equality of finite signed measures on the Polish past-path space, tested against all bounded continuous functions. Thus the continuous integrable process M in (FS2) is a martingale in the raw natural filtration. It extends to the usual augmentation: it is closed by its integrable terminal value, and right limits of conditional expectations agree with its continuous paths.

Conversely, under (FS2), martingale increments contribute zero, so $L(H) = \mathbb{E}\int Hb\,dt$. Pointwise convex duality gives

$$\sup_{h\in\mathbb{R}}\{phb - (p-1)|h|^q\} = |b|^p.$$

Approximate the optimizing predictable variable $\operatorname{sgn}(b)|b|^{p-1}$ in L^q by members of $\mathcal{H}_T$ to obtain equality (FS3). Uniqueness follows from uniqueness of the continuous semimartingale decomposition. □

Proposition 29.2 (Lower semicontinuity and common-tensor selection)**.** *Let $\mathcal{K}$ be a compact parameter space of normalized affine tensors. For each state f in an arbitrary index set and each horizon T, suppose $A \mapsto P^A_{f,T}$ is weakly continuous on $C([0,T],\mathbb{R})$, and, for each deterministic time t, the family of random variables W_t is uniformly integrable over these laws. Then $A \mapsto \mathcal{J}_{p,T}(P^A_{f,T})$ is lower semicontinuous.*

Fix a deterministic finite envelope $B_{f,T}$ for every desired state/horizon pair. If every finite collection of elementary signed tests, from any finite collection of state/horizon pairs, admits one common $A \in \mathcal{K}$ for which all the individual test expressions in (FS1) are at most their prescribed $B_{f,T}$, then there exists one $A_* \in \mathcal{K}$ such that

$$\mathcal{J}_{p,T}(P^{A_*}_{f,T}) \le B_{f,T} \quad \text{for every desired } (f,T). \tag{FS5}$$

In particular all these drivers simultaneously have absolutely continuous L^p drift with the corresponding envelopes. The same conclusion holds if finite collections can only be satisfied up to an arbitrary common error $\eta > 0$.

Proof. For fixed H, its penalty expectation is a bounded continuous function of the law. Each increment expectation is continuous as well: its integrand is continuous, and its absolute value is at most a fixed multiple of $|W_{t_i}| + |W_{t_{i+1}}|$; uniform integrability allows truncation. Therefore each elementary test expression is continuous in A. Their supremum is lower semicontinuous.

The constraint that one test expression is at most $B_{f,T}$ is closed in $\mathcal{K}$. The simultaneous finite-test hypothesis is exactly the finite-intersection property for these closed sets. Compactness gives a point in their entire intersection, and taking the supremum over tests gives (FS5). In the approximate version, compactness and continuity first let $\eta \downarrow 0$ for each fixed finite collection, giving the exact finite-intersection property. No countability of the state index set and no dense-state extension are used. □

One possible compact tensor class is $\{Q = Q^{\mathsf{T}} > 0 : \det Q = 1,\ \lambda^{-1} I \le Q \le \lambda I\}$. The common ellipticity bound λ must be proved or assumed. The finite envelopes must also be specified in advance; the claim that each finite test collection has some finite but uncontrolled budget is not enough for the compactness conclusion.

29.1 Physical affine-coordinate continuity for a fixed state

The law-continuity premise above can be proved pathwise for changes of affine coordinates. The transported coefficient is essential. This is not a continuity assertion as the physical state itself varies.

Proposition 29.3 (Uniform driver continuity under asymptotically conformal changes)**.** *Let K_t, $0 \le t \le T_0$, be a continuously driven half-plane Loewner chain with driver W_t and $K_t \subset B_R$. Let $F_n : \mathbb{H} \to \mathbb{H}$ be quasiconformal homeomorphisms fixing $0, 1, \infty$, with maximal dilatations tending to one. Suppose the image hull chains admit their continuous capacity parametrizations. Write*

$$s_n(t) = \tfrac{1}{2}\,\mathrm{hcap}(F_n K_t), \qquad U_n(t) = \widehat{g}^{\,n}_{s_n(t)}(F_n \gamma(t)),$$

where tips are interpreted as prime ends. Then

$$\sup_{t \le T_0} |s_n(t) - t| \longrightarrow 0, \qquad \sup_{t \le T_0} |U_n(t) - W_t| \longrightarrow 0. \tag{FS6}$$

Consequently their own-capacity drivers converge uniformly to W on every $[0,T]$ with $T < T_0$.

Proof. Reflect F_n across the real line. Normalized quasiconformal compactness gives $F_n \to \mathrm{id}$ uniformly in spherical distance, and hence uniformly on compact subsets of the plane. The image hulls all lie in a fixed disk $B_{R'}$. At each fixed t, their complements converge in the Caratheodory kernel sense to $\mathbb{H} \setminus K_t$: any compact subset persists by uniform convergence of F_n^{-1}, and any purported larger kernel neighborhood would contain points $F_n(z_n) \in F_n K_t$ converging to a point of K_t. Hydrodynamic maps converge locally uniformly. Since all hulls are uniformly bounded, the maps reflect conformally across the real line outside a fixed disk; convergence on a sufficiently large fixed circle gives convergence of their $1/z$ coefficients, hence $s_n(t) \to t$. Monotonicity in t and a finite-grid sandwich against the continuous limit give uniform clock convergence.

Set $f_t = g_t^{-1}$ and consider

$$Q_{n,t} = \widehat{g}^n_{s_n(t)} \circ F_n \circ f_t.$$

This is a quasiconformal self-homeomorphism of the half-plane, fixing infinity, with dilatation tending to one uniformly in t. Uniform hydrodynamic bounds for hulls in a fixed disk give, for every sufficiently large fixed Y,

$$\sup_{t \le T_0} |Q_{n,t}(iY) - iY| \le CR^2/Y + o_n(1), \tag{FS7}$$

enlarging R by a fixed factor if necessary. Indeed $f_t(iY) = iY + O(R^2/Y)$ uniformly in t, F_n converges uniformly on the compact set of these values, and the transformed hydrodynamic map changes its argument by $O(R^2/Y)$.

For any sequence t_n, the images of the three distinct sphere points $\infty, iY, 2iY$, with one sufficiently large fixed Y, remain respectively infinity and in separated compact neighborhoods of $iY, 2iY$. Normalized quasiconformal compactness therefore keeps the reflected Q_{n,t_n} away from degenerate limits. Any subsequential normalized quasiconformal limit is therefore a conformal sphere homeomorphism fixing infinity and preserving $\mathbb{H}$, hence $z \mapsto az + b$ with $a > 0$ and $b \in \mathbb{R}$. The estimate (FS7), for arbitrarily large fixed Y, forces $a = 1, b = 0$. Thus $Q_{n,t} \to \mathrm{id}$ uniformly on bounded closed half-plane sets, uniformly in t. Since W_t remains bounded, $U_n(t) = Q_{n,t}(W_t)$ proves the driver assertion in (FS6). Uniform inverse-clock convergence on $[0, T]$, followed by uniform continuity of W, proves the final statement. □

For a fixed physical state $f_0 : \mathbb{H} \to D_0$, choose one third marked prime end $f_0(1)$ to normalize its chordal charts. If physical affine maps $A_n \to \mathrm{id}$, the comparison $F_n = \varphi_n \circ A_n \circ f_0$, where $\varphi_n : A_n D_0 \to \mathbb{H}$ sends the three corresponding ends to $0, 1, \infty$, has coefficient

$$\mu_{F_n}(z) = \mu_{A_n} \frac{\overline{f_0'(z)}}{f_0'(z)}. \tag{FS8}$$

Its norm tends to zero, so the preceding proposition applies. At a general base affine map A_0, use the reference domain $A_0 D$ and the relative maps $A_n A_0^{-1}$. For each deterministic convergent sequence of affine maps, this gives samplewise convergence of normalized driving paths on horizons strictly inside the available capacity lifetime, on the common probability-one event of admissibility for that countable sequence. Bounded continuous functionals then give weak law continuity by dominated convergence. Admissibility almost surely for each individual affine map suffices for this argument; a universal event for an uncountable tensor class is not asserted. No domain Feller assertion is used. Uniform integrability of increments remains a separate requirement, for example from uniform crossing tails over a compact ellipticity class.

Corollary 29.4 (The selection continuity assumptions under uniform crossings)**.** *Suppose a fixed physical-state law has the uniform avoidable-quad crossing upper bound used earlier, and its curves are continuously Loewner admissible after every affine map in a compact normalized ellipticity*

class $\mathcal{K}$, with infinite capacity lifetime in each such chart. Normalize each initial chart by three corresponding prime ends. Then the associated own-capacity driver laws are weakly continuous in $A \in \mathcal{K}$ and have uniformly integrable coordinate increments on each finite capacity horizon. Thus the lower semicontinuity premise of Proposition 29.2 holds for that state.

Proof. Weak continuity is the preceding deterministic argument, coupled through the fixed physical curve sample. If all affine maps have quasiconformal dilatation at most K_*, modulus distortion shows that a transformed quad of modulus at least K_*M pulls back to one of modulus at least M. Avoidability and the crossing event are topological, so the same probability upper bound survives uniformly over the class, with the enlarged modulus threshold. The established driving upper-tail theorem consequently supplies uniform exponential tails for the driving supremum divided by $\sqrt{T}$, with constants depending only on the original crossing constants and K_*. This gives uniform integrability, indeed all finite moments, for the finitely many coordinate increments in each elementary test. □

Status and role. These propositions are compactness and stochastic duality tools. They do not prove the simultaneous finite-test bound or a common ellipticity class from the exploration axioms. They show that a regularity theorem may select its affine tensor as part of its conclusion. It is unnecessary to know the tensor first, but it remains necessary to prove an estimate uniform over finite collections of tests. Mere L^p drift, without the additional locality/slit control in the rigidity theorems, is not claimed to finish classification.

30 Strong restart excludes a shared smooth random coordinate medium

A natural possible counterexample is to sample a translation-invariant smooth random geometry and run SLE_6 in that geometry. The following argument excludes this class. In fact the prior on the coordinate map need not be stationary: translation covariance of the resulting family is enough. This result does not assume an analogous representation for the unknown exploration family.

Lemma 30.1 (The literal time-zero restart makes the initial germ trivial)**.** *Let P_f be one law in the family, with deterministic initial state f, and suppose strong DMP (KR1) holds at $\tau = 0$ in its usual right-continuous natural driver filtration, with $f_0 = f$. Then $\mathcal{F}_0^f$ is P_f-trivial. In particular every Borel ordered initial-trace-germ observable is almost surely constant.*

Proof. For each bounded Borel functional F of the entire future driver, the time-zero restart identity says

$$\mathbb{E}_f[F(W) \mid \mathcal{F}_0^f] = \mathbb{E}_f F(W). \tag{RM1}$$

The usual initial sigma-field is contained in the completed sigma-field of the full driver. Apply (RM1) to a full-driver version of any event in $\mathcal{F}_0^f$: its indicator equals its probability almost surely.

For completeness an ordered initial-trace-germ statistic belongs to $\mathcal{F}_0^f$. It is measurable from the prefix stopped on every source circle of radius r. The corresponding capacity stopping times decrease to zero almost surely as $r \downarrow 0$, by continuity and strict initial Loewner growth. For any fixed $t > 0$, the statistic is therefore $\mathcal{F}_t^f$-measurable, by taking the union of events on which those stopping times are at most t. Right continuity gives its measurability in $\mathcal{F}_0^f$. □

This is a deduction from the *literal* time-zero strong restart convention. It is not justified merely by DMP at positive times in an unaugmented filtration. The independent-color germ theorem proved earlier remains useful without assuming a limiting strong DMP theorem.

Lemma 30.2 (Endpoint tensor decoding only needs a first-order error)**.** *Lemma 8.2 remains valid with its quadratic error replaced by*

$$\Phi(z) = Lz + o(|z|), \qquad z \to 0 \tag{RM2}$$

uniformly on its closed half-neighborhood. Consequently the same universal physical endpoint decoder recovers $[Dh(p)^{\mathsf{T}} Dh(p)]$ *from a sample with law*

$$(h^{-1})_* \operatorname{SLE}_6(hD; hp, hq) \tag{RM3}$$

whenever D *is a bounded smooth Jordan domain and* h *is an orientation-preserving* $C^{1,\alpha}_{\mathrm{loc}}$ *diffeomorphism of the plane,* $0 < \alpha < 1$*.*

Proof. In the proof of Lemma 8.2, the scaled error on a fixed bounded preimage disk is now at most $\omega(Cr_n)$, for a deterministic modulus $\omega(s) \to 0$, in place of Cr_n. For every fixed radial tolerance $\delta > 0$, it is eventually smaller than δ. The same stopping sandwich therefore bounds the observable difference by $\omega(Cr_n) + h_\delta(Y_n)$. Cesàro averaging removes the first term; ergodicity bounds the second by $\mathbb{E}h_\delta(Y)$; let $\delta \downarrow 0$ as in the original proof. No rate of the first-order error is needed.

To apply this to (RM3), choose a conformal map $\phi : \mathbb{H} \to hD$ sending $0, \infty$ to hp, hq. The image boundary is $C^{1,\alpha}$ and hence Dini-smooth. The classical boundary differentiability theorem gives

$$\phi(z) = hp + cz + o(|z|), \qquad c \neq 0,$$

for unrestricted approach in the closed half-plane; see [1, Theorem 1]. Composition with the differentiable map h^{-1} gives a first-order local embedding into D. After rotating the known physical tangent half-plane to $\mathbb{H}$ by Q, its derivative is $L = QDh(p)^{-1}M_c$, where M_c is complex multiplication by c. The endpoint decoder recovers

$$[(L^{-1})^{\mathsf{T}} L^{-1}] = [QDh(p)^{\mathsf{T}} Dh(p) Q^{\mathsf{T}}],$$

so conjugating back by Q gives the asserted tensor in physical axes. The same Borel decoder is used for every map h. □

Theorem 30.3 (A common smooth random-coordinate prior collapses to one affine metric)**.** *Let* Π *be a probability measure on orientation-preserving global* $C^{1,\alpha}_{\mathrm{loc}}$ *diffeomorphisms of* $\mathbb{R}^2$*, and consider the family*

$$P_{D,p,q} = \int (h^{-1})_* \operatorname{SLE}_6(hD; hp, hq)\, \Pi(dh). \tag{RM4}$$

The same prior Π *is used for every deterministic marked domain. Suppose the family is Borel, has the standing continuous Loewner admissibility, is physically translation covariant, and obeys the time-zero strong restart identity of Lemma 30.1. It suffices to impose these assumptions for translations of one smooth bounded Jordan domain with a regular marked source.*

Then there is one deterministic positive-definite projective tensor T_* *such that, for* Π*-almost every* h*,*

$$h(x) = a_h Ax + b_h, \tag{RM5}$$

where $A = T_*^{1/2}$ *after choosing determinant-one normalization,* $a_h \neq 0$ *acts by complex multiplication, and* b_h *is a translation. Consequently (RM4) is the pure fixed-affine SLE*$_6$ *family with metric* T_**; the random similarities* a_h, b_h *disappear from its law.*

Proof. Take a reference disk D_0 whose source is 0, and translate it to $D_p = D_0 + p$ with translated target. For a fixed p, condition first on h. The endpoint decoder of the preceding lemma recovers

$$T_h(p) = [Dh(p)^{\mathsf{T}} Dh(p)] \tag{RM6}$$

almost surely under the conditional curve law. Integrating over the common prior shows that the decoder under P_{D_p,p,q_p} has exactly the Π-distribution of $T_h(p)$. It is an initial-germ statistic, so Lemma 30.1 makes this distribution a point mass, say at $t(p)$.

The physical tangent frames of the translated disks are translates of the same frame. Physical translation covariance and the universal decoder therefore give $t(p) = t(0) = T_*$ for every deterministic p. For all rational-coordinate points simultaneously, (RM6) equals T_* outside one Π-null set. Continuity of Dh extends this equality to all $x \in \mathbb{R}^2$. Thus

$$Dh(x)^{\mathsf{T}} Dh(x) = c_h(x) T_*, \qquad c_h(x) > 0. \tag{RM7}$$

With $A = T_*^{1/2}$, the derivative of $k_h(y) = h(A^{-1}y)$ is an orientation-preserving Euclidean similarity at every point. Its C^1 coordinate functions satisfy the Cauchy–Riemann equations, so k_h is an entire conformal map. It is a global homeomorphism and hence proper; an entire proper injective map is a polynomial of degree one. This gives (RM5). SLE similarity covariance now makes every conditional law in (RM4) equal to $A_*^{-1}\,\mathrm{SLE}_6(AD; Ap, Aq)$. □

Scope of this obstruction. No RSW estimate, independent-color coupling, or stationarity assumption on the random chart is used. The theorem rules out a smooth random coordinate medium sampled from one common prior as a source of additional families. It does not establish a smooth-coordinate representation for the unknown family. It leaves open rough coordinate media and rules using a different prior for each source/domain, including possible source-germ assignments whose restart consistency remains unresolved. Full natural-filtration strong DMP at zero is essential to this argument.

31 Signed calibration energies force affine compactness

Compactness of the unknown affine structure can follow from signed calibration bounds, without an affine-SLE representation. Two remote analytic deletions control both driving characteristics. Two physical orientations then exclude degenerating anisotropy. The bounds used are four full signed energies in candidate-dependent auxiliary charts; finite collections of scalar tests do not supply them automatically.

31.1 Two opposite deletions control both characteristics

Fix $A^{\pm} = \{\pm 10 + iy : 0 \le y \le 1\}$ in $\mathbb{H}$. There is a deterministic $h_0 > 0$ such that every continuous driver $W_0 = 0$, stopped at

$$\theta = h_0 \wedge \inf\{t : |W_t| \ge 1\}, \tag{TC1}$$

has evolving deletion maps $h_t^{\pm}$ with uniformly bounded analytic jets near the driving interval and

$$0 < m \le p_t^{\pm} = (h_t^{\pm})'(W_t) \le P, \qquad |p_t^{+} q_t^{-} - p_t^{-} q_t^{+}| \ge d_0 > 0, \quad q_t^{\pm} = (h_t^{\pm})''(W_t). \tag{TC2}$$

All constants are deterministic and independent of the driver. At time zero the hydrodynamic maps are $h_0^{\pm}(z) = \pm 10 + \sqrt{(z \mp 10)^2 + 1}$, with the square-root branch asymptotic to $z \mp 10$. On $[-1, 1]$

their first derivatives are positive, whereas $q_0^+ < 0 < q_0^-$. The determinant is therefore bounded away from zero. For drivers bounded by one, the Loewner ODE on a fixed neighborhood of each remote slit gives $g_t = \mathrm{id} + O(t)$ there with all required derivatives, uniformly over drivers. The image slits remain uniformly separated from the driving interval and converge in this analytic parameterization to the original slits. Hydrodynamic uniformization and reflection give uniform convergence of the deletion-map jets near $[-1, 1]$. Decreasing h_0 preserves (TC2).

Put $u_\pm(t) = \int_0^{t\wedge\theta} (p_s^\pm)^2 ds$, and let $\widehat{W}^\pm$ be the corresponding deleted drivers in their own capacities, frozen after $u_\pm(\theta)$ and centered at zero. Then

$$\widehat{W}^\pm_{u_\pm(t)} = h^\pm_{t\wedge\theta}(W_{t\wedge\theta}) - h^\pm_0(0). \tag{TC3}$$

The same physical prefix is retained throughout. Its reconstruction from either driver identifies the stopped natural trace filtrations and makes these increasing clocks stopping-time changes. The same statements hold with any additional observable terminal prefix stop included in θ.

Proposition 31.1 (Two remote signed energies regularize both characteristics)**.** *Let $1 < p < \infty$. Suppose the two stopped own-capacity drivers have continuous martingale decompositions*

$$\widehat{W}^\pm_u = \widehat{M}^\pm_u + \int_0^{u\wedge u_\pm(\theta)} \widehat{b}^\pm_v dv, \qquad \mathbb{E}\int_0^{u_\pm(\theta)} |\widehat{b}^\pm_v|^p dv \le B. \tag{TC4}$$

Then $X_t = W_{t\wedge\theta}$ is a continuous semimartingale with absolutely continuous drift and bracket,

$$X_t = M_t + \int_0^{t\wedge\theta} b_s ds, \qquad \langle M\rangle_t = \int_0^{t\wedge\theta} v_s ds, \qquad \mathbb{E}\int_0^\theta (|b_s|^p + |v_s - 6|^p) ds \le C_p B. \tag{TC5}$$

For $0 < h \le h_0$ and $0 < \delta < 1$ one consequently has

$$\mathbb{P}\{\sup_{t\le h} |X_t| > \delta\} \le C_p \delta^{-2}\{h + B^{1/p} h^{1-1/p}\} + C_p \delta^{-p} B h^{p-1}. \tag{TC6}$$

Constants depend only on the fixed slits, h_0, and p.

Proof. Pull the first deleted driver back through its stopping-time clock. It is a semimartingale in the original trace filtration. The inverse analytic field $(h_t^+)^{-1}$ is adapted, absolutely continuous in t, and has bounded required spatial derivatives on the localized region. Its finite-variation random-field Itô formula, applied to (TC3), makes X a continuous semimartingale. Denote its canonical drift and bracket measures by dA, dC, without assuming their absolute continuity.

For an analytic deletion the time derivative at the moving driving point contributes $-3q_t^\pm dt$. Uniqueness of the canonical decomposition and Itô's formula therefore give, on the active interval $[0, \theta]$,

$$p_t^\pm dA_t + \tfrac{1}{2} q_t^\pm (dC_t - 6dt) = (p_t^\pm)^2 \widehat{b}^\pm_{u_\pm(t)} dt. \tag{TC7}$$

The determinant in (TC2) makes this a uniformly invertible system of two signed-measure equations. Thus $dA = b\, dt$, $dC = v\, dt$, and

$$|b_t| + |v_t - 6| \le C(|\widehat{b}^+_{u_+(t)}| + |\widehat{b}^-_{u_-(t)}|).$$

Changing variables with $m^2 dt \le du_\pm \le P^2 dt$ proves (TC5). It also gives $\mathbb{E}\langle M\rangle_h \le 6h + C_p B^{1/p} h^{1-1/p}$. The martingale is square integrable by this bracket bound. Doob's inequality controls its supremum; Hölder's inequality gives $\mathbb{E}\sup_{t\le h} |\int_0^{t\wedge\theta} b_s ds|^p \le C_p B h^{p-1}$. Split at $\delta/2$ to obtain (TC6). □

By Proposition 29.1, condition (TC4) is exactly $\mathcal{J}_{p,P^2 h_0}(\mathrm{Law}(\widehat{W}^\pm)) \le B$, with the drivers frozen after their matched stops. No signed energy of the original driver is assumed. Locality is relevant when comparison laws are identified with the corresponding deleted-domain members of an exploration family.

31.2 Nonflat calibration prefixes exclude diverging anisotropy

For the moment suppose the physical half-plane states $D_1 = \mathbb{H}, D_2 = i\mathbb{H}$ are admissible, with source zero, target infinity, and third normalization marks $1, i$. Rotate their traces back to $\mathbb{H}$ and let K^1, K^2 be their filled capacity-one hulls. If each is almost surely not contained in a line, there are $R_0 < \infty$, $0 < \delta < 1/8$, and $\eta > 0$ such that for $j = 1, 2$,

$$\mathbb{P}\left\{K^j \subset B_{R_0}, \quad \inf_{|\beta|\le\sqrt{2}} \sup_{z\in K^j} |\operatorname{Re} z + \beta \operatorname{Im} z| \ge 4\delta\right\} \ge \eta. \tag{TC8}$$

Indeed the supremum is continuous in β; its minimum on this compact interval vanishes only for a hull contained in one of the corresponding lines. Boundedness of the hulls and monotone choices of R_0, δ prove (TC8).

The conditional lower oscillation bound supplies this nonflatness. A positive-capacity straight-ray driver has form $c\sqrt{t}$. On $[1/2, 1]$, for $|c| \le C$, its oscillation on each of n equal intervals is $O(C/n)$, below the lower-tail threshold of order $n^{-1/2}$ for large n. Conditioning successively bounds the probability of all such small oscillations by $(1 - p_0)^n$. Let $n \to \infty$, then take a countable union over integer C.

Theorem 31.2 (Model-independent coercivity of baseline slit energies)**.** *Assume the two calibration laws satisfy (TC8) and continuous affine Loewner admissibility. Let $\mathcal{A}$ be any collection of orientation-preserving determinant-one linear candidates. In each candidate's transformed flat-state chart, normalized by the third mark, suppose both remote slit comparisons satisfy (TC4), with the same $p > 1$ and $B < \infty$. Then the condition numbers of all $A \in \mathcal{A}$ are uniformly bounded. Their normalized conformal tensors lie in a compact ellipticity class. The bound depends only on p, B, the fixed slits, and the calibration constants.*

Proof. Choose $e_j \in \{1, i\}$ with $|Ae_j| \ge \|A\|/\sqrt{2}$. Its normalized transformed chart is the half-plane self-map

$$B_A(z) = \frac{A(e_j z)}{Ae_j} = x + \beta y + icy, \quad z = x + iy, \qquad |\beta| \le \sqrt{2}, \quad \|B_A\| \le \sqrt{2}, \quad c = |Ae_j|^{-2} \le \frac{2}{\kappa(A)}, \tag{TC9}$$

where $\kappa(A) = \|A\|\|A^{-1}\| = \|A\|^2$ because $\det A = 1$. On the event in (TC8), $B_A K^j$ has horizontal extent at least 4δ, radius at most $\sqrt{2}R_0$, and height at most cR_0.

A half-plane hull of radius at most R and height at most $H \le R$ has hcap $\le CRH$. The geometric capacity theorem [19, Theorem 1] bounds capacity by a constant times the area of its tangent-disk union; that union lies in a rectangle of width at most $2R + 2H$ and height $2H$. Thus, on this event,

$$s_A(1) = \tfrac{1}{2}\operatorname{hcap}(B_A K^j) \le CR_0^2/\kappa(A). \tag{TC10}$$

Choose $h_* \le h_0$ such that $2\sqrt{h_*} \le \delta$ and the right side of (TC6), at threshold δ, is smaller than η. If $\kappa(A) \ge CR_0^2/h_*$, then $s_A(1) \le h_*$ on the calibration event. The deterministic Loewner hull-radius estimate

$$\sup_{z\in B_A K^j} |z| \le \sup_{u\le s_A(1)} |W_u^A| + 2\sqrt{s_A(1)}$$

forces the driver to reach magnitude at least 3δ by time h_*. Its version stopped on exiting $[-1, 1]$ therefore has supremum greater than δ. This occurs with probability at least η, contradicting (TC6). Hence the stated condition-number bound holds. □

This half-plane statement uses admissible unbounded states explicitly. The original bounded-domain formulation does not silently supply them. The next version uses only bounded physical calibration states.

Corollary 31.3 (Coercivity using two bounded physical prefixes)**.** *Take two fixed bounded half-disks $D_j \subset e_j\mathbb{H}$, $e_1 = 1, e_2 = i$, with flat source zero, remote targets, and radii larger than $2r > 0$. Stop each physical exploration at its first exit from $B_r(0)$, rotate back by e_j^{-1}, and fill relative to the auxiliary half-plane $\mathbb{H}$; call the resulting hull K^j. It has finite positive auxiliary capacity time T_j and a continuous auxiliary driver. The original bounded-state conditional lower oscillation bound implies that K^j is almost surely not contained in a line, hence satisfies (TC8).*

For each candidate A form the transformed prefix in the auxiliary chart $B_{A,j}(z) = A(e_j z)/(Ae_j)$, with terminal capacity $T_{A,j}$. Extend its driver constantly after $T_{A,j}$ and use

$$\theta_{A,j} = h_0 \wedge \inf\{t : |W_t^{A,j}| \geq 1\} \wedge T_{A,j}. \tag{TC11}$$

If the two corresponding slit-comparison drivers in each orientation, frozen at $u_\pm(\theta_{A,j})$, satisfy (TC4) uniformly over A, then the same condition-number bound follows. No unbounded ensemble state or infinite capacity lifetime is required.

Proof. The initial actual chordal chart is analytic on the smaller source half-disk and reflects across its flat boundary. Local analytic Loewner comparison therefore gives continuous auxiliary half-plane driving and a strictly increasing continuous clock through the stopped prefix. Its auxiliary capacity is positive by strict hull growth and finite by boundedness. Its radius is at most r: filling a path in the convex source half-disk cannot add points outside that half-disk.

A line-supported prefix would be a straight ray segment initially. Its image under the analytic actual chart is an analytic simple slit, whose driver is continuously differentiable on every strictly positive interior capacity interval. The lower oscillation bound excludes a locally Lipschitz driver on any nontrivial interval: partition a rational interval into n pieces, apply the same $(1 - p_0)^n$ estimate, and take countable unions over rational intervals and integer Lipschitz constants. This proves nonflatness and (TC8).

The terminal auxiliary time is observable in its driver filtration:

$$T_{A,j} = \inf\{u : \sup_{z \in K_u} |B_{A,j}^{-1} z| \geq r\}.$$

Before the trace reaches the physical circle, its filled physical hull stays in the convex source half-disk. After constant extension, the artificial Loewner hulls retain the terminal hull, so the same first-contact formula still holds. Deleted terminal stops are observable by the inverse geometric reconstruction.

The deletion identities and (TC5)–(TC6) thus remain valid with the additional terminal stop in (TC11), with the time measure active only up to that stop. The constant extension has no further driver movement. On the calibration event the thin-hull estimate now gives $T_{A,j} \leq CR_0^2/\kappa(A)$; the radius estimate at that terminal time gives precisely the preceding contradiction. □

The auxiliary target infinity is a normalization of local hull observables, not a newly asserted target of the bounded-state family. These four energies are observables of two fixed stopped physical laws in candidate-dependent auxiliary charts. They are not the canonical drivers of four fixed physical deleted states. Locality can identify suitable comparison laws when its hypotheses apply, but the stated uniform signed bounds must themselves be proved.

Corollary 31.4 (Polynomial properness of the calibration energy)**.** *In the bounded-prefix setup put*

$$\mathcal{B}_p(A) = \max_{j=1,2,\,\sigma=\pm} \mathcal{J}_{p,P^2h_0}(\mathrm{Law}(\widehat{W}^{A,j,\sigma})).$$

There is a constant C independent of A, with extended values allowed, such that

$$\kappa(A)^{p-1} \le C(1+\mathcal{B}_p(A)). \tag{TC12}$$

Proof. For $B = \mathcal{B}_p(A) < \infty$ choose $h_* = c(1+B)^{-1/(p-1)}$. A sufficiently small fixed $c > 0$ ensures $h_* \le h_0$, $2\sqrt{h_*} \le \delta$, and (TC6) is smaller than η, since

$$B^{1/p} h_*^{1-1/p} \le c^{(p-1)/p}, \qquad B h_*^{p-1} \le c^{p-1}.$$

The preceding argument gives $\kappa(A) \le CR_0^2/h_*$. Raising to the power $p-1$ proves (TC12). Infinite B is immediate. □

Use in common-tensor selection and the remaining gap. An energy-bounded calibration class has compact closure in the nondegenerate normalized tensor space. Its baseline class need not itself be closed. One may apply the earlier lower-semicontinuity and finite-intersection selection argument on its compact closure, provided every finite collection of target tests has a candidate in this same baseline class with the same prescribed envelopes. The coercivity result does not prove that simultaneous signed feasibility. Its four bounds are full energies, not four scalar constraints, and none is claimed as a consequence of unsigned RSW alone.

References

[1] S. E. Warschawski, *On differentiability at the boundary in conformal mapping*, Proc. Amer. Math. Soc. **12** (1961), no. 4, 614–620. https://doi.org/10.1090/S0002-9939-1961-0131524-8.

[2] O. Schramm and D. B. Wilson, *SLE coordinate changes*, New York J. Math. **11** (2005), 659–669. https://nyjm.albany.edu/j/2005/11-31.pdf.

[3] S. Sheffield, *Exploration trees and conformal loop ensembles*, arXiv:math/0609167. https://arxiv.org/abs/math/0609167.

[4] A. Kemppainen and S. Smirnov, *Random curves, scaling limits and Loewner evolutions*, Ann. Probab. **45** (2017), 698–779. https://arxiv.org/abs/1212.6215.

[5] K. Murali Rao, *Quasi-Martingales*, Math. Scand. **24** (1969), 79–92. https://doi.org/10.7146/math.scand.a-10921.

[6] G. F. Lawler, *Schramm–Loewner Evolution*, Park City lecture notes (2007), Lectures 3–4. https://math.uchicago.edu/~lawler/utah.pdf.

[7] Y. Yuan, *Topological characterisations of Loewner traces*, Indiana Univ. Math. J. **71** (2022), 1027–1046. https://arxiv.org/abs/2003.05535.

[8] J. Miller and S. Sheffield, *Imaginary geometry III: reversibility of SLE_κ for $\kappa \in (4,8)$*, Ann. of Math. **184** (2016), 455–486. https://annals.math.princeton.edu/2016/184-2/p03.

[9] S. Aumann, *Singularity of nearcritical percolation exploration paths*, ALEA Lat. Am. J. Probab. Math. Stat. **11** (2014), 229–252, Theorem 2.1 and Corollaries 2.2 and 5.1. https://alea.impa.br/articles/v11/11-08.pdf.

[10] S. Lalley, *Lecture 10: Change of measure and the Girsanov theorem*, Theorems 1–2. https://www.stat.uchicago.edu/~lalley/Courses/390/Lecture10.pdf.

[11] D. Zhan, *Ergodicity of the tip of an SLE curve*, Probab. Theory Related Fields **164** (2016), 333–360. https://arxiv.org/abs/1310.2573.

[12] G. F. Lawler and M. A. Rezaei, *Minkowski content and natural parameterization for the Schramm–Loewner evolution*, Ann. Probab. **43** (2015), 1082–1120. https://arxiv.org/abs/1211.4146.

[13] D. Zhan, *Decomposition of Schramm–Loewner evolution along its curve*, Stochastic Process. Appl. **129** (2019), 129–152. https://arxiv.org/abs/1509.05015.

[14] J. Miller and S. Sheffield, *Imaginary geometry IV: interior rays, whole-plane reversibility, and space-filling trees*, Probab. Theory Related Fields **169** (2017), 729–869, Theorem 1.20. https://arxiv.org/abs/1302.4738.

[15] D. Zhan, *SLE loop measures*, Ann. Inst. Henri Poincare Probab. Stat. **57** (2021), 701–738. https://arxiv.org/abs/1702.08026.

[16] D. Zhan, *Optimal Hölder continuity and dimension properties for SLE with Minkowski content parametrization*, Probab. Theory Related Fields **175** (2019), 447–466. https://arxiv.org/abs/1706.05603.

[17] L. Hörmander, *Hypoelliptic second order differential equations*, Acta Math. **119** (1967), 147–171, Theorem 1.1. https://doi.org/10.1007/BF02392081.

[18] *Research record: locality, terminal information, and planar exploration laws*, `progress.tex` in this repository, including the original-locality investigation and the corrected-locality continuation (27 October 2026).

[19] S. Lalley, G. Lawler and H. Narayanan, *A geometric interpretation of half-plane capacity*, Electron. Commun. Probab. **14** (2009), 566–571, Theorem 1. https://www.math.uchicago.edu/~lawler/LLN.pdf.

[20] L. Ahlfors and L. Bers, *Riemann's mapping theorem for variable metrics*, Ann. of Math. (2) **72** (1960), 385–404. https://www.cimat.mx/~mmoreno/teaching/fall10/AhlforsBers.pdf.

[21] J. Sung and Y. Wang, *Quasiconformal deformation of the chordal Loewner driving function and first variation of the Loewner energy*, Math. Ann. **390** (2024), 4789–4812. https://doi.org/10.1007/s00208-024-02866-0.